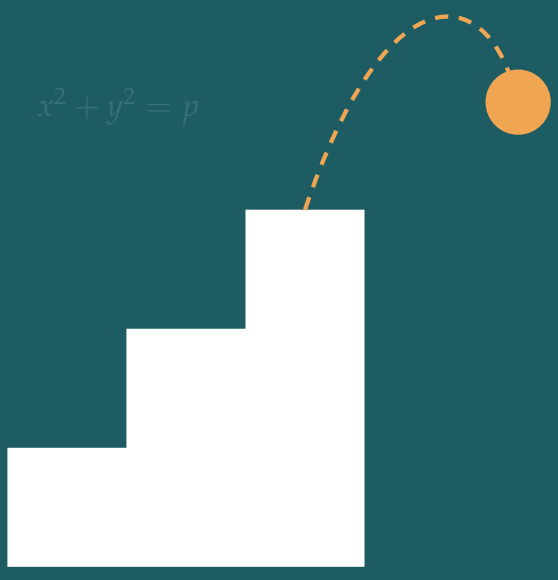
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2016 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2016 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2016 Worked Solutions

Worked solutions to all 37 questions of the 2016 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

First published online November 2026.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 2016, Question 1

Pure • Polynomial Identities and Factorisation

STEP I 2016, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q1

The key idea

The quotient $q_n(x)$ is really a polynomial: $x+1$ divides $x^{2n+1}+1$ because $2n+1$ is odd, and the quotient is an alternating geometric sum. So part (i) is a matter of expanding p_n and comparing coefficients. The pay-off in part (ii) comes from the shape of p_n: when $2n+1$ times x is a perfect square, p_n is a difference of two squares, and so it factorises.

Solution

Part (i)

For $x \neq -1$, summing the geometric series with first term 1 and ratio $-x$ gives

$$1 - x + x^2 - \cdots + x^{2n} = \sum_{k=0}^{2n} (-x)^k = \frac{1-(-x)^{2n+1}}{1+x} = \frac{1+x^{2n+1}}{1+x},$$

using $(-x)^{2n+1} = -x^{2n+1}$ as $2n+1$ is odd. So

$$q_n(x) = x^{2n} - x^{2n-1} + x^{2n-2} - \cdots - x + 1,$$

with all $2n+1$ coefficients equal to ± 1, alternating in sign. We now expand p_n in each case.

$n = 1$.

$$p_1(x) = (x+1)^2 - 3x = x^2 + 2x + 1 - 3x = x^2 - x + 1 = q_1(x).$$

$n = 2$.

$$\begin{aligned} p_2(x) &= (x+1)^4 - 5x(x^2+x+1) \\ &= (x^4 + 4x^3 + 6x^2 + 4x + 1) - (5x^3 + 5x^2 + 5x) \\ &= x^4 - x^3 + x^2 - x + 1 = q_2(x). \end{aligned}$$

$n = 3$. First $(x^2+x+1)^2 = x^4+2x^3+3x^2+2x+1$, so

$$\begin{aligned} p_3(x) &= (x+1)^6 - 7x(x^2+x+1)^2 \\ &= (x^6+6x^5+15x^4+20x^3+15x^2+6x+1) - (7x^5+14x^4+21x^3+14x^2+7x) \\ &= x^6 - x^5 + x^4 - x^3 + x^2 - x + 1 = q_3(x). \end{aligned}$$

So $p_n(x) \equiv q_n(x)$ for $x \neq -1$ when $n = 1, 2, 3$, as required.

$n = 4$. One value of x is enough to break an identity. At $x = 1$,

$$p_4(1) = 2^8 - 9 \cdot 1 \cdot 3^3 = 256 - 243 = 13, \qquad q_4(1) = \frac{1+1}{1+1} = 1,$$

so $p_4 \not\equiv q_4$. (Expanding fully gives $p_4(x) = x^8 - x^7 + x^6 + 2x^5 + 7x^4 + 2x^3 + x^2 - x + 1$, which goes wrong from the x^5 term onwards.)

Part (ii)

Ask yourself Which n and which x should we use, and how can p_n possibly factorise?

Matching $\dfrac{x^{2n+1}+1}{x+1}$ to the given numbers fixes x and n. The form of p_n then helps: $(x+1)^{2n} = ((x+1)^n)^2$ is always a square, and $(x^2+x+1)^{n-1}$ is a square when n is odd. So if $(2n+1)x$ is also a perfect square, $p_n(x)$ is a difference of two squares.

(a) Take $n = 1$ and $x = 300$, so $2n+1 = 3$ and $3x = 900 = 30^2$. By part (i),

$$\begin{aligned} \frac{300^3+1}{301} = q_1(300) = p_1(300) &= 301^2 - 3 \cdot 300 \\ &= 301^2 - 30^2 \\ &= (301-30)(301+30) = 271 \times 331. \end{aligned}$$

Neither factor is 1. (Directly, $300^2 - 300 + 1 = 89\,701 = 271 \times 331$.)

(b) Since $7^{49} = (7^7)^7$, take $n = 3$ (so $2n+1 = 7$) and $x = 7^7$. Then $(2n+1)x = 7 \cdot 7^7 = 7^8 = (7^4)^2$, a perfect square, and $n - 1 = 2$ is even. By part (i),

$$\begin{aligned} \frac{7^{49}+1}{7^7+1} = q_3(7^7) = p_3(7^7) &= (7^7+1)^6 - 7^8\,(7^{14}+7^7+1)^2 \\ &= [(7^7+1)^3]^2 - [7^4(7^{14}+7^7+1)]^2, \end{aligned}$$

so

$$\frac{7^{49}+1}{7^7+1} = \Big[(7^7+1)^3 - 7^4(7^{14}+7^7+1)\Big]\Big[(7^7+1)^3 + 7^4(7^{14}+7^7+1)\Big].$$

The second factor is clearly greater than 1. For the first, $(7^7+1)^3 > 7^{21}$ while

$$7^4(7^{14}+7^7+1) = 7^{18} + 7^{11} + 7^4 < 7^{19},$$

so the first factor exceeds $7^{21} - 7^{19} = 48 \cdot 7^{19} > 1$. Neither factor is 1, and we have the required factorisation. (Multiplying out the brackets, the factors could also be written $7^{21} + 3 \cdot 7^{14} + 3 \cdot 7^7 + 1 \mp (7^{18} + 7^{11} + 7^4)$.)

Checks

- Put $x = 1$ in part (i): $p_1(1) = 4 - 3 = 1$, $p_2(1) = 16 - 15 = 1$, $p_3(1) = 64 - 63 = 1$, all equal to $q_n(1) = 1$, while $p_4(1) = 13$.
- In (a), $271 \times 331 = 89\,701 = 300^2 - 300 + 1$.
- In (b), a computer check of the exact integers confirms that the product of the two factors equals $(7^{49} + 1)/(7^7 + 1)$; the factors are about 5.57×10^{17} and 5.60×10^{17}.

Common slips

- Forgetting to say why neither factor is 1; in (b) this needs a short size estimate.
- In (b), using $n = 24$ or $x = 7$: the exponent 49 must be written as 7×7, giving $x = 7^7$ and $2n + 1 = 7$.

Beyond STEP

The trick here is a special case of Aurifeuillean factorisation: identities such as $x^4 + 4y^4 = (x^2 + 2y^2 - 2xy)(x^2 + 2y^2 + 2xy)$, which split numbers of the form $a^k \pm 1$ further than the obvious algebraic factors would suggest. They come from cyclotomic polynomials, which appear in first-year algebra and number theory courses, and they were used in the hunt to factor large numbers of the form $2^k \pm 1$. Davenport's *The Higher Arithmetic* is a good place to read more about factorisation and primes.

WORKED SOLUTION

STEP I 2016, Question 4

Pure • Curvature of Curves

STEP I 2016, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q4

The key idea

The expression for κ looks like a hard second-order differential equation, but part (i) is there to show that it is an exact derivative. Writing $p = f'(x)$, the curvature is the x-derivative of $p/\sqrt{1+p^2}$, so a constant curvature can be integrated once at sight. A second integration is then routine, and the answer turns out to be a familiar shape.

Solution

Part (i)

Using the product rule on $z(1+z^2)^{-1/2}$,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}z}\left(\frac{z}{(1+z^2)^{1/2}}\right) &= (1+z^2)^{-1/2} - z\cdot z(1+z^2)^{-3/2} \\ &= \frac{(1+z^2)-z^2}{(1+z^2)^{3/2}} = \frac{1}{(1+z^2)^{3/2}}.\end{aligned}$$

Part (ii)

Let κ be a non-zero constant and write $p = f'(x)$, so $f''(x) = \dfrac{\mathrm{d}p}{\mathrm{d}x}$. By part (i) and the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{p}{\sqrt{1+p^2}}\right) = \frac{1}{(1+p^2)^{3/2}}\frac{\mathrm{d}p}{\mathrm{d}x} = \frac{f''(x)}{\left(1+(f'(x))^2\right)^{3/2}} = \kappa.$$

Integrating with respect to x, there is a constant, which we write as $-\kappa a$ (possible since $\kappa \neq 0$), with

$$\frac{p}{\sqrt{1+p^2}} = \kappa(x-a). \qquad (*)$$

Ask yourself How do we solve $(*)$ for p without losing track of signs?

Squaring is safe if we remember what it throws away. Since $\sqrt{1+p^2} > 0$, $(*)$ says that p has the same sign as $\kappa(x-a)$, and it also says $|\kappa(x-a)| < 1$, because $|p| < \sqrt{1+p^2}$. Squaring, $p^2 = \kappa^2(x-a)^2(1+p^2)$, so

$$p^2 = \frac{\kappa^2(x-a)^2}{1-\kappa^2(x-a)^2}, \qquad \text{and, choosing the sign as above,} \qquad p = \frac{\kappa(x-a)}{\sqrt{1-\kappa^2(x-a)^2}}.$$

Conversely, this p does satisfy $(*)$, since $1+p^2 = 1/\left(1-\kappa^2(x-a)^2\right)$.

Now integrate again. Since $\dfrac{\mathrm{d}}{\mathrm{d}x}\sqrt{1-\kappa^2(x-a)^2} = \dfrac{-\kappa^2(x-a)}{\sqrt{1-\kappa^2(x-a)^2}}$, we get, for some constant b,

$$y = b - \frac{1}{\kappa}\sqrt{1-\kappa^2(x-a)^2}, \qquad |x-a| < \frac{1}{|\kappa|}.$$

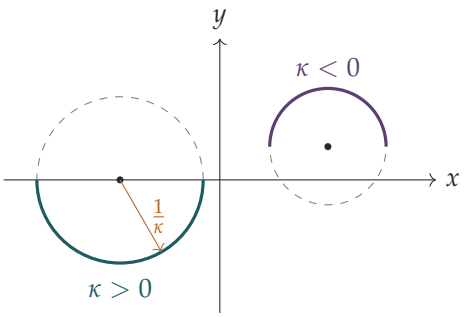
Rearranging, $\kappa(y-b) = -\sqrt{1-\kappa^2(x-a)^2}$, and squaring gives

$$(x-a)^2 + (y-b)^2 = \frac{1}{\kappa^2}.$$

So every curve of constant non-zero signed curvature κ is part of a circle of radius $1/|\kappa|$, centre (a,b), and every such arc arises: reversing the steps, each function above has signed curvature exactly κ. More precisely, $\kappa(y-b) \le 0$, so:

- if $\kappa > 0$, the curve is the lower half $y = b - \sqrt{\kappa^{-2} - (x-a)^2}$, lying below its centre and bending upwards ($\mathrm{f}'' > 0$);
- if $\kappa < 0$, the curve is the upper half $y = b + \sqrt{\kappa^{-2} - (x-a)^2}$, lying above its centre and bending downwards ($\mathrm{f}'' < 0$).

The endpoints $x = a \pm 1/|\kappa|$ are excluded, since there the gradient is infinite; a curve $y = \mathrm{f}(x)$ can only be a graph, so it can never be more than an open semicircle (or a piece of one).



Geometrical significance

For these curves $|\kappa|$ is the reciprocal of the radius of the circle, so a larger $|\kappa|$ means a smaller, more tightly bending circle. The sign of κ records which way the curve bends:

$\kappa > 0$ when the centre of the circle lies above the curve (the curve is concave up), and $\kappa < 0$ when the centre lies below it.

Checks

- Take $\kappa = \frac{1}{2}$, $a = b = 0$: the curve is $y = -\sqrt{4 - x^2}$, the lower half of the circle of radius 2. At $x = 1$, $y = -\sqrt{3}$, $y' = x/\sqrt{4 - x^2} = 1/\sqrt{3}$ and $y'' = 4/(4 - x^2)^{3/2} = 4/3^{3/2}$, so $1 + y'^2 = \frac{4}{3}$ and $\kappa = \dfrac{4/3^{3/2}}{(4/3)^{3/2}} = \frac{1}{2}$, as it should be.
- A straight line has $f'' = 0$, so $\kappa = 0$: the limit of a circle as the radius $1/|\kappa| \to \infty$, which is why the question excludes $\kappa = 0$.
- Solving $f'' = \kappa(1 + f'^2)^{3/2}$ step by step by computer from random starting values, the points stayed on the predicted circle to six decimal places.

Common slips

- Forgetting the constant of integration at the first stage, which loses the freedom to move the centre horizontally.
- Taking the wrong square root for p, which gives the semicircle on the wrong side of the centre and the wrong sign of κ.
- Saying only "κ is the radius": it is the reciprocal of the radius, with a sign showing the direction of bending.

Beyond STEP

Curvature is the starting point of the differential geometry of curves. For a curve parametrised by arc length s, $\kappa = \mathrm{d}\psi/\mathrm{d}s$, where ψ is the angle the tangent makes with the x-axis; here $p/\sqrt{1 + p^2} = \sin\psi$, so $(*)$ is really the statement that ψ changes at a constant rate along the curve. In three dimensions curvature is joined by torsion, and the Frenet–Serret formulae show that these two functions determine a curve up to rigid motion. Pressley's *Elementary Differential Geometry* is a friendly first-year-level introduction.

WORKED SOLUTION

STEP I 2016, Question 5

Pure • Touching Circles

STEP I 2016, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q5

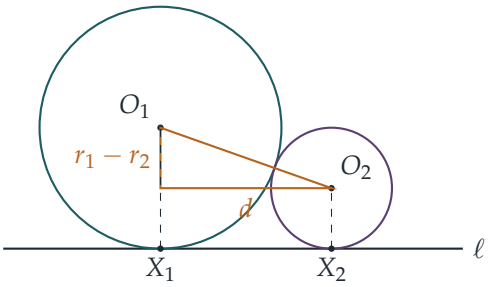
The key idea

Everything rests on one fact: if two circles touch each other and both touch the same straight line, the distance between their points of contact with the line is $2\sqrt{r_1 r_2}$. That comes from a single right-angled triangle. After that, the substitution $x = 1/\sqrt{a}$, $y = 1/\sqrt{b}$, $z = 1/\sqrt{c}$ turns $(**)$ into a symmetric quartic which factorises into four linear factors, and one of those factors is exactly $(*)$.

Solution

Part (i)

Two touching circles on a line. Take two circles of radii r_1 and r_2 touching each other externally, both on the same side of a line ℓ and both tangent to it, at points X_1 and X_2. The radii to X_1 and X_2 are perpendicular to ℓ, so the centres O_1, O_2 lie at heights r_1 and r_2 above ℓ. Since the circles touch externally, $O_1O_2 = r_1 + r_2$.



Drop the perpendicular from O_2 to the radius O_1X_1 (shown with $r_1 \geq r_2$). This gives a right-angled triangle with hypotenuse $r_1 + r_2$, vertical side $|r_1 - r_2|$ and horizontal side $d = X_1X_2$. By Pythagoras,

$$d^2 = (r_1 + r_2)^2 - (r_1 - r_2)^2 = 4r_1r_2, \qquad \text{so} \qquad X_1X_2 = 2\sqrt{r_1 r_2}.$$

Applying it three times. In the diagram each pair of circles touches and all three touch the line at P, Q, R, with Q between P and R. So

$$PQ = 2\sqrt{ab}, \qquad QR = 2\sqrt{bc}, \qquad PR = 2\sqrt{ac},$$

and $PQ + QR = PR$ gives $\sqrt{ab} + \sqrt{bc} = \sqrt{ac}$. Dividing by $\sqrt{abc} > 0$,

$$\frac{1}{\sqrt{c}} + \frac{1}{\sqrt{a}} = \frac{1}{\sqrt{b}},$$

which is $(*)$.

Deducing $(**)$. Write $x = \dfrac{1}{\sqrt{a}}$, $y = \dfrac{1}{\sqrt{b}}$, $z = \dfrac{1}{\sqrt{c}}$, so that $\dfrac{1}{a} = x^2$ and so on, and $(*)$ says $y = x + z$. Then $(**)$ reads $2(x^4 + y^4 + z^4) = (x^2 + y^2 + z^2)^2$; expanding the right-hand side and rearranging, it is equivalent to

$$F := x^4 + y^4 + z^4 - 2x^2y^2 - 2y^2z^2 - 2z^2x^2 = 0.$$

? **Ask yourself** How can this quartic be factorised?

Group it as a difference of two squares. Since $(x^2 + y^2 - z^2)^2 = x^4 + y^4 + z^4 + 2x^2y^2 - 2y^2z^2 - 2z^2x^2$,

$$\begin{aligned} F &= (x^2 + y^2 - z^2)^2 - (2xy)^2 \\ &= \left(x^2 - 2xy + y^2 - z^2\right)\left(x^2 + 2xy + y^2 - z^2\right) \\ &= \left((x-y)^2 - z^2\right)\left((x+y)^2 - z^2\right) \\ &= (x - y - z)(x - y + z)(x + y - z)(x + y + z). \end{aligned}$$

By $(*)$, $x - y + z = 0$, so $F = 0$ and $(**)$ holds.

Part (ii)

Now a, b, c are any positive numbers with $b < c < a$ satisfying $(**)$. Define x, y, z as before; they are positive, and because $t \mapsto 1/\sqrt{t}$ is decreasing on $t > 0$, the condition $b < c < a$ gives

$$y > z > x > 0.$$

As shown in part (i), $(**)$ is equivalent to

$$(x - y - z)(x - y + z)(x + y - z)(x + y + z) = 0,$$

so one of the four factors vanishes. Three of them cannot:

- $x + y + z > 0$, since all three are positive;
- $x + y - z > 0$, since $y > z$ and $x > 0$;
- $x - y - z < 0$, since $x < y$ and $z > 0$.

Hence $x - y + z = 0$, that is $y = x + z$, or

$$\frac{1}{\sqrt{b}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{c}},$$

which is $(*)$. (Only the fact that b is the smallest of the three was needed.)

Checks

- Take $a = 4$, $c = 1$. Then $(*)$ gives $1/\sqrt{b} = \frac{1}{2} + 1 = \frac{3}{2}$, so $b = \frac{4}{9}$. Now $2\left(\frac{1}{16} + \frac{81}{16} + 1\right) = \frac{49}{4}$ and $\left(\frac{1}{4} + \frac{9}{4} + 1\right)^2 = \left(\frac{7}{2}\right)^2 = \frac{49}{4}$, as $(**)$ requires.
- Why is the ordering needed in (ii)? With $a = 4$, $c = 1$, the relation $(**)$ is a quadratic in $1/b$ with two roots: $b = \frac{4}{9}$ and $b = 4$ (from $1/\sqrt{b} = |x - z| = \frac{1}{2}$). The second has b largest rather than smallest, and it satisfies $(**)$ but not $(*)$: here it is a and b that are tied, through $1/\sqrt{c} = 1/\sqrt{a} + 1/\sqrt{b}$. Swapping the roles of the letters corresponds to swapping which circle sits in the middle.
- A computer check on thousands of random configurations of three touching circles on a line confirmed both $(*)$ and $(**)$.

Common slips

- Using the distance between the centres, $r_1 + r_2$, as the distance between the points of contact: the centres are at different heights.
- In part (ii), concluding $(*)$ from the factorisation without ruling out the other three factors; this is exactly where $b < c < a$ is used.

Beyond STEP

Relation $(**)$ is the degenerate case of Descartes' circle theorem: for four mutually touching circles with curvatures $k_i = 1/r_i$, one has $2\sum k_i^2 = \left(\sum k_i\right)^2$, and a straight line counts as a circle of curvature 0. Repeatedly fitting new circles into the gaps produces the Apollonian gasket, where integer curvatures lead to surprising number theory. The factorisation of F is also the one behind Heron's formula, $16\Delta^2 = -F$ with x, y, z the sides of a triangle. For more on circle geometry and inversion, which makes these results transparent, see Needham, *Visual Complex Analysis*.

WORKED SOLUTION

STEP I 2016, Question 6

Pure • Vector Geometry

STEP I 2016, Question 6

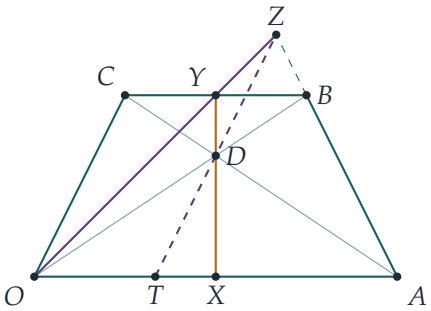
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q6

The key idea

Every point in the figure is a meeting point of two lines, so we write each line in vector form and equate. Because **a** and **b** are not parallel, any vector can be written as $\alpha\mathbf{a} + \beta\mathbf{b}$ in only one way, so equating two expressions for the same point lets us compare coefficients of **a** and **b** separately. Working in the basis **a**, **b** throughout keeps every step to two short linear equations.

Solution



The forms of c and x

Since CB is parallel to OA, the vector $\overrightarrow{BC} = \mathbf{c} - \mathbf{b}$ is a scalar multiple of **a**, say $\mathbf{c} - \mathbf{b} = k\mathbf{a}$, which gives $\mathbf{c} = k\mathbf{a} + \mathbf{b}$. Going round the quadrilateral $O \to A \to B \to C$, the side BC must run in the opposite direction to OA (if it ran the same way, the sides AB and CO would cross and $OABC$ would not be a genuine quadrilateral), and $C \neq B$. So

$$k < 0.$$

Any negative value is possible; $k = -1$ is the parallelogram.

Since X lies on OA, **x** is a multiple of **a**, say $\mathbf{x} = m\mathbf{a}$, and X strictly between O and A

means

$$0 < m < 1.$$

Also $\mathbf{a}$ and $\mathbf{b}$ are not parallel, as O, A, B are not collinear.

The point Z

Finding D. Points of OB are $\lambda\mathbf{b}$; points of AC are

$$\mathbf{a} + \mu(\mathbf{c} - \mathbf{a}) = (1 - \mu + \mu k)\mathbf{a} + \mu\mathbf{b}.$$

Comparing coefficients, $\lambda = \mu$ and $1 - \mu(1-k) = 0$. As $k < 0$, $1 - k > 0$, so $\mu = \frac{1}{1-k}$ and

$$\mathbf{d} = \frac{\mathbf{b}}{1-k}.$$

Finding Y. Points of BC are $\mathbf{b} + t\mathbf{a}$. Points of XD are

$$m\mathbf{a} + s(\mathbf{d} - m\mathbf{a}) = m(1-s)\mathbf{a} + \frac{s}{1-k}\,\mathbf{b}.$$

The $\mathbf{b}$-coefficients give $s = 1 - k$, and then the $\mathbf{a}$-coefficient is $m(1-(1-k)) = mk$. So

$$\mathbf{y} = \mathbf{b} + mk\,\mathbf{a}.$$

Finding Z. Points of OY are $\rho(\mathbf{b} + mk\mathbf{a})$; points of AB are $\mathbf{a} + \sigma(\mathbf{b} - \mathbf{a}) = (1-\sigma)\mathbf{a} + \sigma\mathbf{b}$. Comparing coefficients, $\rho = \sigma$ and $\rho mk = 1 - \sigma$, so $\rho(mk+1) = 1$. Provided $mk + 1 \neq 0$,

$$\mathbf{z} = \frac{\mathbf{b} + mk\,\mathbf{a}}{mk+1},$$

as required. (If $mk = -1$, which can happen when $k < -1$, then $\mathbf{y} = \mathbf{b} - \mathbf{a}$, so OY is parallel to AB and Z does not exist.)

Ask yourself Is Z on the side AB itself?

No: $\sigma = \frac{1}{mk+1} > 1$ because $-1 < mk < 0$, so Z lies on AB extended beyond B, as in the figure. That doesn't matter, since the question concerns the lines.

The point T and the two relations

Points of DZ are $\mathbf{d} + \tau(\mathbf{z} - \mathbf{d})$. The $\mathbf{b}$-coefficient is

$$\frac{1}{1-k} + \tau\left(\frac{1}{mk+1} - \frac{1}{1-k}\right) = \frac{1}{1-k} - \tau\,\frac{k(1+m)}{(mk+1)(1-k)},$$

using $(1-k) - (mk+1) = -k(1+m)$. For T on OA this must vanish, so $\tau = \dfrac{mk+1}{k(1+m)}$ (recall $k \neq 0$). The $\mathbf{a}$-coefficient is then

$$\tau \cdot \frac{mk}{mk+1} = \frac{mk+1}{k(1+m)} \cdot \frac{mk}{mk+1} = \frac{m}{1+m}.$$

Hence

$$\mathbf{t} = \frac{m}{1+m}\,\mathbf{a}.$$

Remarkably, k has disappeared: T depends only on where X is.

Now write $a = |\mathbf{a}| = OA$. Since $0 < \frac{m}{1+m} < 1$, T lies between O and A, so

$$OT = \frac{ma}{1+m}, \qquad TA = a - \frac{ma}{1+m} = \frac{a}{1+m}, \qquad OX = ma.$$

Therefore

$$OT \times OA = \frac{ma^2}{1+m} = ma \times \frac{a}{1+m} = OX \times TA,$$

and

$$\frac{1}{OX} + \frac{1}{OA} = \frac{1}{ma} + \frac{1}{a} = \frac{1+m}{ma} = \frac{1}{OT},$$

which are both results required.

Checks

Try the trapezium in the figure: $\mathbf{a} = (4,0)$, $\mathbf{b} = (3,2)$, $k = -\frac{1}{2}$, $m = \frac{1}{2}$. Then $D = (2, \frac{4}{3})$, $Y = (2,2)$, $Z = (\frac{8}{3}, \frac{8}{3})$ and $T = (\frac{4}{3}, 0)$; you can confirm that T, D, Z are collinear, and $\frac{1}{OT} = \frac{3}{4} = \frac{1}{2} + \frac{1}{4} = \frac{1}{OX} + \frac{1}{OA}$. A computer check on 20,000 random trapezia agreed with every formula above to within 10^{-10}. The limiting case $m \to 1$ ($X \to A$) gives $OT = \frac{1}{2}OA$, consistent with $\frac{1}{OT} = \frac{2}{OA}$.

Common slips

- Taking $k > 0$: with $\mathbf{c} = \mathbf{b} + k\mathbf{a}$ and $k > 0$ the "quadrilateral" crosses itself.
- Equating coefficients without saying why it is allowed: it relies on $\mathbf{a}$ and $\mathbf{b}$ not being parallel.
- Using the same parameter letter for two different lines, which forces a false coincidence.

Beyond STEP

The fact that T does not depend on k (nor on the angle between $\mathbf{a}$ and $\mathbf{b}$) shows this is really a theorem of affine, and indeed projective, geometry: only incidences and ratios along a line matter. The relation $\frac{1}{OT} = \frac{1}{OX} + \frac{1}{OA}$ is the sort of statement that cross-ratios and harmonic ranges handle elegantly, and the same construction, repeated, places points at $\frac{m}{1+2m}\mathbf{a}$, $\frac{m}{1+3m}\mathbf{a}$, and so on. For more, see Coxeter and Greitzer, *Geometry Revisited*, or Coxeter, *Projective Geometry*.

WORKED SOLUTION

STEP I 2016, Question 7

Pure • Factorisation Modulo Four

STEP I 2016, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q7

This question has a full guided treatment in *Beyond STEP*, Section 1.7, *When Factorisation Fails*: https://beyondstep.keithmcnulty.org/ch01/factorisation-fails.html

The key idea

Everything follows from working modulo 4. Odd numbers are $\equiv 1$ or $\equiv 3 \equiv -1 \pmod 4$, so a product of odd numbers lies in S or T according to whether it has an even or odd number of factors from T. That one parity rule does parts (ii), (iii) and (iv)(a). Part (iv)(b) asks you to see that S-primes need not be primes, and that loses unique factorisation.

Solution

Part (i)

Every odd positive integer leaves remainder 1 or 3 on division by 4, and no even number does. So $S \cup T$ is the set of all *odd positive integers*. No integer leaves two different remainders, so $S \cap T$ is the *empty set*.

Part (ii)

Take $4a+1$ and $4b+1$ in S, with $a, b \geq 0$ integers. Then

$$(4a+1)(4b+1) = 16ab + 4a + 4b + 1 = 4(4ab + a + b) + 1,$$

which is a positive integer leaving remainder 1, so it is in S.

For T, the answer is *no*. For example $3 \in T$ but $3 \times 3 = 9 = 4 \times 2 + 1 \in S$, and $S \cap T$ is empty. In fact this always happens:

$$(4a+3)(4b+3) = 16ab + 12a + 12b + 9 = 4(4ab + 3a + 3b + 2) + 1 \in S.$$

We will also need the mixed product:

$$(4a+1)(4b+3) = 16ab + 12a + 4b + 3 = 4(4ab + 3a + b) + 3 \in T.$$

In short, writing T-numbers as $\equiv -1 \pmod 4$: a product of odd numbers is $\equiv (-1)^k \pmod 4$, where k is the number of factors from T. So the product is in S if k is even and in T if k is odd.

Part (iii)

Let $n \in T$ be composite, and write $n = p_1 p_2 \cdots p_r$ as a product of primes. Since n is odd, every p_i is odd, so by part (i) each p_i is in S or in T.

Suppose, for contradiction, that every p_i is in S. Applying part (ii) repeatedly, $p_1 p_2 \in S$, then $(p_1 p_2)p_3 \in S$, and so on, so $n \in S$. But $n \in T$ and $S \cap T = \varnothing$, which is a contradiction. Hence at least one prime factor of n lies in T. (In parity terms: $n \equiv -1$ needs an odd number, so at least one, of factors $\equiv -1$.)

Part (iv)(a)

Note first that $1 \notin T$, so every element of T is at least 3.

Every $n \in T$ is T-prime or a product of T-primes. We use strong induction on n. If n is T-prime, we are done. Otherwise $n = t_1 t_2 \cdots t_k$ with $k \geq 2$ and each $t_i \in T$. Each $t_i \geq 3$, so each $t_i \leq n/3 < n$, and by the induction hypothesis each t_i is T-prime or a product of T-primes. Substituting these into $n = t_1 \cdots t_k$ writes n as a product of T-primes.

> **Ask yourself** We have a product of T-primes, but why must there be an *odd* number of them?

Every T-prime is in T, so it is $\equiv -1 \pmod 4$. A product of m of them is therefore $\equiv (-1)^m \pmod 4$, by the rule at the end of part (ii). If m were even, the product would be $\equiv 1$, so it would lie in S, not T. Since the product is $n \in T$, m is odd. So every $n \in T$ is either T-prime or the product of an odd number of T-primes, as required.

Note that T-primes need not be primes: $15 = 3 \times 5$ is T-prime, because $5 \notin T$ and the only factorisation into numbers bigger than 1 is 3×5.

Part (iv)(b)

The same idea as the remark about 15 works in S: a product of two primes from T lies in S but has no factorisation within S. Take

$$9 = 3 \times 3, \qquad 21 = 3 \times 7, \qquad 49 = 7 \times 7.$$

Each is in S. Each is S-prime: its only divisors other than 1 and itself are 3 or 7, which are in T, not S. (For 21, the divisors are $1, 3, 7, 21$.) Now

$$441 = 9 \times 49 = 21 \times 21,$$

and $441 = 4 \times 110 + 1 \in S$. These are two genuinely different factorisations of 441 into S-primes: the first uses 9 and 49, the second uses 21 twice. So 441 is an example.

Ask yourself How would you have found this?

In the ordinary integers, $441 = 3^2 \times 7^2$, and the primes $3, 7$ are in T. Inside S we cannot split off a single 3 or 7, so the T-primes must be paired up. With two 3s and two 7s there are two ways to pair them: $(3 \cdot 3)(7 \cdot 7)$ or $(3 \cdot 7)(3 \cdot 7)$. Any p^2q^2 with distinct primes $p, q \in T$ works in the same way, such as $9 \times 121 = 33 \times 33$.

Checks

- Try small cases: $5 \times 9 = 45 \in S$, $3 \times 7 = 21 \in S$, $5 \times 7 = 35 \in T$.
- For (iii): $15 = 3 \times 5$ and $35 = 5 \times 7$ are in T, and each has a prime factor (3, 7) in T.
- By computer, for every $n \in T$ below 3000, every factorisation into T-primes has an odd number of factors. The T-primes start $3, 7, 11, 15, 19, 23, 31, 35, \ldots$. Note $27 = 3 \times 3 \times 3$ is not T-prime: three factors, which is odd.

Common slips

- Claiming that the product of two members of T is in T without testing a case: $3 \times 3 = 9$ settles it at once.
- Forgetting that an X-prime need not be a prime. This is the whole point of (iv)(b): 9 and 21 are S-prime.
- In (iv)(a), only proving that the number of factors is odd and forgetting to show that a factorisation into T-primes exists at all.

Beyond STEP

The set S is closed under multiplication, and its "primes" do not factorise uniquely. This is a classic toy model (sometimes called the Hilbert numbers) of what goes wrong in rings such as $\mathbb{Z}[\sqrt{-5}]$, where $6 = 2 \times 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$. Repairing unique factorisation in such rings led to ideals and algebraic number theory. Davenport's *The Higher Arithmetic* explains why unique factorisation in $\mathbb{Z}$ is a theorem that needs proof, and Allenby's *Rings, Fields and Groups* covers rings where it fails.

WORKED SOLUTION

STEP I 2016, Question 8

Pure • Generating Functions

STEP I 2016, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q8

This question has a full guided treatment in *Beyond STEP*, Section 9.4, *Sequences as Functions*: https://beyondstep.keithmcnulty.org/ch09/generating-functions.html

The key idea

A generating function packages a whole sequence into a single function, and operations on the sequence become algebra on the function. In part (i) the binomial series for negative powers of $1-x$ produces the coefficients we want, and we build n^2 out of sequences we already know. In part (ii) we turn a recurrence into an equation for f by multiplying by x^n and summing: shifting the index of a sequence is the same as multiplying its generating function by x.

Solution

Part (i)

By the binomial series with index -2,

$$(1-x)^{-2} = \sum_{n\geq 0} \binom{-2}{n}(-x)^n = 1 + 2x + 3x^2 + 4x^3 + \cdots = \sum_{n\geq 0}(n+1)x^n,$$

since $\binom{-2}{n}(-1)^n = \frac{2\cdot 3\cdots(n+1)}{n!} = n+1$. Multiplying by x,

$$x(1-x)^{-2} = \sum_{n\geq 0}(n+1)x^{n+1} = \sum_{n\geq 0} nx^n,$$

where re-indexing the sum lets us include $n=0$ harmlessly (its term is 0). So $u_n = n$ has generating function $x(1-x)^{-2}$, as required.

Similarly, with index -3,

$$\binom{-3}{n}(-1)^n = \frac{3\cdot 4\cdots(n+2)}{n!} = \frac{(n+1)(n+2)}{2},$$

so

$$(1-x)^{-3} = \sum_{n\geq 0} \frac{(n+1)(n+2)}{2} x^n.$$

Multiplying by x and re-indexing as before,

$$x(1-x)^{-3} = \sum_{n\geq 0} \frac{n(n+1)}{2} x^n,$$

so the sequence is $u_n = \frac{1}{2}n(n+1)$: that is, $0, 1, 3, 6, 10, \ldots$, the triangular numbers.

Ask yourself How can we get n^2 from the two sequences we now know?

Generating functions are linear: if u_n and v_n have generating functions f and g, then $\alpha u_n + \beta v_n$ has generating function $\alpha \mathrm{f} + \beta \mathrm{g}$. Since

$$n^2 = 2 \cdot \tfrac{1}{2}n(n+1) - n,$$

the generating function for $u_n = n^2$ is

$$\frac{2x}{(1-x)^3} - \frac{x}{(1-x)^2} = \frac{2x - x(1-x)}{(1-x)^3} = \frac{x(1+x)}{(1-x)^3}.$$

Part (ii)(a)

For each $n \geq 1$ we have $u_n x^n = k u_{n-1} x^n$. Summing over all $n \geq 1$:

$$\sum_{n\geq 1} u_n x^n = kx \sum_{n\geq 1} u_{n-1} x^{n-1}$$
$$\mathrm{f}(x) - u_0 = kx\, \mathrm{f}(x),$$

because the left-hand sum is $\mathrm{f}(x)$ without its constant term, and the right-hand sum, with $m = n-1$, is $\sum_{m\geq 0} u_m x^m = \mathrm{f}(x)$. As $u_0 = a$, this is $\mathrm{f}(x) = a + kx\,\mathrm{f}(x)$, as required. Hence

$$\mathrm{f}(x) = \frac{a}{1-kx}.$$

(As a check, expanding gives $a + akx + ak^2x^2 + \cdots$, and indeed $u_n = ak^n$.)

Part (ii)(b)

Multiply $u_n = u_{n-1} + u_{n-2}$ by x^n and sum over $n \geq 2$ (the recurrence only holds from $n = 2$, so this is where we must start):

$$\sum_{n\geq 2} u_n x^n = x \sum_{n\geq 2} u_{n-1} x^{n-1} + x^2 \sum_{n\geq 2} u_{n-2} x^{n-2}.$$

Ask yourself Which terms of f are missing from each of these three sums?

The left-hand side lacks the terms $u_0 + u_1 x$; the first sum on the right runs over $u_1 x + u_2 x^2 + \cdots$ and so lacks only u_0; the second sum is all of $\mathrm{f}(x)$. Therefore

$$\mathrm{f}(x) - u_0 - u_1 x = x\big(\mathrm{f}(x) - u_0\big) + x^2\,\mathrm{f}(x).$$

With $u_0 = 0$ and $u_1 = 1$ this becomes $\mathrm{f}(x) - x = x\,\mathrm{f}(x) + x^2\,\mathrm{f}(x)$, so

$$\mathrm{f}(x) = \frac{x}{1 - x - x^2}.$$

This is the generating function of the Fibonacci numbers $0, 1, 1, 2, 3, 5, 8, \ldots$.

Checks

- Expanding $x(1+x)(1-x)^{-3} = (x + x^2)(1 + 3x + 6x^2 + 10x^3 + \cdots)$ gives $x + 4x^2 + 9x^3 + 16x^4 + \cdots$, the squares, as it should.
- Long division of x by $1 - x - x^2$ gives $x + x^2 + 2x^3 + 3x^4 + 5x^5 + \cdots$.
- At $x = 0.1$: the sum $\sum n^2 (0.1)^n$ and $0.11/0.9^3 = 0.150891\ldots$ agree, and $\sum u_n (0.1)^n$ for the Fibonacci numbers matches $0.1/0.89 = 10/89 = 0.112359\ldots$. These were confirmed by computer, along with the first 30 coefficients of every generating function in the question.

Common slips

- Getting the binomial coefficients for negative indices wrong: the signs from $\binom{-r}{n}$ and from $(-x)^n$ cancel, so every coefficient is positive.
- In (ii)(b), summing from $n = 0$ or $n = 1$, where the recurrence does not hold, or forgetting to subtract the initial terms u_0 and $u_1 x$.
- Writing the triangular numbers as $\frac{1}{2}n(n-1)$: check $u_1 = 1$.

Beyond STEP

Factorising $1 - x - x^2 = (1 - \phi x)(1 - \psi x)$, with $\phi, \psi = \frac{1}{2}(1 \pm \sqrt{5})$, and using partial fractions together with part (ii)(a) gives Binet's formula $u_n = (\phi^n - \psi^n)/\sqrt{5}$. The same method solves any linear recurrence with constant coefficients, and generating functions are a central tool in combinatorics and probability (where they become probability generating functions). Grimmett and Welsh, *Probability: An Introduction*, develops the probability side; Wilf's *generatingfunctionology* is a lively, freely available book on the combinatorial side.

WORKED SOLUTION

STEP I 2016, Question 9

Mechanics • Limiting Friction Equilibrium

STEP I 2016, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q9

The key idea

Take moments about A: the two forces at the wall then drop out and the normal reaction at the rail comes straight out in terms of θ, with no friction involved. Resolving horizontally and vertically finishes the job, once we know which way each friction force acts. That is the real content of the question: friction opposes the motion the rod is about to make, so we must decide whether the end A is about to slide down the wall or up it.

Solution

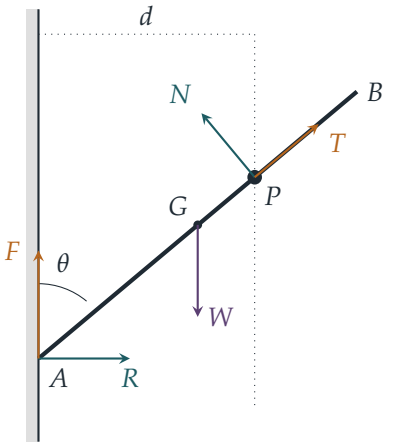
Model the rod as a uniform rigid rod of weight W, with centre G at distance a from A, and the rail as a fixed line meeting the rod at a single point P. Since the horizontal distance from A to the rail is d and the rod makes angle θ with the vertical,

$$AP = \frac{d}{\sin\theta}.$$

The condition $a\sin\theta < d$ says that G lies between A and P.

The forces are:

- at A: a normal reaction R, horizontal and away from the wall, and friction F along the wall;
- at P: a normal reaction N perpendicular to the rod, and friction T along the rod;
- the weight W at G.



The arrows for F and T show the directions for the first case below.

Moments about A

The moment of N about A is $N \cdot AP$, since N is perpendicular to the rod; T acts along the rod and has no moment about A, and neither do R and F. The weight has lever arm $a\sin\theta$. So

$$N\frac{d}{\sin\theta} = Wa\sin\theta \quad\Longrightarrow\quad N = \frac{Wa\sin^2\theta}{d}. \tag{1}$$

This holds whichever way the rod is about to slip.

Which way does it slip?

> **Ask yourself** Is A about to slide down the wall or up it?

Suppose first that there were no friction at all. Resolving vertically would need $N\sin\theta = W$, and with (1) this means $a\sin^3\theta = d$. Here $a\sin^3\theta < a\sin\theta < d$, so by (1) $N\sin\theta < W$: the normal reactions alone cannot hold the rod up, and friction must supply the rest. The rod is about to slip *downwards*: A slides down the wall, θ decreases, $AP = d/\sin\theta$ grows, and so the rod slides down through the rail towards the wall. Friction opposes this, so F acts up the wall and T acts up the rod (towards B), as drawn. In limiting equilibrium at both contacts,

$$F = \mu R, \qquad T = \lambda N.$$

Resolving

The unit vector up the rod is $(\sin\theta, \cos\theta)$ and the direction of N is $(-\cos\theta, \sin\theta)$ (towards the wall and upwards). Resolving horizontally:

$$R - N\cos\theta + \lambda N\sin\theta = 0 \quad\Longrightarrow\quad R = N(\cos\theta - \lambda\sin\theta).$$

(For the wall to push, we need $R \geq 0$, so this case requires $\tan\theta \leq 1/\lambda$.) Resolving vertically:

$$\begin{aligned} W &= \mu R + N\sin\theta + \lambda N\cos\theta \\ &= N\big(\mu\cos\theta - \lambda\mu\sin\theta + \sin\theta + \lambda\cos\theta\big) \\ &= N\big((\lambda+\mu)\cos\theta + (1-\lambda\mu)\sin\theta\big). \end{aligned}$$

Substituting N from (1) and dividing by $W\sin^2\theta$,

$$d\,\mathrm{cosec}^2\theta = a\big((\lambda+\mu)\cos\theta + (1-\lambda\mu)\sin\theta\big),$$

which is the required result.

The case $a\sin\theta > d$

Now G lies beyond the rail. Equation (1) is unchanged, as is the geometry, so the only question is again the direction of slipping.

Ask yourself What changes now that G is beyond the rail?

The argument above showed that downward slipping happens when $a\sin^3\theta < d$, and upward slipping when $a\sin^3\theta > d$. With $a\sin\theta < d$ only the first was possible. With $a\sin\theta > d$ the second becomes possible: the overhanging weight beyond the rail makes $N\sin\theta > W$, so the rod tends to slide down through the rail *away* from the wall, with A moving up the wall. This is the new case, and in it both friction forces are reversed: $F = \mu R$ acts down the wall and $T = \lambda N$ acts down the rod. Replacing λ by $-\lambda$ and μ by $-\mu$ in the working above,

$$R = N(\cos\theta + \lambda\sin\theta), \qquad W = N\big((1-\lambda\mu)\sin\theta - (\lambda+\mu)\cos\theta\big),$$

and with (1),

$$d\,\mathrm{cosec}^2\theta = a\big((1-\lambda\mu)\sin\theta - (\lambda+\mu)\cos\theta\big).$$

Here $R > 0$ automatically. (If instead $a\sin^3\theta < d < a\sin\theta$, the rod is still on the point of slipping downwards and the first result applies unchanged, since its derivation never used where G was.)

Checks

- With no friction ($\lambda = \mu = 0$) both results reduce to $d = a\sin^3\theta$, the frictionless balance found above.
- Writing $\lambda = \tan\alpha$ and $\mu = \tan\beta$ (the angles of friction), the right-hand sides become $a\sin(\theta+\alpha+\beta)/(\cos\alpha\cos\beta)$ and $a\sin(\theta-\alpha-\beta)/(\cos\alpha\cos\beta)$: the two cases differ only in the sign of the friction angles, as they should.
- With $\lambda = \mu = \frac{1}{2}$ and $\theta = \frac{1}{3}\pi$, the two cases give $d/a = \frac{3}{8} + \frac{9\sqrt{3}}{32} \approx 0.862$ and $d/a = \frac{9\sqrt{3}}{32} - \frac{3}{8} \approx 0.112$. Substituting each back into the force and moment equations, with the friction directions above, gives zero resultant and zero moment.

Common slips

- Taking the rail's normal reaction to be vertical. It is perpendicular to the rod, because the rod touches the rail at a single point.
- Using the length d instead of $d / \sin\theta$ as the distance from A to the rail in the moment equation.
- Fixing the friction directions without asking which way the rod would actually move.

Beyond STEP

The rod is stable for a whole interval of angles θ, between the downward and upward limiting cases: friction gives a range of equilibria, not a single one. Describing this kind of set (the friction cone at each contact, and the set of states the contacts can hold) is the starting point of contact mechanics and of the analysis of grasping in robotics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and rigid-body statics carefully at first-year level.

WORKED SOLUTION

STEP I 2016, Question 10

Mechanics • Collisions and Restitution

STEP I 2016, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q10

The key idea

There are three collisions, and each one is handled by the same two equations: conservation of momentum and Newton's law of restitution. The first two collisions happen at the same time but independently, so we can work them out separately to find the velocities of B and C just before they meet. The condition that B ends up at rest then gives one equation linking λ and e.

Solution

We model the four bodies as particles on a smooth table, so no horizontal forces act except in the collisions, and every collision is direct (along the line). Take the direction from A towards D as positive, so A starts with velocity u and D with velocity $-u$.



Part (i)

A **hits** B**.** Let A and B have velocities v_A and v_B afterwards. Then

$$\begin{aligned} \text{momentum:} \quad & \lambda m u = \lambda m v_A + m v_B, \\ \text{restitution:} \quad & v_B - v_A = eu. \end{aligned}$$

Substituting $v_B = v_A + eu$ into the first equation gives $(\lambda + 1)v_A = (\lambda - e)u$, so

$$v_A = \frac{(\lambda - e)u}{\lambda + 1}, \qquad v_B = v_A + eu = \frac{\lambda(1 + e)u}{\lambda + 1}.$$

D **hits** C**.** Let C and D have velocities v_C and v_D afterwards. Then

$$\begin{aligned} \text{momentum:} \quad & -mu = mv_C + mv_D, \\ \text{restitution:} \quad & v_D - v_C = e\big(0-(-u)\big) = eu. \end{aligned}$$

(Before the impact C and D approach at speed u; afterwards D is on the right of C, so they separate at rate $v_D - v_C$.) Solving these two equations:

$$v_C = -\tfrac{1}{2}(1+e)u, \qquad v_D = -\tfrac{1}{2}(1-e)u.$$

So C moves towards B with speed $\frac{1}{2}(1+e)u$, and D follows it to the left with the smaller speed $\frac{1}{2}(1-e)u$.

B **hits** C**.** Now B moves right at v_B and C moves left, so they approach with speed $v_B - v_C$. After the collision B is at rest; let C have velocity w. Then

$$\begin{aligned} \text{momentum:} \quad & mv_B + mv_C = 0 + mw, \\ \text{restitution:} \quad & w - 0 = e(v_B - v_C). \end{aligned}$$

Eliminating w gives $v_B + v_C = e(v_B - v_C)$, that is,

$$(1-e)v_B = -(1+e)v_C.$$

Substituting the velocities found above,

$$(1-e)\cdot\frac{\lambda(1+e)u}{\lambda+1} = (1+e)\cdot\tfrac{1}{2}(1+e)u.$$

We may divide by $(1+e)u$, since $u \neq 0$ and $1+e>0$:

$$\begin{aligned} 2\lambda(1-e) &= (1+e)(\lambda+1) \\ 2\lambda - 2\lambda e &= \lambda + 1 + e\lambda + e \\ \lambda - 1 &= e(3\lambda+1), \end{aligned}$$

and as $3\lambda+1>0$,

$$e = \frac{\lambda-1}{3\lambda+1},$$

as required.

For the deduction,

$$\frac{1}{3} - e = \frac{(3\lambda+1)-3(\lambda-1)}{3(3\lambda+1)} = \frac{4}{3(3\lambda+1)} > 0,$$

since $\lambda > 1$. Hence $e < \frac{1}{3}$. (Also $e>0$ because $\lambda>1$, so the collisions are genuinely not perfectly inelastic.)

Part (ii)

After the third collision C has velocity

$$w = v_B + v_C = (1+e)u\left(\frac{\lambda}{\lambda+1} - \frac{1}{2}\right) = \frac{(1+e)(\lambda-1)u}{2(\lambda+1)},$$

which is positive because $\lambda>1$: C now moves to the right, back towards D, which is still moving left with speed $\frac{1}{2}(1-e)u$.

Ask yourself Which equation does "the same speed" give, and how do we get rid of e?

The speeds are equal when

$$\frac{(1+e)(\lambda-1)}{2(\lambda+1)} = \frac{1-e}{2}.$$

From part (i), $1+e = \dfrac{4\lambda}{3\lambda+1}$ and $1-e = \dfrac{2(\lambda+1)}{3\lambda+1}$. Substituting and cancelling the common factor $\dfrac{1}{3\lambda+1}$:

$$\begin{aligned}\frac{4\lambda(\lambda-1)}{\lambda+1} &= 2(\lambda+1)\\ 2\lambda^2-2\lambda &= \lambda^2+2\lambda+1\\ \lambda^2-4\lambda-1 &= 0,\end{aligned}$$

so $\lambda = 2\pm\sqrt{5}$. Only $\lambda = 2+\sqrt{5}$ satisfies $\lambda > 1$ (the other root is negative). Then

$$e = \frac{1+\sqrt{5}}{7+3\sqrt{5}} = \frac{(1+\sqrt{5})(7-3\sqrt{5})}{49-45} = \frac{4\sqrt{5}-8}{4} = \sqrt{5}-2.$$

So

$$\lambda = 2+\sqrt{5}, \qquad e = \sqrt{5}-2.$$

Checks

- $e = \sqrt{5}-2 \approx 0.236$, which is indeed between 0 and $\frac{1}{3}$, as part (i) demands.
- With $\lambda = 2+\sqrt{5} \approx 4.236$, $u = 1$ and $e \approx 0.236$, running the three collisions numerically gives $v_B = 1$ (exactly, as it happens) and $v_C \approx -0.618$ before the third collision; afterwards B is at rest and C moves right at about 0.382, while D moves left at about 0.382. The common speed is $\frac{1}{2}(1-e)u = \frac{1}{2}(3-\sqrt{5})u$.
- Limiting case: as $\lambda \to 1$, $e \to 0$. With equal masses and $e = 0$, B and C meet with equal and opposite velocities, so coalescing leaves them at rest, as it should.

Common slips

- Getting the sign wrong in the restitution equation for the collision between C and D: write every velocity with the same positive direction and use "separation $= e\times$ approach" carefully.
- Forgetting that C arrives at the third collision moving *towards* B, so the approach speed is $v_B - v_C = v_B + \frac{1}{2}(1+e)u$, not $v_B - \frac{1}{2}(1+e)u$.
- Keeping the root $\lambda = 2-\sqrt{5}$, which is negative.

Beyond STEP

A collision with restitution is the simplest model of energy loss: the kinetic energy lost in a direct collision is proportional to $1 - e^2$, and $e = 1$ is the elastic case in which energy is conserved. In a first-year course these ideas lead to the centre-of-mass frame, where any two-body collision looks symmetric and the algebra above becomes almost trivial. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops collisions this way.

WORKED SOLUTION

STEP I 2016, Question 11

Mechanics • Projectile Maximum Range

STEP I 2016, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

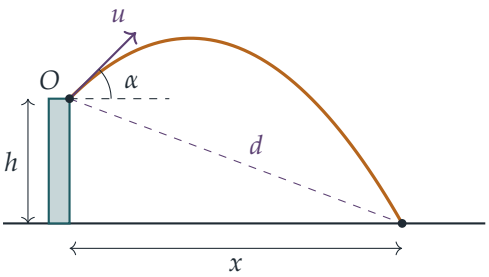
step.maths.org/questions/16-s1-q11

The key idea

Write down the horizontal and vertical motion, eliminate the time, and convert the squares of $\sin\alpha$ and $\cos\alpha$ into double angles. For the greatest range, treat the landing equation as defining x implicitly as a function of α and differentiate it: at the maximum $\mathrm{d}x/\mathrm{d}\alpha = 0$, and the rest is trigonometric algebra in the single unknown $\cos 2\alpha$.

Solution

We model P as a particle moving under gravity alone (no air resistance), with the plain horizontal. Take axes with origin at the base of the tower, x horizontal in the direction of firing and y vertically upwards, so that O is the point $(0, h)$.



The landing equation

At time t after firing,

$$X = ut\cos\alpha, \qquad Y = h + ut\sin\alpha - \tfrac{1}{2}gt^2.$$

The tower stands upright and P lands at some distance $x > 0$, so $\cos\alpha \neq 0$ and $t = \dfrac{x}{u\cos\alpha}$ at landing, where $Y = 0$:

$$0 = h + x\tan\alpha - \frac{gx^2}{2u^2\cos^2\alpha}.$$

Multiply through by $2\cos^2\alpha$ and use $2\cos^2\alpha = 1 + \cos 2\alpha$ and $2\sin\alpha\cos\alpha = \sin 2\alpha$:

$$\frac{gx^2}{u^2} = 2h\cos^2\alpha + 2x\sin\alpha\cos\alpha = h(1+\cos 2\alpha) + x\sin 2\alpha,$$

which is the required equation.

The greatest horizontal distance

The angle α can range over $-\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ (firing downwards is allowed). For each such α the landing distance x is a positive, smoothly varying function of α, and $x \to 0$ as $\alpha \to \pm\frac{\pi}{2}$, because the horizontal speed $u\cos\alpha$ tends to 0 while the time of flight stays bounded. So x has a greatest value at some interior α, and there $\mathrm{d}x/\mathrm{d}\alpha = 0$.

Differentiate the landing equation with respect to α, writing $x' = \mathrm{d}x/\mathrm{d}\alpha$:

$$\frac{2gx}{u^2}\,x' = -2h\sin 2\alpha + x'\sin 2\alpha + 2x\cos 2\alpha.$$

Setting $x' = 0$ leaves

$$x\cos 2\alpha = h\sin 2\alpha.$$

Here $\cos 2\alpha \neq 0$: otherwise $h\sin 2\alpha = 0$ with $h > 0$ would force $\sin 2\alpha = 0$ as well, which is impossible. So at the maximum

$$x = h\tan 2\alpha,$$

as required.

> **Ask yourself** We now have two equations in x and α. How do we get rid of x neatly?

Substitute $x = h\tan 2\alpha$ into the landing equation and write $c = \cos 2\alpha$, $s = \sin 2\alpha$. The right-hand side becomes

$$h(1+c) + h\,\frac{s^2}{c} = \frac{h\,(c + c^2 + s^2)}{c} = \frac{h(1+c)}{c},$$

while the left-hand side is

$$\frac{gh^2}{u^2}\cdot\frac{s^2}{c^2} = \frac{gh^2(1-c)(1+c)}{u^2c^2}.$$

Equating, and dividing by $h(1+c)/c$, which is non-zero since $c \neq -1$ (we showed $c \neq 0$, and $c = -1$ would give $\alpha = \pm\frac{\pi}{2}$):

$$\frac{gh(1-c)}{u^2c} = 1 \quad\Longrightarrow\quad gh - ghc = u^2c \quad\Longrightarrow\quad \cos 2\alpha = \frac{gh}{u^2+gh}.$$

Since this is the only possible value at a stationary point, and $\tan 2\alpha = x/h > 0$ picks out $0 < 2\alpha < \frac{\pi}{2}$, there is exactly one stationary point and it is the maximum. (For the record, $\tan 2\alpha = \dfrac{u\sqrt{u^2+2gh}}{gh}$, so the greatest distance is $x = \dfrac{u\sqrt{u^2+2gh}}{g}$.)

The greatest distance from O

The landing point is at $(x, 0)$, so its distance from $O = (0, h)$ is

$$d = \sqrt{x^2 + h^2}.$$

With h fixed, d increases with x, so d is greatest exactly when x is greatest. At that point $x = h\tan 2\alpha$, so

$$d = \sqrt{h^2\tan^2 2\alpha + h^2} = h\sec 2\alpha = h \cdot \frac{u^2+gh}{gh} = \frac{u^2}{g} + h,$$

using $\sec 2\alpha > 0$. This is the required greatest distance.

Checks

- With $h = 0$ we get $\cos 2\alpha = 0$, so $\alpha = 45°$, and greatest range u^2/g: the familiar results on level ground.
- Take $u = 10$, $g = 10$, $h = 5$ (SI units). Then $\cos 2\alpha = \frac{50}{150} = \frac{1}{3}$, the greatest range is $\sqrt{100 \cdot 200}/10 = 10\sqrt{2} \approx 14.14$, and $d = \sqrt{200 + 25} = 15 = \frac{100}{10} + 5$, as predicted. A computer search over α confirms that 14.14 is the largest landing distance.
- As h grows, $\cos 2\alpha \to 1$, so the best angle tends to 0: from a very tall tower you should fire almost horizontally.

Common slips

- Writing $x = h\tan 2\alpha$ and then forgetting that it holds only at the maximum, not for every α.
- Dividing by $1 + \cos 2\alpha$ or $\cos 2\alpha$ without saying why they are non-zero.
- Maximising d separately instead of noticing that $d^2 = x^2 + h^2$ is maximised at the same angle as x.

Beyond STEP

The answer $d = u^2/g + h$ has a lovely geometric explanation. All the trajectories with speed u from O touch a single *envelope*, the "parabola of safety", which has its focus at O and its directrix a height u^2/g above O. The furthest landing point lies on this envelope, so by the focus–directrix property its distance from O equals its distance

below the directrix, namely $u^2/g + h$. Envelopes of families of curves are a standard topic in first-year calculus and mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good next step for projectile problems treated with this kind of care.

WORKED SOLUTION

STEP I 2016, Question 12

Probability and Statistics • Binomial Symmetry

STEP I 2016, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q12

This question has a full guided treatment in *Beyond STEP*, Section 12.6, *One More Coin*: https://beyondstep.keithmcnulty.org/ch12/one-more-coin.html

The key idea

Parts (i) and (ii) are direct calculations with binomial probabilities, and both give the same strikingly simple answer. Part (iii) explains why. Split Bob's tosses into the first n and one extra toss: for the first n tosses Bob and Alice are on an equal footing, so symmetry does the work, and the extra toss breaks ties fairly.

Solution

Throughout, let A be the number of heads Alice gets and B the number Bob gets. With m tosses of a fair coin, the number of heads X has

$$\mathrm{P}(X = k) = \binom{m}{k}\frac{1}{2^m}, \qquad k = 0, 1, \ldots, m,$$

and Alice's and Bob's tosses are independent, so we can multiply probabilities for A and B.

Part (i)

Alice tosses twice and Bob three times, so

k	0	1	2	3
$\mathrm{P}(A = k)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$	
$\mathrm{P}(B = k)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Condition on Alice's score. Given $A = a$, we need $B > a$, so we need the tail probabilities $\mathrm{P}(B > 0) = \frac{7}{8}$, $\mathrm{P}(B > 1) = \frac{1}{2}$, $\mathrm{P}(B > 2) = \frac{1}{8}$. Hence

$$\begin{aligned}\mathrm{P}(B > A) &= \tfrac{1}{4} \cdot \tfrac{7}{8} + \tfrac{1}{2} \cdot \tfrac{1}{2} + \tfrac{1}{4} \cdot \tfrac{1}{8} \\ &= \frac{7+8+1}{32} = \frac{1}{2}.\end{aligned}$$

Part (ii)

Now $\mathrm{P}(A = a) = \frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$ for $a = 0, 1, 2, 3$, and $\mathrm{P}(B = k) = \frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16}$ for $k = 0, \ldots, 4$. The tail probabilities are

$$\mathrm{P}(B > 0) = \tfrac{15}{16}, \quad \mathrm{P}(B > 1) = \tfrac{11}{16}, \quad \mathrm{P}(B > 2) = \tfrac{5}{16}, \quad \mathrm{P}(B > 3) = \tfrac{1}{16}.$$

So

$$\begin{aligned}\mathrm{P}(B > A) &= \tfrac{1}{8} \cdot \tfrac{15}{16} + \tfrac{3}{8} \cdot \tfrac{11}{16} + \tfrac{3}{8} \cdot \tfrac{5}{16} + \tfrac{1}{8} \cdot \tfrac{1}{16} \\ &= \frac{15+33+15+1}{128} = \frac{64}{128} = \frac{1}{2}.\end{aligned}$$

Once again the answer is exactly $\frac{1}{2}$.

Part (iii)

First, a fact about the equal-tosses game. When Alice and Bob each toss n times, exactly one of "Bob more", "same", "Alice more" happens. The situation is symmetric between the two players (they do identical experiments), so "Alice more" also has probability p_2. Therefore

$$2p_2 + p_1 = 1. \qquad (*)$$

Now let Alice toss n times and Bob $n+1$ times. Let B' be the number of heads in Bob's *first* n tosses, and let H be the event that his last toss is a head. Then $B = B' + 1$ if H happens and $B = B'$ otherwise.

? **Ask yourself** How does comparing B with A depend on comparing B' with A?

Since B' and A are whole numbers and B is either B' or $B' + 1$:

- if $B' > A$, then certainly $B > A$, whatever the last toss;
- if $B' = A$, then $B > A$ exactly when the last toss is a head;
- if $B' < A$, then $B' \leq A - 1$, so $B \leq A$ and Bob does not win.

The first n tosses of Bob against Alice's n tosses is exactly the equal-tosses game, so $\mathrm{P}(B' > A) = p_2$ and $\mathrm{P}(B' = A) = p_1$. The last toss is independent of everything else and $\mathrm{P}(H) = \frac{1}{2}$. Hence

$$\mathrm{P}(\text{Bob gets more heads than Alice}) = p_2 + \tfrac{1}{2}p_1.$$

By $(*)$, $p_2 + \frac{1}{2}p_1 = \frac{1}{2}(2p_2 + p_1) = \frac{1}{2}$.

This is the generalisation: *if Alice tosses a fair coin n times and Bob tosses a fair coin $n+1$ times, the probability that Bob gets more heads than Alice is $\frac{1}{2}$, for every $n \geq 0$.* Parts (i) and (ii) are the cases $n = 2$ and $n = 3$.

Checks

- $n = 0$: Alice tosses nothing and Bob tosses once; Bob wins exactly when he gets a head, with probability $\frac{1}{2}$.
- $n = 2$: by hand, $p_1 = \frac{1}{16} + \frac{4}{16} + \frac{1}{16} = \frac{3}{8}$, and from $(*)$ $p_2 = \frac{5}{16}$. Then $p_2 + \frac{1}{2}p_1 = \frac{5}{16} + \frac{3}{16} = \frac{1}{2}$, agreeing with part (i).
- Exact enumeration by computer confirms $\mathrm{P}(B > A) = \frac{1}{2}$ for every n up to 29, and a simulation of part (ii) with 200 000 trials gave 0.4997.

Common slips

- Assuming $p_2 = \frac{1}{2}$ in the equal-tosses game: ties have positive probability, so $p_2 = \frac{1}{2}(1 - p_1) < \frac{1}{2}$.
- Forgetting the case $B' = A$ in part (iii), or counting it fully rather than with the factor $\frac{1}{2}$ for the extra toss.
- Using tail probabilities $\mathrm{P}(B \geq a)$ instead of $\mathrm{P}(B > a)$.

Beyond STEP

The neat answer comes from a symmetry argument plus conditioning on one extra step, a pattern that runs right through probability, from random walks to ballot problems. A slicker version: because Bob has one more toss, he *fails* to get more heads than Alice exactly when he gets more tails than her. Swapping heads and tails shows "more heads" and "more tails" have equal probability, and they are complementary, so each has probability $\frac{1}{2}$. Grimmett and Welsh, *Probability: An Introduction*, develops these conditioning and symmetry techniques.

WORKED SOLUTION

STEP I 2016, Question 13

Probability and Statistics • Exponential Order Statistics

STEP I 2016, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q13

The key idea

For the first and second arrivals, ask when they *haven't* happened yet. The first e-mail arrives after time t exactly when all n do, and the second arrives after t exactly when at most one has arrived by then. Both events are easy to count using independence, and differentiating the survival probability gives the density.

Solution

Let $X_1, \ldots, X_n$ be the arrival times. They're independent, and each has

$$\mathrm{P}(X_i > t) = \int_t^\infty \lambda \mathrm{e}^{-\lambda s}\,\mathrm{d}s = \mathrm{e}^{-\lambda t}, \qquad \mathrm{P}(X_i \le t) = 1 - \mathrm{e}^{-\lambda t}$$

for $t \ge 0$. We'll use the standard integral $\int_0^\infty t\,k\mathrm{e}^{-kt}\,\mathrm{d}t = \frac{1}{k}$ for $k > 0$, which is the mean of an exponential distribution with parameter k. (Integrate by parts: $\left[-t\mathrm{e}^{-kt}\right]_0^\infty + \int_0^\infty \mathrm{e}^{-kt}\,\mathrm{d}t = 0 + \frac{1}{k}$.)

Part (i)

The first arrival happens later than t if and only if every e-mail arrives later than t. By independence,

$$\mathrm{P}(T > t) = \prod_{i=1}^{n} \mathrm{P}(X_i > t) = \left(\mathrm{e}^{-\lambda t}\right)^n = \mathrm{e}^{-n\lambda t}.$$

So the cumulative distribution function is $F(t) = 1 - \mathrm{e}^{-n\lambda t}$ for $t \ge 0$. Differentiating gives the density

$$f(t) = F'(t) = n\lambda\,\mathrm{e}^{-n\lambda t} \qquad (t \ge 0),$$

as required. This is an exponential density with parameter $n\lambda$, so by the standard integral with $k = n\lambda$,

$$\mathrm{E}(T) = \frac{1}{n\lambda}.$$

Part (ii)

Let S be the arrival time of the second e-mail (so we need $n \geq 2$). The second arrival comes later than t if and only if *at most one* e-mail has arrived by time t: either none has, or exactly one has.

Ask yourself How many ways can exactly one e-mail have arrived by time t?

Any one of the n e-mails could be the early one. That one has arrived (probability $1 - \mathrm{e}^{-\lambda t}$) and the other $n-1$ haven't (probability $\mathrm{e}^{-\lambda t}$ each). By independence, and since the n cases are mutually exclusive,

$$\mathrm{P}(S > t) = \mathrm{e}^{-n\lambda t} + n\left(1 - \mathrm{e}^{-\lambda t}\right)\mathrm{e}^{-(n-1)\lambda t}.$$

Expanding the second term gives $n\mathrm{e}^{-(n-1)\lambda t} - n\mathrm{e}^{-n\lambda t}$, so

$$\mathrm{P}(S > t) = n\,\mathrm{e}^{-(n-1)\lambda t} - (n-1)\,\mathrm{e}^{-n\lambda t}.$$

As a quick sanity check, this equals 1 at $t = 0$ and tends to 0 as $t \to \infty$.

The density g of S is the derivative of $1 - \mathrm{P}(S > t)$:

$$\begin{aligned} g(t) &= -\frac{\mathrm{d}}{\mathrm{d}t}\left(n\,\mathrm{e}^{-(n-1)\lambda t} - (n-1)\,\mathrm{e}^{-n\lambda t}\right) \\ &= n(n-1)\lambda\,\mathrm{e}^{-(n-1)\lambda t} - n(n-1)\lambda\,\mathrm{e}^{-n\lambda t} \\ &= n(n-1)\lambda\left(\mathrm{e}^{-(n-1)\lambda t} - \mathrm{e}^{-n\lambda t}\right) \qquad (t \geq 0). \end{aligned}$$

For the expectation, write g as a combination of exponential densities:

$$g(t) = n \cdot (n-1)\lambda\,\mathrm{e}^{-(n-1)\lambda t} \;-\; (n-1) \cdot n\lambda\,\mathrm{e}^{-n\lambda t}.$$

Using the standard integral with $k = (n-1)\lambda$ and $k = n\lambda$ (valid because $n \geq 2$, so $k > 0$ in both cases),

$$\begin{aligned} \mathrm{E}(S) = \int_0^\infty t\,g(t)\,\mathrm{d}t &= \frac{n}{(n-1)\lambda} - \frac{n-1}{n\lambda} \\ &= \frac{1}{\lambda} \cdot \frac{n^2 - (n-1)^2}{n(n-1)} = \frac{1}{\lambda} \cdot \frac{2n-1}{n(n-1)}. \end{aligned}$$

Since $\dfrac{1}{n-1} + \dfrac{1}{n} = \dfrac{n + (n-1)}{n(n-1)} = \dfrac{2n-1}{n(n-1)}$, this is

$$\mathrm{E}(S) = \frac{1}{\lambda}\left(\frac{1}{n-1} + \frac{1}{n}\right),$$

which is the required result.

Ask yourself Why is the answer a sum of two simple terms?

The exponential distribution is memoryless. The first arrival comes after an exponential time with rate $n\lambda$, with mean $\frac{1}{n\lambda}$. After that, the remaining $n-1$ e-mails start afresh, as though the clock had just been reset, so the further wait is exponential with rate $(n-1)\lambda$ and mean $\frac{1}{(n-1)\lambda}$. Adding the two waits gives the answer at once, and it's a good way to remember it.

Checks

- With $n = 2$ and $\lambda = 1$, the second arrival is the later of two independent exponential times. Its mean should be $1 + \frac{1}{2} = 1.5$, and a computer simulation of 10^5 trials gave 1.501.
- The density g integrates to 1: $\int_0^\infty g = n - (n-1) = 1$.
- With $n = 5$ and $\lambda = 1$, the formula gives $\mathrm{E}(S) = \frac{1}{4} + \frac{1}{5} = 0.45$, which agrees with a numerical evaluation of $\int_0^\infty t\, g(t)\, \mathrm{d}t$.

Common slips

- Forgetting the factor n when exactly one e-mail has arrived. Any of the n could be the early one.
- Writing $\mathrm{P}(S > t)$ as the probability that *exactly* one has arrived, which leaves out the case where none has.

Beyond STEP

T and S are the first two *order statistics* of an exponential sample. The memoryless argument above extends to show that the kth arrival has mean $\frac{1}{\lambda}\left(\frac{1}{n} + \frac{1}{n-1} + \cdots + \frac{1}{n-k+1}\right)$, and this is the basis of the Poisson process and of queueing theory. Blitzstein and Hwang, *Introduction to Probability*, covers order statistics and the memoryless property clearly.

WORKED SOLUTION

STEP II 2016, Question 1

Pure • Parametric Curves and Tangents

> **STEP II 2016, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/16-s2-q1

The key idea

Work with the parameters rather than the coordinates. The right angle at O turns into a single, very simple relation between the parameters p and q of the two points, and the tangent at a general point has a neat equation in t. Once the two tangents are intersected, the coordinates of the meeting point come out as symmetric functions of p and q, and the relation between p and q is exactly what is needed to eliminate them.

Solution

The tangent at a general point

At the point with parameter t,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t} = \frac{3t^2}{2t} = \frac{3t}{2} \qquad (t \neq 0).$$

So the tangent at (t^2, t^3) is $y - t^3 = \frac{3t}{2}(x - t^2)$, that is,

$$2y = 3tx - t^3. \qquad (*)$$

(At $t = 0$ this gives $y = 0$, which is the tangent at the cusp, but we shall not need that point.)

The right angle

Let $P = (p^2, p^3)$ and $Q = (q^2, q^3)$. For $\angle POQ$ to make sense, neither point is O, so $p \neq 0$ and $q \neq 0$. The angle is a right angle exactly when $\overrightarrow{OP} \cdot \overrightarrow{OQ} = 0$:

$$p^2q^2 + p^3q^3 = p^2q^2(1 + pq) = 0.$$

Since $p^2q^2 \neq 0$, this forces

$$pq = -1.$$

In particular $p \neq q$ (their product is negative), so the tangents at P and Q have different gradients $\frac{3p}{2} \neq \frac{3q}{2}$ and really do meet in a single point.

Where the tangents meet

Subtracting the two versions of $(*)$, with $t = p$ and $t = q$:

$$3(p-q)x = p^3 - q^3 = (p-q)(p^2+pq+q^2).$$

Dividing by $p - q \neq 0$ and using $pq = -1$,

$$3x = p^2 + pq + q^2 = p^2 + q^2 - 1.$$

Substituting back into $(*)$ with $t = p$:

$$\begin{aligned} 2y &= p(p^2+pq+q^2) - p^3 \\ &= p^2q + pq^2 = pq(p+q) = -(p+q). \end{aligned}$$

Ask yourself We have x and y in terms of two parameters. How do we get a single equation in x and y?

Square y: the cross term $2pq$ that appears is a known constant, and what remains is $p^2 + q^2$, which is exactly what $3x$ contains. Indeed,

$$4y^2 = (p+q)^2 = p^2 + q^2 + 2pq = p^2 + q^2 - 2 = (3x+1) - 2 = 3x - 1.$$

So the tangents at P and Q meet on C_2: $4y^2 = 3x - 1$, as required.

(In fact every point of C_2 arises: $2y = -(p+q) = \frac{1}{p} - p$ takes every real value as p runs through the positive reals, and x is then fixed by the equation of C_2.)

Do C_1 and C_2 meet?

Substitute $x = t^2$, $y = t^3$ into the equation of C_2:

$$4t^6 = 3t^2 - 1 \iff 4u^3 - 3u + 1 = 0, \quad \text{where } u = t^2 \geq 0.$$

Clearly $u = -1$ is a root, and dividing out the factor $u + 1$ gives

$$4u^3 - 3u + 1 = (u+1)(4u^2 - 4u + 1) = (u+1)(2u-1)^2.$$

Since $u \geq 0$, the only possibility is $u = \frac{1}{2}$, so $t = \pm\frac{1}{\sqrt{2}}$. Hence the curves *do* meet, at exactly two points

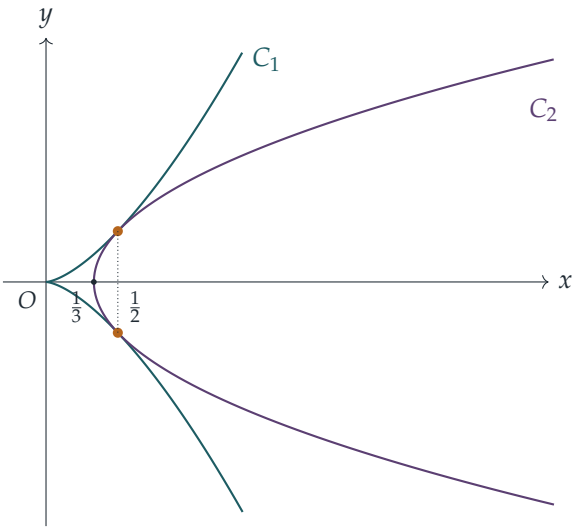
$$\left(\tfrac{1}{2},\ \tfrac{1}{2\sqrt{2}}\right) \quad \text{and} \quad \left(\tfrac{1}{2},\ -\tfrac{1}{2\sqrt{2}}\right).$$

The repeated root tells us more: the curves touch there rather than cross. We can confirm this directly. At $t = \frac{1}{\sqrt{2}}$ the gradient of C_1 is $\frac{3t}{2} = \frac{3}{2\sqrt{2}}$. Differentiating $4y^2 = 3x - 1$ implicitly gives $8y\frac{\mathrm{d}y}{\mathrm{d}x} = 3$, so at $y = \frac{1}{2\sqrt{2}}$ the gradient of C_2 is $\frac{3}{8y} = \frac{3}{2\sqrt{2}}$, the same. By symmetry in the x-axis the same holds at the other point.

Since $4u^3 - 3u + 1 = (u+1)(2u-1)^2 \geq 0$ for $u \geq 0$, the curve C_1 never goes inside the parabola: C_2 lies between the two branches of C_1 except at the two points of contact.

Sketch

C_1 satisfies $y^2 = x^3$: it lies in $x \geq 0$, is symmetric in the x-axis, and has a cusp at O (both branches have gradient $\frac{3t}{2} \to 0$, so they leave O along the positive x-axis). C_2 is the parabola $x = \frac{1}{3}(4y^2 + 1)$, with vertex $(\frac{1}{3}, 0)$, opening to the right and also symmetric in the x-axis. The parabola sits inside the cusp, between the two branches of C_1, and touches each branch once, at $x = \frac{1}{2}$.



Checks

- Take $p = 1$, $q = -1$: $P = (1, 1)$, $Q = (1, -1)$, and $\overrightarrow{OP} \cdot \overrightarrow{OQ} = 1 - 1 = 0$. The tangents are $2y = 3x - 1$ and $2y = -3x + 1$, meeting at $(\frac{1}{3}, 0)$, the vertex of C_2.
- Take $p = 2$, $q = -\frac{1}{2}$: the tangents $2y = 6x - 8$ and $2y = -\frac{3}{2}x + \frac{1}{8}$ meet at $x = \frac{13}{12}$, $y = -\frac{3}{4}$, and $4y^2 = \frac{9}{4} = 3 \cdot \frac{13}{12} - 1$.
- The point of contact $(\frac{1}{2}, \frac{1}{2\sqrt{2}}) \approx (0.5, 0.35355)$ satisfies $4y^2 = \frac{1}{2} = 3x - 1$, and both gradients there are ≈ 1.06066. A computer check of 10^5 random pairs with $pq = -1$ found every intersection on C_2 to within rounding error.

Common slips

- Forgetting to exclude $p = 0$ or $q = 0$: the factor p^2q^2 must be non-zero before dividing to get $pq = -1$.
- Dividing by $p - q$ without noting why it is non-zero.
- Stopping at "$4u^3 - 3u + 1 = 0$ has a root" without checking that it gives $u = t^2 \geq 0$; the root $u = -1$ does not correspond to any point of C_1.

Beyond STEP

The curve $y^2 = x^3$ is the classic *cuspidal cubic*, and the trick of meeting a curve with a parametrised curve and reading tangency from a repeated root is the starting point of algebraic geometry: intersection multiplicities count how many times curves meet, and Bézout's theorem says that a cubic and a conic meet in six points counted properly. Here the six are the double roots $t = \pm\frac{1}{\sqrt{2}}$ (four in all) and the complex pair $t = \pm i$. A friendly introduction is Kirwan, *Complex Algebraic Curves*.

WORKED SOLUTION

STEP II 2016, Question 2

Pure • Factor Theorem

STEP II 2016, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q2

The key idea

Think of $(*)$ as a polynomial in c whose coefficients happen to involve a and b. The factor theorem says $c-(a+b)$ is a factor exactly when substituting $c=a+b$ gives zero. Once one factor is found, the symmetry of $(*)$ in a, b, c hands us two more for free, and a degree count finishes the factorisation. The two equations are then disguised special cases: the job is to choose a, b, c (and later d, e) to match them.

Solution

The factor $a+b-c$

Write $F(a,b,c)$ for $(*)$. Regarded as a polynomial in c, F has the factor $c-(a+b)$, and hence $a+b-c$, if and only if $F(a,b,a+b)=0$. Put $p=a+b$ and $c=p$. Then $a+b+c=2p$, and using $a^2+b^2=p^2-2ab$ and $a^3+b^3=p^3-3abp$,

$$a^2+b^2+c^2=2p^2-2ab, \qquad a^3+b^3+c^3=2p^3-3abp.$$

So

$$\begin{aligned} F(a,b,a+b) &= 8p^3-6(2p)(2p^2-2ab)+8(2p^3-3abp) \\ &= 8p^3-24p^3+24abp+16p^3-24abp=0, \end{aligned}$$

and $a+b-c$ is a factor of $(*)$, as required.

Factorising $(*)$ completely

F is unchanged by any rearrangement of a, b, c. Swapping b and c turns the factor $a+b-c$ into $a+c-b$; swapping a and c turns it into $b+c-a$. So all three of

$$a+b-c, \qquad b+c-a, \qquad c+a-b$$

are factors. They are distinct linear polynomials, none a constant multiple of another, so their product divides F. Every term of F has degree 3, so the remaining factor has

degree 0: it is a constant k, and

$$F(a,b,c) = k(a+b-c)(b+c-a)(c+a-b).$$

To find k, put $a = b = c = 1$: the left-hand side is $27 - 6\cdot 3\cdot 3 + 8\cdot 3 = -3$ and the product is 1. Hence $k = -3$ and

$$\begin{aligned}(a+b+c)^3 - 6(a+b+c)(a^2+b^2+c^2) + 8(a^3+b^3+c^3)\\ = -3(a+b-c)(b+c-a)(c+a-b).\end{aligned}$$

Part (i)

Divide the middle and last terms of the equation into the shape of $(*)$:

$$(x+1)^3 - 6(x+1)\left(x^2+\tfrac{5}{2}\right) + 8\left(x^3+\tfrac{13}{4}\right) = 0.$$

Ask yourself Which values of a, b, c make this exactly $(*)$?

Try $a = x$ with b, c constants. We need

$$b+c = 1, \qquad b^2+c^2 = \tfrac{5}{2}, \qquad b^3+c^3 = \tfrac{13}{4}.$$

The first two give $2bc = (b+c)^2 - (b^2+c^2) = -\tfrac{3}{2}$, so $bc = -\tfrac{3}{4}$. Then

$$b^3+c^3 = (b+c)^3 - 3bc(b+c) = 1 + \tfrac{9}{4} = \tfrac{13}{4},$$

so the third condition holds automatically. Now b, c are the roots of $t^2 - t - \tfrac{3}{4} = 0$, that is $(2t-3)(2t+1) = 0$, so $b = \tfrac{3}{2}$, $c = -\tfrac{1}{2}$.

With $a = x$, $b = \tfrac{3}{2}$, $c = -\tfrac{1}{2}$ the factorisation gives

$$a+b-c = x+2, \qquad b+c-a = 1-x, \qquad c+a-b = x-2,$$

so the equation is $-3(x+2)(1-x)(x-2) = 0$, and

$$x = -2, \quad x = 1, \quad x = 2.$$

Part (ii)

Write $G(a,b,d,e)$ for the four-variable expression. (Setting $c = d+e$ makes $a+b-d-e$ look like $a+b-c$, which suggests the factor; but it's just as quick to check it directly.) By the factor theorem, regarding G as a polynomial in e, it is enough to show $G = 0$ when $a+b = d+e$. Let $p = a+b = d+e$. Exactly as before, $a^2+b^2 = p^2 - 2ab$, $d^2+e^2 = p^2 - 2de$, $a^3+b^3 = p^3 - 3abp$ and $d^3+e^3 = p^3 - 3dep$, so with $q = ab+de$,

$$\begin{aligned}G &= (2p)^3 - 6(2p)(2p^2 - 2q) + 8(2p^3 - 3pq)\\ &= 8p^3 - 24p^3 + 24pq + 16p^3 - 24pq = 0.\end{aligned}$$

So $a+b-d-e$ is a factor of G.

Now G is unchanged by any rearrangement of a, b, d, e. The factor $a+b-d-e$ comes from splitting the four letters into the pairs $\{a,b\}$ and $\{d,e\}$; the other two ways of splitting them into pairs give the factors

$$a+d-b-e \quad (\text{swap } b,d), \qquad a+e-b-d \quad (\text{swap } b,e).$$

These three linear factors are distinct and none is a multiple of another, and G is homogeneous of degree 3, so $G = k(a+b-d-e)(a+d-b-e)(a+e-b-d)$ for a constant k. Putting $a=1$, $b=d=e=0$ gives $1-6+8=3$ on the left and k on the right, so

$$G = 3(a+b-d-e)(a+d-b-e)(a+e-b-d).$$

For the equation, take $a=x$ and constants b, d, e with

$$b+d+e=6, \qquad b^2+d^2+e^2=14, \qquad b^3+d^3+e^3=36.$$

The numbers 1, 2, 3 work: $1+2+3=6$, $1+4+9=14$, $1+8+27=36$. So with $a=x$, $b=1$, $d=2$, $e=3$ the equation is exactly $G=0$, which factorises as

$$3(x+1-2-3)(x+2-1-3)(x+3-1-2) = 3(x-4)(x-2)x = 0.$$

Hence

$$x=0, \quad x=2, \quad x=4.$$

Checks

- Put $a=b=1$, $c=2$ into $(*)$: $64-6\cdot4\cdot6+8\cdot10=0$, as the factor $a+b-c$ predicts.
- In (i), $x=1$ gives $8-3\cdot2\cdot7+2\cdot17=8-42+34=0$.
- In (ii), $x=0$ gives $216-6\cdot6\cdot14+8\cdot36=216-504+288=0$, and $x=4$ gives $1000-6\cdot10\cdot30+8\cdot100=0$.
- The factorisations were also confirmed by computer at a few thousand random rational values of the variables.

Common slips

- Forgetting the constant: $(*)$ is -3 times the product of the factors, not the product itself. Always fix k with a convenient substitution.
- In (ii), listing more than three factors. Swapping a and d, say, gives $d+b-a-e$, which is just $-(a+e-b-d)$; the three genuinely different factors come from the three ways of pairing up four letters.
- Expanding everything by brute force. It works, but it is long and error-prone; matching the equation to the identity is the point of the question.

Beyond STEP

The factor theorem works over any field, and here we used it for polynomials in several variables by treating all but one as constants. Symmetry plus unique factorisation of polynomials is a powerful combination: the theory of symmetric polynomials says that every symmetric polynomial can be written in terms of the elementary ones $a+b+c$, $ab+bc+ca$, abc, which is the structure hiding behind the conditions on b, c in (i). Incidentally, the product $(a+b-c)(b+c-a)(c+a-b)$ is exactly what appears in Heron's formula for the area of a triangle. For more on polynomial rings and factorisation, see Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP II 2016, Question 3

Pure • Roots of Polynomials

STEP II 2016, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q3

The key idea

Everything rests on one identity: $\mathrm{f}_n(x) = \mathrm{f}_{n-1}(x) + \dfrac{x^n}{n!}$. Combined with part (i), it says that at a root a of f_n the slope is $\mathrm{f}_n'(a) = -a^n/n!$, so we know the gradient at every root without knowing the roots. Since all roots have the same sign, all those gradients have the same sign, and a curve cannot cross the axis upwards twice in a row without crossing downwards in between.

Solution

Throughout, write $\mathrm{f}_n(x) = \displaystyle\sum_{k=0}^{n} \frac{x^k}{k!}$.

Part (i)

Differentiating term by term, the constant term 1 disappears and, for $1 \le k \le n$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{x^k}{k!}\right) = \frac{kx^{k-1}}{k!} = \frac{x^{k-1}}{(k-1)!}.$$

So

$$\mathrm{f}_n'(x) = \sum_{k=1}^{n} \frac{x^{k-1}}{(k-1)!} = \sum_{j=0}^{n-1} \frac{x^j}{j!} = \mathrm{f}_{n-1}(x),$$

as required.

Part (ii)

If $x \ge 0$, every term of $\mathrm{f}_n(x)$ is non-negative and the first term is 1, so $\mathrm{f}_n(x) \ge 1 > 0$. Hence no root can satisfy $a \ge 0$, and any real root has $a < 0$. (For $n = 0$, $\mathrm{f}_0 = 1$ has no roots at all, which is consistent.)

Part (iii)

From the definition, $\mathrm{f}_n(x) = \mathrm{f}_{n-1}(x) + \dfrac{x^n}{n!}$. If $\mathrm{f}_n(a) = 0$, then by part (i)

$$\mathrm{f}_n'(a) = \mathrm{f}_{n-1}(a) = \mathrm{f}_n(a) - \frac{a^n}{n!} = -\frac{a^n}{n!},$$

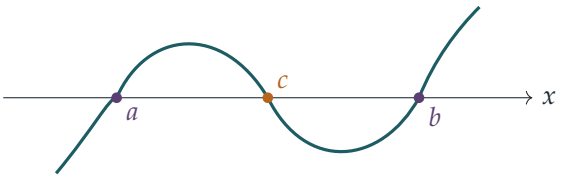
and likewise $\mathrm{f}_n'(b) = -b^n/n!$. Therefore

$$\mathrm{f}_n'(a)\,\mathrm{f}_n'(b) = \frac{(ab)^n}{(n!)^2} > 0,$$

because $a < 0$ and $b < 0$ by part (ii), so $ab > 0$. This is the required inequality. In particular neither gradient is zero, and the two have the same sign.

Ask yourself Why does equal sign of the two gradients force another root in between?

Take $a < b$ and suppose both gradients are positive (the negative case is the mirror image). Since $\mathrm{f}_n'(a) > 0$, the curve crosses the axis upwards at a, so $\mathrm{f}_n(x) > 0$ for x slightly bigger than a. Since $\mathrm{f}_n'(b) > 0$, it also crosses upwards at b, so $\mathrm{f}_n(x) < 0$ for x slightly smaller than b. A continuous function that is positive at some point and negative at a later point must vanish in between (intermediate value theorem), so $\mathrm{f}_n(c) = 0$ for some c with $a < c < b$.



The curve rises through a and rises through b, so it must come down through the axis somewhere between them.

At most one real root. Suppose $(*)$ had two or more distinct real roots. It is a polynomial of degree n, so it has finitely many roots, and we may choose two of them, $a < b$, that are *consecutive*, with no root strictly between them. But the argument above produces a root c strictly between a and b, a contradiction. So $(*)$ has at most one real root. (The argument used $n \geq 2$ as in the question; for $n = 1$ the only root is -1, and for $n = 0$ there is none, so the conclusion holds for every n.)

n odd. Now f_n has odd degree with positive leading coefficient $1/n!$, so $\mathrm{f}_n(x) \to -\infty$ as $x \to -\infty$ and $\mathrm{f}_n(x) \to +\infty$ as $x \to +\infty$. By the intermediate value theorem it has at least one real root, and so **exactly one**.

n even. For $n = 0$ there are no roots. For even $n \geq 2$, $\mathrm{f}_n(x) \to +\infty$ as $x \to \pm\infty$, so f_n attains a global minimum at some point x_0, where $\mathrm{f}_n'(x_0) = \mathrm{f}_{n-1}(x_0) = 0$. Since $\mathrm{f}_{n-1}(0) = 1 \neq 0$, $x_0 \neq 0$, and so

$$\mathrm{f}_n(x_0) = \mathrm{f}_{n-1}(x_0) + \frac{x_0^n}{n!} = \frac{x_0^n}{n!} > 0,$$

because n is even. The minimum value is positive, so $\mathrm{f}_n(x) > 0$ for all x and $(*)$ has **no real roots**.

(Alternatively: a root a would be simple, since $\mathrm{f}_n'(a) \neq 0$, so f_n would change sign there; but an even-degree polynomial with positive leading coefficient that changes sign exactly once would have to be negative for all large x or all very negative x, which is impossible.)

Checks

- $n = 1$: $1 + x = 0$ has the single root -1. $n = 2$: $1 + x + \frac{1}{2}x^2 = \frac{1}{2}\big((x+1)^2 + 1\big) > 0$, no roots. Both fit.
- $n = 3$: the root is about -1.596, and there $\mathrm{f}_3'(a) = \mathrm{f}_2(a) \approx 0.678 = -a^3/3!$, as predicted.
- A computer check of the sign changes of f_n for every $n \leq 25$ found exactly one root for odd n and none for even n.

Common slips

- Showing the product of gradients is positive without saying *why* $ab > 0$: it needs part (ii).
- Concluding "at most one root" from a root between any two roots without choosing the two roots to be consecutive; that choice is what produces the contradiction.
- For even n, forgetting to check $x_0 \neq 0$, which is needed for $x_0^n/n! > 0$ to be strict.

Beyond STEP

The polynomials f_n are the Taylor polynomials of e^x, and part (i) is the finite shadow of $(\mathrm{e}^x)' = \mathrm{e}^x$. Since e^x never vanishes, it is natural that the f_n have few real roots; their complex roots, scaled by $1/n$, cluster on a curve called the Szegő curve as $n \to \infty$. The tools used here, the intermediate value theorem and the existence of a minimum for a continuous function, are proved carefully in a first analysis course; see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2016, Question 4

Pure • Range of Rational Function

STEP II 2016, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q4

The key idea

Clearing the fraction turns the definition of y into a quadratic equation in x, and "x is real" becomes "the discriminant is non-negative". To get rid of θ we need an upper bound for $y\cos\theta - \sin\theta$ that holds for every θ; writing it as a dot product of $(y, -1)$ with the unit vector $(\cos\theta, \sin\theta)$ gives exactly $\sqrt{y^2+1}$. In part (ii) every inequality in the chain must be an equality, and that pins down both x and θ.

Solution

Part (i)

First note that the denominator never vanishes: as a quadratic in x, $x^2 + x\cos\theta + 1$ has discriminant $\cos^2\theta - 4 < 0$, so it is always positive. Multiplying up is therefore safe:

$$y(x^2 + x\cos\theta + 1) = x^2 + x\sin\theta + 1 \iff (y-1)x^2 + (y\cos\theta - \sin\theta)\,x + (y-1) = 0.$$

If $y \neq 1$ this is a genuine quadratic in x, and it has a real root x if and only if its discriminant is non-negative:

$$(y\cos\theta - \sin\theta)^2 - 4(y-1)^2 \geq 0.$$

If $y = 1$ the required inequality reads $(\cos\theta - \sin\theta)^2 \geq 0$, which is true anyway. So for real x,

$$(y\cos\theta - \sin\theta)^2 \geq 4(y-1)^2,$$

as required.

Ask yourself How can we remove θ? We want something bigger than $(y\cos\theta - \sin\theta)^2$ that doesn't depend on θ.

The expression $y\cos\theta - \sin\theta$ is the scalar product of $(y, -1)$ with the unit vector $(\cos\theta, \sin\theta)$, so its size is at most $|(y,-1)| = \sqrt{y^2+1}$. Equivalently, by the Cauchy–

Schwarz inequality, or directly from

$$(y^2+1)(\cos^2\theta+\sin^2\theta)-(y\cos\theta-\sin\theta)^2=(y\sin\theta+\cos\theta)^2\geq 0,$$

we have $(y\cos\theta-\sin\theta)^2\leq y^2+1$. Combining this with the first result,

$$y^2+1\geq(y\cos\theta-\sin\theta)^2\geq 4(y-1)^2,$$

which is the second inequality.

Expanding, $y^2+1\geq 4y^2-8y+4$, that is,

$$3y^2-8y+3\leq 0.$$

The roots of $3y^2-8y+3=0$ are $y=\dfrac{8\pm\sqrt{64-36}}{6}=\dfrac{4\pm\sqrt{7}}{3}$, and since the quadratic has positive leading coefficient it is non-positive exactly between its roots. Hence

$$\frac{4-\sqrt{7}}{3}\leq y\leq\frac{4+\sqrt{7}}{3}.$$

Part (ii)

Let $y=\frac{1}{3}(4+\sqrt{7})$, so $y-1=\frac{1}{3}(1+\sqrt{7})>0$. Then

$$y^2+1=\frac{16+8\sqrt{7}+7}{9}+1=\frac{32+8\sqrt{7}}{9},$$
$$4(y-1)^2=\frac{4(1+2\sqrt{7}+7)}{9}=\frac{32+8\sqrt{7}}{9}.$$

These are equal, and $2(y-1)>0$, so taking the positive square root gives $\sqrt{y^2+1}=2(y-1)$, as required. (Of course: y is a root of $3y^2-8y+3=0$, which is the equation $y^2+1=4(y-1)^2$.)

Ask yourself What does this equality tell us about the chain of inequalities in part (i)?

The two ends of the chain $y^2+1\geq(y\cos\theta-\sin\theta)^2\geq 4(y-1)^2$ are now equal, so both inequalities are equalities.

The discriminant is zero, so the quadratic in x has a repeated root

$$x=-\frac{y\cos\theta-\sin\theta}{2(y-1)}.$$

The bound on θ is attained, so $y\sin\theta+\cos\theta=0$, giving

$$\tan\theta=-\frac{1}{y}=-\frac{3}{4+\sqrt{7}}=-\frac{3(4-\sqrt{7})}{16-7}=\frac{\sqrt{7}-4}{3}.$$

(Here $\cos\theta \neq 0$, since otherwise $\sin\theta = 0$ too.) Moreover $(y\cos\theta - \sin\theta)^2 = y^2 + 1 = 4(y-1)^2$, so $y\cos\theta - \sin\theta = \pm 2(y-1)$ and therefore

$$x = \mp 1.$$

Concretely, $(\cos\theta, \sin\theta)$ is parallel to $(y, -1)$. If it points the same way, $\cos\theta > 0$ and $\sin\theta < 0$, then $y\cos\theta - \sin\theta = +2(y-1)$ and $x = -1$. If it points the opposite way, $\cos\theta < 0$ and $\sin\theta > 0$, then $x = 1$. In both cases

$$x = \pm 1, \qquad \tan\theta = \frac{\sqrt{7} - 4}{3}.$$

The two cases are really one: replacing x by $-x$ and θ by $\theta + \pi$ leaves y unchanged.

Checks

- At $x = 1$ the formula is $y = \dfrac{2 + \sin\theta}{2 + \cos\theta}$. Taking θ in the second quadrant with $\tan\theta = \frac{1}{3}(\sqrt{7} - 4) \approx -0.4514$ gives $\theta \approx 2.7176$, and then $y \approx 2.21525$, which matches $\frac{1}{3}(4 + \sqrt{7}) \approx 2.21525$.
- Sampling many random real x and θ by computer, the values of y stayed between 0.45142 and 2.21524, just inside the bounds $\frac{1}{3}(4 \mp \sqrt{7}) \approx 0.451416$ and 2.215250.
- A neat coincidence to notice: $\tan\theta = -\frac{1}{3}(4 - \sqrt{7})$ is minus the lower bound, because the two bounds are reciprocals (their product is $\frac{1}{9}(16 - 7) = 1$).

Common slips

- Forgetting the case $y = 1$, where the equation in x is not a quadratic and the discriminant argument does not apply directly.
- Taking $\sqrt{y^2 + 1} = -2(y - 1)$ or not checking signs: the positive root is correct here only because $y > 1$.
- Giving only one value of x; both $x = 1$ and $x = -1$ occur, with θ in different quadrants.

Beyond STEP

The step $(y\cos\theta - \sin\theta)^2 \leq (y^2 + 1)(\cos^2\theta + \sin^2\theta)$ is the two-dimensional case of the Cauchy–Schwarz inequality $|\mathbf{u}\cdot\mathbf{v}| \leq |\mathbf{u}|\,|\mathbf{v}|$, with equality exactly when the vectors are parallel. This inequality holds in every inner-product space and is one of the most used tools in first-year analysis and linear algebra. Strang's *Introduction to Linear Algebra* develops it geometrically; Abbott's *Understanding Analysis* shows it at work in analysis.

WORKED SOLUTION

STEP II 2016, Question 5

Pure • Binomial Coefficient Identities

STEP II 2016, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q5

The key idea

All three identities come from one trick: write a function as a product in two ways and compare the coefficient of a single power of x. A sum of products of binomial coefficients, with indices adding to a fixed total, is exactly what you get when you multiply two power series and read off one coefficient. The question's convention that $\binom{p}{q} = 0$ outside $p \geq q \geq 0$ is what lets the sums run over a tidy range of j without fuss about which terms exist.

Solution

Part (i)

For $|x| < 1$ the binomial series gives the coefficient of x^n in $(1-x)^{-N}$ as

$$\frac{(-N)(-N-1)\cdots(-N-n+1)}{n!}(-1)^n = \frac{N(N+1)\cdots(N+n-1)}{n!}$$
$$= \frac{(N+n-1)!}{n!\,(N-1)!} = \binom{N+n-1}{n}.$$

The n minus signs from the factors cancel the n minus signs from $(-x)^n$. So

$$(1-x)^{-N} = \sum_{n=0}^{\infty} \binom{N+n-1}{n} x^n \qquad (|x| < 1).$$

(When $n = 0$ the coefficient is $\binom{N-1}{0} = 1$. This is fine because $N \geq 1$.)

Now $(1-x)^{-1}(1-x)^{-N} = (1-x)^{-(N+1)}$. By the expansion above with $N+1$ in place of N, the coefficient of x^n on the right is $\binom{N+n}{n}$.

On the left, $(1-x)^{-1} = \sum_{i \geq 0} x^i$ has every coefficient equal to 1. A term x^j from $(1-x)^{-N}$ must pair with x^{n-j} from $(1-x)^{-1}$, and j runs from 0 to n. So the coefficient

of x^n in the product is

$$\sum_{j=0}^{n} 1 \cdot \binom{N+j-1}{j}.$$

Both power series converge for $|x| < 1$ and represent the same function there, so their coefficients agree. That gives

$$\sum_{j=0}^{n} \binom{N+j-1}{j} = \binom{N+n}{n},$$

as required.

Part (ii)

Use $(1+x)^{m+n} = (1+x)^m (1+x)^n$, where all three are polynomials. With the question's convention we can write

$$(1+x)^m = \sum_{j \geq 0} \binom{m}{j} x^j, \qquad (1+x)^n = \sum_{k \geq 0} \binom{n}{k} x^k,$$

since the coefficients with $j > m$ or $k > n$ are zero. The coefficient of x^r in the product comes from the pairs with $j + k = r$, that is $k = r - j$ with $0 \leq j \leq r$:

$$\sum_{j=0}^{r} \binom{m}{j} \binom{n}{r-j}.$$

Ask yourself What about the terms where $j > m$ or $r - j > n$? Those powers don't appear in the polynomials.

Those terms contain a factor $\binom{m}{j}$ with $j > m$ or $\binom{n}{r-j}$ with $r - j > n$, so they are zero by the convention. They contribute nothing, and they also correspond to no actual product of terms, so the sum over $0 \leq j \leq r$ is exactly right. The coefficient of x^r in $(1+x)^{m+n}$ is $\binom{m+n}{r}$. Polynomials that are equal have equal coefficients, so

$$\binom{m+n}{r} = \sum_{j=0}^{r} \binom{m}{j} \binom{n}{r-j}.$$

Part (iii)

The factor $(-1)^j \binom{m+j-1}{j}$ suggests part (i) with $-x$ in place of x. Replacing x by $-x$ in the expansion from part (i), with m in place of N, gives

$$(1+x)^{-m} = \sum_{j=0}^{\infty} (-1)^j \binom{m+j-1}{j} x^j \qquad (|x| < 1).$$

Ask yourself Which product should we use?

The other factor $\binom{N+m}{n-j}$ is a coefficient of $(1+x)^{N+m}$, and the answer $\binom{N}{n}$ is a coefficient of $(1+x)^N$. Since m is a positive integer, for $|x|<1$ we have

$$(1+x)^{N+m}(1+x)^{-m} = (1+x)^N.$$

On the left, $(1+x)^{N+m} = \sum_{k\geq 0}\binom{N+m}{k}x^k$ (using the convention for $k > N+m$). The coefficient of x^n comes from x^j in $(1+x)^{-m}$ times x^{n-j} in $(1+x)^{N+m}$, for $0 \leq j \leq n$:

$$\sum_{j=0}^{n}\binom{N+m}{n-j}(-1)^j\binom{m+j-1}{j}.$$

On the right, the coefficient of x^n in $(1+x)^N$ is $\binom{N}{n}$. When $n > N$ this is 0 by the convention, which is also correct, because $(1+x)^N$ has degree N. The two sides are equal on $|x|<1$, so their power series coefficients agree:

$$\sum_{j=0}^{n}(-1)^j\binom{N+m}{n-j}\binom{m+j-1}{j} = \binom{N}{n}.$$

Checks

- Part (i) with $N=2$, $n=3$: $1+2+3+4 = 10 = \binom{5}{3}$.
- Part (ii) with $m=n=2$, $r=2$: $1\cdot 1 + 2\cdot 2 + 1\cdot 1 = 6 = \binom{4}{2}$.
- Part (iii) with $m=2$, $N=3$, $n=2$: $\binom{5}{2}\binom{1}{0} - \binom{5}{1}\binom{2}{1} + \binom{5}{0}\binom{3}{2} = 10 - 10 + 3 = 3 = \binom{3}{2}$. With $n > N$ the alternating sum should come out as 0. For example, $m=1$, $N=1$, $n=2$ gives $\binom{2}{2} - \binom{2}{1} + \binom{2}{0} = 1 - 2 + 1 = 0$.

All three identities were also confirmed by computer for every $m, N \leq 9$ and $n \leq 24$.

Common slips

- Losing track of the signs in the coefficient of $(1-x)^{-N}$. There are two lots of $(-1)^n$, and they cancel.
- In part (iii), forgetting that the series for $(1+x)^{-m}$ is infinite and holds only for $|x|<1$. Comparing coefficients is valid because two power series that agree on an interval have the same coefficients.

Beyond STEP

This is the method of *generating functions*: encode a sequence as the coefficients of a power series, and multiplying series turns into the convolution sums seen here. Part (ii)

is Vandermonde's identity, and part (i) is often called the hockey-stick identity. Both also have neat counting proofs, which are worth finding. Wilf's *generatingfunctionology* (free online) and Graham, Knuth and Patashnik, *Concrete Mathematics*, develop this toolkit much further.

WORKED SOLUTION

STEP II 2016, Question 6

Pure • Chebyshev Polynomial Equations

STEP II 2016, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q6

The key idea

Every part is a matter of substituting a proposed function into the differential equation and checking the condition at $x = 1$; the uniqueness assumption then does the rest. The trick in (iii) and (iv) is never to expand y_n (we don't know it explicitly), but to use the equation $(1 - x^2)y_n'^2 = n^2(1 - y_n^2)$ itself to eliminate the derivative.

Solution

Throughout, y_k satisfies

$$(1 - x^2)\big(y_k'(x)\big)^2 = k^2\big(1 - y_k(x)^2\big), \qquad y_k(1) = 1,$$

which is just $(*)$ rearranged. To show a function equals y_k, it is enough (by the assumed uniqueness) to show it satisfies $(*)$ with that value of k and takes the value 1 at $x = 1$.

Part (i)

With $k = 1$ the equation is

$$(1 - x^2)\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + y^2 = 1.$$

For $y = x$ we have $y' = 1$, and the left-hand side is $(1 - x^2) + x^2 = 1$, as required. Also $y(1) = 1$, so $y_1(x) = x$.

Part (ii)

With $k = 2$ the equation is

$$(1 - x^2)\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + 4y^2 = 4.$$

For $y = 2x^2 - 1$ we have $y' = 4x$, so

$$\begin{aligned}(1-x^2)(4x)^2 + 4(2x^2-1)^2 &= 16x^2 - 16x^4 + 16x^4 - 16x^2 + 4\\ &= 4.\end{aligned}$$

Since $y(1) = 2 - 1 = 1$, we get $y_2(x) = 2x^2 - 1$.

Part (iii)

Write $y = y_n(x)$ for short. Differentiating $z = 2y^2 - 1$ by the chain rule gives $z' = 4yy'$, so

$$(1-x^2)z'^2 = 16y^2\,(1-x^2)y'^2 = 16y^2 \cdot n^2(1-y^2),$$

using the equation for y_n. Also

$$4n^2z^2 = 4n^2(2y^2-1)^2 = 4n^2(4y^4 - 4y^2 + 1).$$

Adding,

$$(1-x^2)z'^2 + 4n^2z^2 = 16n^2y^2 - 16n^2y^4 + 16n^2y^4 - 16n^2y^2 + 4n^2 = 4n^2,$$

which is the required equation.

? **Ask yourself** What does this equation have to do with y_{2n}?

It is exactly $(*)$ with $k = 2n$, since $(2n)^2 = 4n^2$. Moreover $z(1) = 2y_n(1)^2 - 1 = 2 - 1 = 1$. So z is a solution of $(*)$ with $k = 2n$ taking the value 1 at $x = 1$, and by uniqueness

$$y_{2n}(x) = 2\big(y_n(x)\big)^2 - 1.$$

As a check, $n = 1$ gives $y_2 = 2x^2 - 1$, agreeing with (ii), and $n = 2$ gives $y_4 = 2(2x^2-1)^2 - 1 = 8x^4 - 8x^2 + 1$.

Part (iv)

Write $u = y_m(x)$, so $v = y_n(u)$. By the chain rule,

$$v'(x) = y_n'(u)\,y_m'(x).$$

Hence, using first the equation for y_m (at the point x) and then the equation for y_n (at the point u),

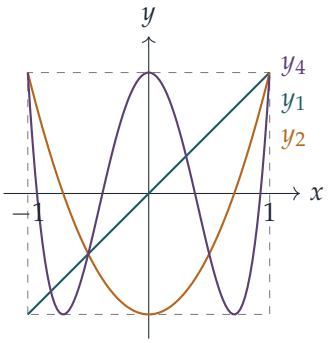
$$\begin{aligned}(1-x^2)v'^2 &= y_n'(u)^2\,(1-x^2)y_m'(x)^2\\ &= y_n'(u)^2 \cdot m^2(1-u^2)\\ &= m^2\,(1-u^2)y_n'(u)^2\\ &= m^2 \cdot n^2\big(1 - y_n(u)^2\big) = m^2n^2(1-v^2).\end{aligned}$$

So $(1-x^2)v'^2 + (mn)^2v^2 = (mn)^2$, which is $(*)$ with $k = mn$. Finally

$$v(1) = y_n\big(y_m(1)\big) = y_n(1) = 1,$$

so by uniqueness $v(x) = y_{mn}(x)$, as required.

The equation for y_n is applied at the point u rather than x: it holds for every value of its variable, so we may use it with u in place of x. This is the step that makes (iv) work.



All three curves pass through $(1,1)$ and stay between -1 and 1 on $[-1,1]$, as the equation forces: $(1-x^2)y'^2 \geq 0$ there, so $k^2y^2 \leq k^2$.

Checks

- Part (iii) with $n = 2$ predicts $y_4(x) = 8x^4 - 8x^2 + 1$. Substituting directly, $y_4' = 32x^3 - 16x$ and $(1-x^2)(32x^3-16x)^2 + 16(8x^4-8x^2+1)^2$ simplifies to 16. At $x = 0.3$, $y_4 = 0.3448$.
- Part (iv) with $m = n = 2$: $y_2(y_2(x)) = 2(2x^2-1)^2 - 1$, which is the same y_4 as from (iii), as it must be.
- A computer check confirmed with exact arithmetic that the polynomials built this way satisfy $(*)$ for every k up to 24, and that $y_n(y_m(x)) = y_{mn}(x)$ for $m, n \leq 4$.

Common slips

- Forgetting the condition at $x = 1$. Satisfying $(*)$ alone isn't enough: $-y_k$ also satisfies it.
- Trying to expand y_n explicitly in (iii) or (iv). Nothing explicit is known about y_n; use its differential equation to replace $(1-x^2)y_n'^2$.
- In (iv), using the equation for y_n with x instead of $u = y_m(x)$, which gives the factor $(1-x^2)$ in the wrong place.

Beyond STEP

The functions y_k are the Chebyshev polynomials T_k: putting $x = \cos\theta$ turns $(*)$ into $(\mathrm{d}y/\mathrm{d}\theta)^2 = k^2(1-y^2)$, solved by $y = \cos k\theta$, so $T_k(\cos\theta) = \cos k\theta$. Part (iii) is then the

double-angle formula and part (iv) says $\cos\big(n(m\theta)\big) = \cos(mn\theta)$. A word of caution: the uniqueness you were allowed to assume is not quite true, since the constant $y = 1$ also satisfies $(*)$ and $y(1) = 1$. Equations like this one, where the derivative is only determined up to sign, are exactly where uniqueness theorems need care; Boyce and DiPrima, *Elementary Differential Equations*, discuss existence and uniqueness of solutions.

WORKED SOLUTION

STEP II 2016, Question 7

Pure • Definite Integrals by Symmetry

STEP II 2016, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q7

The key idea

Reflecting the interval $[0, a]$ about its midpoint, $x \mapsto a - x$, doesn't change the integral, but it can change the integrand into something that combines beautifully with the original. For trigonometric integrands on $[0, \frac{1}{2}\pi]$ this swaps sin and cos; on $[0, \frac{1}{4}\pi]$ the compound-angle formula for $\tan(\frac{1}{4}\pi - x)$ does the work. The final part looks new, but an integration by parts turns it into part (iii).

Solution

The result $(*)$

In $\int_0^a f(a-x)\,\mathrm{d}x$ substitute $u = a - x$, so $\mathrm{d}u = -\mathrm{d}x$; when $x = 0$, $u = a$, and when $x = a$, $u = 0$. Then

$$\int_0^a f(a-x)\,\mathrm{d}x = \int_a^0 f(u)\,(-\mathrm{d}u) = \int_0^a f(u)\,\mathrm{d}u = \int_0^a f(x)\,\mathrm{d}x,$$

since the name of the integration variable doesn't matter. This is $(*)$.

Part (i)

Call the integral I. With $a = \frac{1}{2}\pi$ we have $\sin(\frac{1}{2}\pi - x) = \cos x$ and $\cos(\frac{1}{2}\pi - x) = \sin x$, so $(*)$ gives

$$I = \int_0^{\frac{1}{2}\pi} \frac{\cos x}{\sin x + \cos x}\,\mathrm{d}x.$$

Adding this to the original form of I,

$$2I = \int_0^{\frac{1}{2}\pi} \frac{\sin x + \cos x}{\cos x + \sin x}\,\mathrm{d}x = \int_0^{\frac{1}{2}\pi} 1\,\mathrm{d}x = \frac{\pi}{2},$$

so

$$I = \frac{\pi}{4}.$$

(The denominator $\cos x + \sin x$ is positive on $[0, \frac{1}{2}\pi]$, so nothing goes wrong.)

Part (ii)

Now $a = \frac{1}{4}\pi$, and the trick of part (i) no longer swaps sin and cos. Instead, dividing top and bottom by $\cos x$ (positive on this interval), the integrand is

$$\frac{\sin x}{\cos x + \sin x} = \frac{\tan x}{1 + \tan x}.$$

Let $t = \tan x$. Since $\tan(\frac{1}{4}\pi - x) = \dfrac{1-t}{1+t}$,

$$\frac{\tan(\frac{1}{4}\pi - x)}{1 + \tan(\frac{1}{4}\pi - x)} = \frac{(1-t)/(1+t)}{2/(1+t)} = \frac{1-t}{2} = \frac{1}{2} - \frac{1}{2}\tan x.$$

So by $(*)$, calling the integral J,

$$\begin{aligned} J &= \int_0^{\frac{1}{4}\pi} \Big(\frac{1}{2} - \frac{1}{2}\tan x\Big)\,\mathrm{d}x = \frac{\pi}{8} - \frac{1}{2}\Big[-\ln\cos x\Big]_0^{\frac{1}{4}\pi} \\ &= \frac{\pi}{8} + \frac{1}{2}\ln\frac{1}{\sqrt{2}} = \frac{\pi}{8} - \frac{1}{4}\ln 2. \end{aligned}$$

Ask yourself Is there a way to do part (ii) that doesn't use $(*)$ at all?

Yes. Let $K = \int_0^{\frac{1}{4}\pi} \frac{\cos x}{\cos x + \sin x}\,\mathrm{d}x$. Then $J + K = \frac{1}{4}\pi$, and since the numerator $\cos x - \sin x$ is the derivative of the denominator,

$$K - J = \Big[\ln(\cos x + \sin x)\Big]_0^{\frac{1}{4}\pi} = \ln\sqrt{2} = \tfrac{1}{2}\ln 2.$$

Subtracting, $2J = \frac{1}{4}\pi - \frac{1}{2}\ln 2$, which agrees with the answer above.

Part (iii)

Call the integral L. The same compound-angle formula gives

$$1 + \tan(\tfrac{1}{4}\pi - x) = 1 + \frac{1 - \tan x}{1 + \tan x} = \frac{2}{1 + \tan x}.$$

So by $(*)$,

$$L = \int_0^{\frac{1}{4}\pi} \ln\frac{2}{1+\tan x}\,\mathrm{d}x = \int_0^{\frac{1}{4}\pi} \big(\ln 2 - \ln(1+\tan x)\big)\,\mathrm{d}x = \frac{\pi}{4}\ln 2 - L.$$

Hence $2L = \frac{1}{4}\pi\ln 2$ and

$$L = \frac{\pi}{8}\ln 2.$$

Part (iv)

Divide top and bottom by $\cos^2 x$:

$$\frac{x}{\cos x\,(\cos x + \sin x)} = \frac{x\sec^2 x}{1+\tan x}.$$

Ask yourself What has $\dfrac{\sec^2 x}{1+\tan x}$ got to do with part (iii)?

It is exactly the derivative of $\ln(1+\tan x)$, because $\frac{\mathrm{d}}{\mathrm{d}x}(1+\tan x) = \sec^2 x$; and $1+\tan x > 0$ on $[0, \frac{1}{4}\pi]$. So we integrate by parts with $u = x$ and $\mathrm{d}v = \dfrac{\sec^2 x}{1+\tan x}\,\mathrm{d}x$, $v = \ln(1+\tan x)$:

$$\begin{aligned}\int_0^{\frac{1}{4}\pi} \frac{x\sec^2 x}{1+\tan x}\,\mathrm{d}x &= \Big[x\ln(1+\tan x)\Big]_0^{\frac{1}{4}\pi} - \int_0^{\frac{1}{4}\pi} \ln(1+\tan x)\,\mathrm{d}x \\ &= \frac{\pi}{4}\ln 2 - \frac{\pi}{8}\ln 2 \qquad \text{(using part (iii))} \\ &= \frac{\pi}{8}\ln 2.\end{aligned}$$

So the integral in part (iv) has the same value as the one in part (iii).

Checks

- Part (i): the integrand rises from 0 to 1 and is symmetric about the point $(\frac{1}{4}\pi, \frac{1}{2})$, so its average value is $\frac{1}{2}$ and the integral is half the interval length, $\frac{1}{4}\pi$.
- Part (ii) should be less than half of part (i), because the integrand is below $\frac{1}{2}$ on $[0, \frac{1}{4}\pi]$: indeed $\frac{1}{8}\pi - \frac{1}{4}\ln 2 \approx 0.2194 < 0.3927$.
- Numerical quadrature (Simpson's rule) gives 0.785398, 0.219412, 0.272198 and 0.272198 for the four parts, matching $\frac{1}{4}\pi$, $\frac{1}{8}\pi - \frac{1}{4}\ln 2$ and $\frac{1}{8}\pi\ln 2$ (twice).

Common slips

- Forgetting that the substitution $u = a - x$ swaps the limits; the minus sign from $\mathrm{d}u = -\mathrm{d}x$ swaps them back.
- In part (ii), trying to swap sin and cos as in part (i): on $[0, \frac{1}{4}\pi]$ the reflection is $x \mapsto \frac{1}{4}\pi - x$, not $\frac{1}{2}\pi - x$.
- Writing $\int \tan x\,\mathrm{d}x = \ln\cos x$; it is $-\ln\cos x$ (or $\ln\sec x$).

Beyond STEP

Part (iii) is a classic: the same reflection trick evaluates integrals such as $\int_0^{\frac{1}{2}\pi} \ln \sin x \, \mathrm{d}x = -\frac{1}{2}\pi \ln 2$, which can't be found from an antiderivative in elementary functions. At university these symmetry arguments sit alongside contour integration and differentiation under the integral sign as tools for definite integrals with no elementary antiderivative. For many more examples in the STEP spirit, see Siklos, *Advanced Problems in Mathematics*; for the analysis that justifies substitution and integration by parts, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2016, Question 8

Pure • Approximating Sums by Integrals

STEP II 2016, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q8

The key idea

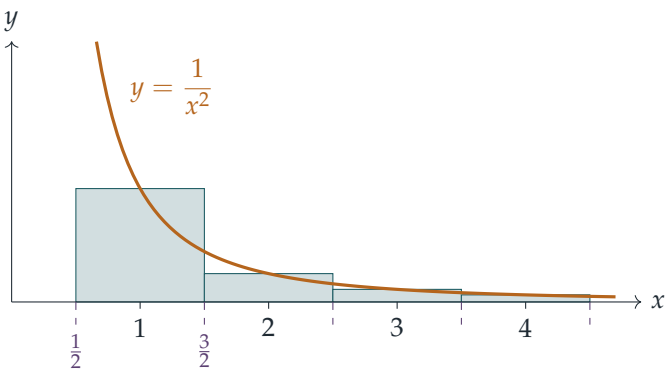
The term $\frac{1}{r^2}$ is the area of a rectangle of width 1 and height $\frac{1}{r^2}$, and if that rectangle is centred on $x = r$ its area is very close to the area under $y = \frac{1}{x^2}$ between $r - \frac{1}{2}$ and $r + \frac{1}{2}$ (the midpoint rule). Gluing these strips together turns a sum into one integral. Then the error in each strip can be computed exactly, and adding up the errors is what produces a second series, $\sum \frac{1}{r^4}$, almost for free.

Solution

The integral and the sketch

For $m > \frac{1}{2}$ the lower limit is positive, so the integrand is continuous on the range and

$$\int_{m-\frac{1}{2}}^{\infty} \frac{1}{x^2}\,\mathrm{d}x = \lim_{X\to\infty}\left[-\frac{1}{x}\right]_{m-\frac{1}{2}}^{X} = \frac{1}{m-\frac{1}{2}} = \frac{2}{2m-1}.$$



The rectangle centred on $x = r$ has width 1 and height $\frac{1}{r^2}$, so its area is $\frac{1}{r^2}$. Over the strip $r - \frac{1}{2} \le x \le r + \frac{1}{2}$ the curve lies above the rectangle's top on the left half and

below it on the right half, and these two small pieces nearly cancel. So

$$\frac{1}{r^2} \approx \int_{r-\frac{1}{2}}^{r+\frac{1}{2}} \frac{1}{x^2}\,\mathrm{d}x.$$

Adding these for $r = m, m+1, \ldots, n$, the strips fit together end to end to cover $[m - \frac{1}{2}, n + \frac{1}{2}]$, which gives $(*)$.

Part (i)

Let $n \to \infty$ in $(*)$. The right-hand side tends to the integral we evaluated, so

$$\sum_{r=m}^{\infty} \frac{1}{r^2} \approx \frac{2}{2m-1}.$$

We can apply this to the whole series, or add the first few terms exactly and approximate only the tail:

$$\begin{aligned}
m = 1: &\quad E \approx \frac{2}{1} = 2,\\
m = 2: &\quad E = 1 + \sum_{r=2}^{\infty} \frac{1}{r^2} \approx 1 + \frac{2}{3} = \frac{5}{3},\\
m = 3: &\quad E = 1 + \frac{1}{4} + \sum_{r=3}^{\infty} \frac{1}{r^2} \approx \frac{5}{4} + \frac{2}{5} = \frac{33}{20}.
\end{aligned}$$

These are the three required approximations. They should improve as m increases, since the curve is flatter further out and the midpoint approximation is better there.

Part (ii)

The integral over one strip is

$$\int_{r-\frac{1}{2}}^{r+\frac{1}{2}} \frac{1}{x^2}\,\mathrm{d}x = \frac{1}{r-\frac{1}{2}} - \frac{1}{r+\frac{1}{2}} = \frac{1}{r^2-\frac{1}{4}} = \frac{4}{4r^2-1}.$$

So the error in approximating $\frac{1}{r^2}$ by this integral is

$$\frac{4}{4r^2-1} - \frac{1}{r^2} = \frac{4r^2-(4r^2-1)}{r^2(4r^2-1)} = \frac{1}{r^2(4r^2-1)} = \frac{1}{4r^4} \cdot \frac{1}{1-\frac{1}{4r^2}}.$$

When r is large, $\frac{1}{4r^2}$ is small, so the last factor is close to 1 and the error is approximately $\frac{1}{4r^4}$, as required. (The integral is always the larger, which fits the picture: $\frac{1}{x^2}$ is convex, so the extra area on the left of each strip beats the shortfall on the right.)

For the final result, add up the errors from $r = m$ onwards. The strip integrals add up to the integral we evaluated first, so

$$\frac{2}{2m-1} - \sum_{r=m}^{\infty} \frac{1}{r^2} = \sum_{r=m}^{\infty} \frac{1}{r^2(4r^2-1)} \approx \frac{1}{4} \sum_{r=m}^{\infty} \frac{1}{r^4}.$$

Ask yourself Why not take $m = 1$, which looks simplest?

With $m = 1$ the left-hand side is $2 - 1.645 = 0.355$, giving $\sum \frac{1}{r^4} \approx 1.42$, which is far too big. The trouble is that "$\approx \frac{1}{4r^4}$" is only good for large r: at $r = 1$ the true error is $\frac{1}{3}$, not $\frac{1}{4}$, and that first term dominates the whole sum. So we should keep the first few terms exact and use the approximation only on the tail. Taking $m = 3$ (the choice that gave $\frac{33}{20}$), the worst factor $\frac{1}{1-1/(4r^2)}$ is $\frac{36}{35}$, within 3% of 1:

$$\frac{1}{4}\sum_{r=3}^{\infty}\frac{1}{r^4} \approx \frac{2}{5} - \left(E - 1 - \frac{1}{4}\right) \approx 0.4 - 0.395 = 0.005,$$

so $\sum_{r=3}^{\infty}\frac{1}{r^4} \approx 0.02$ and

$$\sum_{r=1}^{\infty}\frac{1}{r^4} \approx 1 + \frac{1}{16} + 0.02 = 1.0825 \approx 1.08.$$

This is the required result. Since E is only given to three decimal places, an error of up to 0.0005 in E becomes up to 0.002 here, which still leaves 1.08.

Checks

- The approximations 2, $\frac{5}{3} \approx 1.667$, $\frac{33}{20} = 1.65$ decrease towards $E = \frac{\pi^2}{6} \approx 1.6449$, and each is too big, as the convexity argument predicts.
- At $r = 10$ the exact error $\frac{1}{100\times399} \approx 2.506 \times 10^{-5}$ is very close to $\frac{1}{4\times10^4} = 2.5 \times 10^{-5}$.
- The exact value is $\sum \frac{1}{r^4} = \frac{\pi^4}{90} \approx 1.0823$, which agrees with our 1.0825. Using $m = 2$ instead gives $1 + 4\left(\frac{2}{3} - 0.645\right) \approx 1.087$, noticeably worse; $m = 1$ gives 1.42.

Common slips

- Using $m = 1$ in the last part, where the large-r approximation $\frac{1}{4r^4}$ is poor, and getting 1.42.
- Getting the sign of the error backwards: the integral exceeds $\frac{1}{r^2}$, so $\frac{2}{2m-1}$ is an overestimate of the tail.
- Forgetting to add the exact first terms ($1 + \frac{1}{4}$, and $1 + \frac{1}{16}$) back on.

Beyond STEP

This is the midpoint rule, and the error term $\frac{1}{4r^4}$ is a glimpse of the Euler–Maclaurin formula, which corrects a sum by an integral plus a series of derivative terms; it is how sums like $\sum \frac{1}{r^2}$ are computed to many decimal places. The exact values $\frac{\pi^2}{6}$ and $\frac{\pi^4}{90}$ come

from Fourier series or from Euler's product for $\sin x$. Abbott's *Understanding Analysis* treats series and integrals carefully, and Siklos's *Advanced Problems in Mathematics* has more questions in this spirit.

WORKED SOLUTION

STEP II 2016, Question 9

Mechanics • Momentum and Energy Loss

STEP II 2016, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

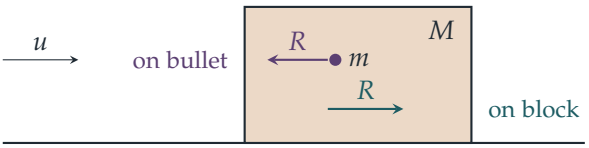
step.maths.org/questions/16-s2-q9

The key idea

The resistance R is an internal force between bullet and block: it decelerates the bullet and, by Newton's third law, pushes the block forward with the same magnitude. So momentum of the pair is conserved, while kinetic energy is not. The energy lost is exactly the work done by R through the *relative* distance the bullet moves inside the block. Applying the work–energy principle to the block alone gives how far the block moves.

Solution

We model the bullet and block as particles moving along one horizontal line, with the resistance R constant while the bullet moves relative to the block. Take the bullet's initial direction as positive.



The vertical forces (weights, normal reaction from the table) balance and play no part. Horizontally, the bullet feels R backwards and the block feels R forwards.

Part (i)

With the block fixed, the bullet has constant deceleration R/m. Using $v^2 = u^2 + 2\alpha s$ with final speed 0 and $\alpha = -R/m$:

$$0 = u^2 - \frac{2R}{m}a \quad \Longrightarrow \quad \boxed{a = \frac{mu^2}{2R}.}$$

(Equivalently, the work done against the resistance, Ra, equals the bullet's initial kinetic energy $\frac{1}{2}mu^2$.)

Part (ii)

Common final velocity. No external horizontal force acts on the system of bullet and block, because the table is smooth. When the bullet is at rest relative to the block, both move with the same velocity V, and conservation of momentum gives

$$mu = (M+m)V \quad \Longrightarrow \quad V = \frac{mu}{M+m}.$$

Ask yourself The resistance acts through a distance b on the bullet's path inside the block, but the bullet itself travels $b+c$ over the table. Which distance goes into the energy equation?

Finding b. Relative to the table, the force R on the bullet does work $-R(b+c)$, since the bullet moves $b+c$; the force R on the block does work $+Rc$, since the block moves c. The net work done on the system is therefore $-Rb$: only the relative displacement counts. By the work–energy principle,

$$\begin{aligned} Rb &= \tfrac{1}{2}mu^2 - \tfrac{1}{2}(M+m)V^2 \\ &= \tfrac{1}{2}mu^2 - \frac{m^2u^2}{2(M+m)} \\ &= \tfrac{1}{2}mu^2\left(1 - \frac{m}{M+m}\right) = \tfrac{1}{2}mu^2 \cdot \frac{M}{M+m}. \end{aligned}$$

Since $\frac{1}{2}mu^2 = Ra$ from part (i), dividing by R gives

$$\boxed{b = \frac{M}{M+m}\,a.}$$

Finding c. Now apply the work–energy principle to the block alone. It starts at rest, ends with speed V, and the only horizontal force on it is R forwards, acting through distance c:

$$Rc = \tfrac{1}{2}MV^2 = \frac{Mm^2u^2}{2(M+m)^2} = \tfrac{1}{2}mu^2 \cdot \frac{mM}{(M+m)^2}.$$

Using $\frac{1}{2}mu^2 = Ra$ again,

$$\boxed{c = \frac{mM}{(M+m)^2}\,a.}$$

An alternative via kinematics. The bullet decelerates at R/m and the block accelerates at R/M. Their velocities agree when $u - Rt/m = Rt/M$, so

$$t = \frac{mMu}{R(M+m)}, \qquad c = \frac{1}{2}\cdot\frac{R}{M}\,t^2 = \frac{m^2Mu^2}{2R(M+m)^2},$$

which agrees with the above, and b follows from the bullet's displacement minus c.

Checks

- As $M \to \infty$ the block behaves as if fixed: $b \to a$ and $c \to 0$, as they should.
- Since $\frac{M}{M+m} < 1$, we have $b < a$: a movable block absorbs less of the bullet's energy as heat, so the bullet does not penetrate as far.
- With $M = 4m$ we get $b = \frac{4}{5}a$ and $c = \frac{4}{25}a$. Stepping the motion forward numerically in small time steps, for several random choices of m, M, u and R, reproduces a, b and c to four significant figures.

Common slips

- Using the bullet's distance over the table, $b + c$, in the energy-loss equation instead of the relative distance b.
- Assuming kinetic energy is conserved in the collision; it is not, and the loss Rb is the heat generated.
- Forgetting that the block feels the force R too (Newton's third law), and so leaving the block's motion unexplained.

Beyond STEP

The fact that internal forces do zero net work only when there is no relative motion is the start of a careful treatment of energy in systems of particles. Working in the centre-of-mass frame makes it clean: the kinetic energy available for dissipation is $\frac{1}{2}\mu u^2$, with reduced mass $\mu = mM/(M+m)$, which is exactly Rb here. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops these ideas thoroughly.

WORKED SOLUTION

STEP II 2016, Question 10

Mechanics • Friction and Equilibrium

STEP II 2016, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q10

The key idea

Only two things act on the triangle: its weight, at the centre of mass G, and the single contact force from the peg at the point of contact P. Two forces in equilibrium must be equal, opposite and along the same line, so the triangle swings round until G is directly below P. The question then becomes: how steeply must the peg's reaction lean away from the normal to BC, and is friction strong enough for that? So the first job is to locate G.

Solution

The model and the centre of mass

Treat the wire as three uniform rods with the same mass per unit length, and the peg as a single point of contact giving a normal reaction R (perpendicular to BC) and a friction force F (along BC), with $F \leq \mu R$.

Let $AB = AC = b$ and let M be the midpoint of BC. Since the angle at A is 2θ, the line AM bisects it and is perpendicular to BC, so

$$BM = MC = b\sin\theta, \qquad AM = b\cos\theta, \qquad BC = 2b\sin\theta.$$

By symmetry G lies on AM. Measure distances from BC towards A. The rods AB and AC each have mass proportional to b, with centres at their midpoints, at distance $\frac{1}{2}b\cos\theta$ from BC; the rod BC has mass proportional to $2b\sin\theta$ and its centre lies on BC. So the distance $d = GM$ is

$$d = \frac{b\cdot\frac{1}{2}b\cos\theta + b\cdot\frac{1}{2}b\cos\theta + 2b\sin\theta\cdot 0}{b+b+2b\sin\theta} = \frac{b\cos\theta}{2(1+\sin\theta)}.$$

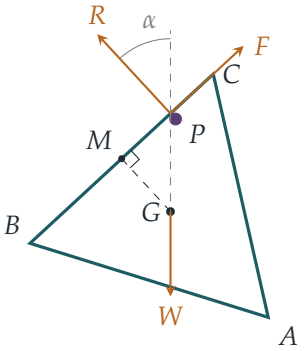
Equilibrium with the peg at a general point

Suppose the peg touches BC at P, with $MP = x$, where $0 \leq x \leq b\sin\theta$ (by symmetry we may take P between M and C). The triangle hangs below the peg, so the peg is

inside the triangle and BC rests on top of it.

Ask yourself The triangle can hang at any angle. How do we know which one it takes?

The forces on the triangle are the weight W at G and the peg's total contact force at P. With only two forces, equilibrium needs them to be equal and opposite (so the contact force is W vertically upwards) and to act along the same line (otherwise they would form a couple). So PG must be vertical: the triangle settles with G directly below P.



Let α be the angle between PG (the vertical) and the normal GM to BC. In the right-angled triangle PMG,

$$\tan\alpha = \frac{MP}{GM} = \frac{x}{d}.$$

The normal to BC makes angle α with the vertical, and BC makes angle α with the horizontal. Resolving perpendicular to and along BC:

$$R = W\cos\alpha, \qquad F = W\sin\alpha.$$

Both are non-negative since $0 \le \alpha < \frac{1}{2}\pi$, and $R > 0$. The triangle stays put without slipping provided $F \le \mu R$, that is,

$$\mu \ge \tan\alpha = \frac{x}{d}.$$

Every point of BC

The right-hand side x/d increases with x, so the condition is hardest to satisfy when P is at an end, $x = b\sin\theta$. Equilibrium is possible at every point of BC exactly when it is possible there:

$$\mu \ge \frac{b\sin\theta}{d} = b\sin\theta \cdot \frac{2(1+\sin\theta)}{b\cos\theta} = 2\tan\theta\,(1+\sin\theta),$$

which is the required condition. If it holds, then $\mu \ge x/d$ for every x in $[0, b\sin\theta]$, so the triangle can rest with the peg at any point of BC.

Checks

- At $x = 0$ (peg at M) we get $\alpha = 0$: the triangle hangs with BC horizontal and needs no friction at all, as symmetry says it should.
- With $\theta = 30^\circ$ the triangle is equilateral, $d = \frac{1}{3}b\cos 30^\circ$, and G should be the centroid, a third of the way up the height $b\cos 30^\circ$ from BC. It is. The condition becomes $\mu \geq 2 \cdot \frac{1}{\sqrt{3}} \cdot \frac{3}{2} = \sqrt{3} \approx 1.732$: at a corner, BC would have to sit at 60° to the horizontal.
- As $\theta \to 0$ the triangle becomes a thin spike, BC shrinks to nothing and the bound tends to 0; as $\theta \to \frac{1}{2}\pi$ it becomes a long flat strip and the bound blows up. Both fit intuition.
- A computer check, finding G by splitting the three rods into many small pieces and resolving forces at 201 points along BC for random θ and b, agreed with $2\tan\theta(1 + \sin\theta)$ to within 10^{-11}.

Common slips

- Using the centroid of the triangular *lamina* (a third of the way up AM) instead of the centre of mass of the *wire*. A wire frame has its mass on the sides, so BC counts with weight $2b\sin\theta$.
- Assuming BC stays horizontal and trying to take moments. The triangle tilts until G is below the peg; the tilt angle is what friction has to cope with.
- Checking only one point of BC. The worst case is the end of BC, and that is what fixes the bound.

Beyond STEP

Writing $\tan\alpha \leq \mu$ says that the total contact force must lie within the *cone of friction*, a cone of half-angle $\arctan\mu$ about the normal. This picture makes many friction problems (ladders, wedges, rough pegs) almost geometric, and it generalises to rigid bodies with several contacts, where the question becomes whether some choice of forces inside each cone can balance the load. Kleppner and Kolenkow, *An Introduction to Mechanics*, treat friction and rigid-body equilibrium with this kind of care.

WORKED SOLUTION

STEP II 2016, Question 11

Mechanics • Projectile Collisions

STEP II 2016, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q11

The key idea

Two particles collide when both coordinates agree at the *same* time, so we get two equations sharing one unknown t; dividing one by the other removes t. In part (ii) the trick is to notice that gravity pulls the bullet and the target down by exactly the same amount, $\frac{1}{2}gt^2$, at every instant. So whether they meet is decided by the gravity-free motion of part (i), and gravity only matters for asking where the target is when they meet.

Solution

Part (i)

Call the first particle P (starting at $(a,0)$) and the second Q (starting at $(0,b)$). Suppose they collide at time t. Equating the x- and y-coordinates,

$$\begin{aligned} a + ut\cos\alpha &= vt\cos\beta, & \text{so} \quad & t\,(v\cos\beta - u\cos\alpha) = a, \\ ut\sin\alpha &= b + vt\sin\beta, & \text{so} \quad & t\,(u\sin\alpha - v\sin\beta) = b. \end{aligned}$$

Since $a > 0$, the first equation shows $t \neq 0$ (the particles start at different points), so $t > 0$ and we may divide the second equation by the first:

$$\frac{u\sin\alpha - v\sin\beta}{v\cos\beta - u\cos\alpha} = \frac{b}{a} = \tan\theta = \frac{\sin\theta}{\cos\theta}.$$

(The denominator is $a/t \neq 0$.) Cross-multiplying,

$$\cos\theta\,(u\sin\alpha - v\sin\beta) = \sin\theta\,(v\cos\beta - u\cos\alpha),$$

and collecting the u terms on the left and the v terms on the right,

$$u(\sin\alpha\cos\theta + \cos\alpha\sin\theta) = v(\sin\beta\cos\theta + \cos\beta\sin\theta).$$

By the compound-angle formula for $\sin(A+B)$ this is

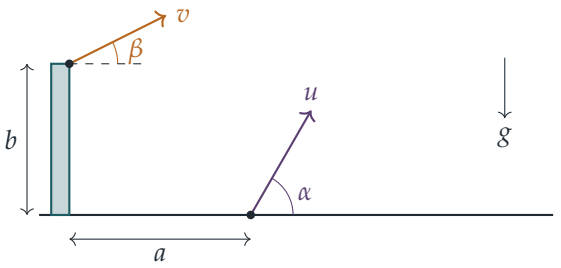
$$u\sin(\theta+\alpha) = v\sin(\theta+\beta),$$

as required. We also record two facts for later: since $a, b, t > 0$,

$$v\cos\beta > u\cos\alpha \qquad \text{and} \qquad u\sin\alpha > v\sin\beta. \tag{$*$}$$

Part (ii)

Model the bullet and the target as particles moving freely under gravity, with no air resistance. Take the foot of the tower as the origin, the x-axis horizontal through the target's launch point (in the direction away from the tower), and the y-axis vertically up. The bullet must be fired towards the target for a hit to be possible, so both move in this vertical plane.



At time t (before either lands), the target and bullet are at

$$T = \left(a + ut\cos\alpha,\ ut\sin\alpha - \tfrac{1}{2}gt^2\right), \qquad B = \left(vt\cos\beta,\ b + vt\sin\beta - \tfrac{1}{2}gt^2\right).$$

These are exactly the positions of part (i), each shifted down by the same amount $\frac{1}{2}gt^2$.

Ask yourself Why can we use part (i) when part (i) had no gravity?

The hit happens when $T = B$. In the y-equation the term $-\frac{1}{2}gt^2$ appears on both sides and cancels, so the collision conditions are word for word those of part (i):

$$t\left(v\cos\beta - u\cos\alpha\right) = a, \qquad t\left(u\sin\alpha - v\sin\beta\right) = b.$$

In particular part (i) and $(*)$ apply, and the time of the hit is

$$t = \frac{b}{u\sin\alpha - v\sin\beta},$$

where the denominator is positive by $(*)$.

The target is still in the air at this moment exactly when its height is positive:

$$ut\sin\alpha - \tfrac{1}{2}gt^2 > 0 \iff t < \frac{2u\sin\alpha}{g},$$

dividing by $\frac{1}{2}t > 0$. Substituting the time of the hit,

$$\frac{b}{u\sin\alpha - v\sin\beta} < \frac{2u\sin\alpha}{g}.$$

Both denominators are positive, so multiplying through by $g(u\sin\alpha - v\sin\beta)$ keeps the direction of the inequality, and we obtain

$$2u\sin\alpha\,(u\sin\alpha - v\sin\beta) > bg,$$

as required. (At the hit the bullet is at the same height as the target, so it too is above the ground.)

Why we need $\alpha > \beta$

We have seen that a hit requires the two particles to collide in the sense of part (i), so

$$u\sin(\theta+\alpha) = v\sin(\theta+\beta),$$

where $\tan\theta = b/a$, and also, from $(*)$, $u\sin\alpha > v\sin\beta$. All four sines here are positive, because θ, α, β are acute, so $\theta+\alpha$ and $\theta+\beta$ lie strictly between 0 and π. Dividing the equation by the inequality reverses it:

$$\frac{\sin(\theta+\alpha)}{\sin\alpha} < \frac{\sin(\theta+\beta)}{\sin\beta}.$$

Expanding, $\dfrac{\sin(\theta+\alpha)}{\sin\alpha} = \sin\theta\cot\alpha + \cos\theta$, and similarly for β, so

$$\sin\theta\cot\alpha < \sin\theta\cot\beta \iff \cot\alpha < \cot\beta,$$

since $\sin\theta > 0$. On $(0, \frac{\pi}{2})$ the cotangent is strictly decreasing, so $\alpha > \beta$.

Physically: relative to the bullet, the target moves in a straight line at constant velocity, because gravity has cancelled. That relative velocity must point from the target's start $(a, 0)$ to the gun at $(0, b)$, so it must be backwards and upwards; by $(*)$ the target must be slower horizontally but faster vertically than the bullet, which forces its path to be steeper, $\tan\alpha > \tan\beta$. (Dividing the two parts of $(*)$ gives the same conclusion directly.)

Checks

Try $u = 20$, $\alpha = 60^\circ$, $v = 15$, $\beta = 30^\circ$ and a hit at $t = 1$, with $g = 9.8$. The collision equations give $a \approx 2.990$ and $b \approx 9.821$, so $\theta \approx 73.06^\circ$. Then $u\sin(\theta+\alpha) \approx 14.612$ and $v\sin(\theta+\beta) \approx 14.612$, agreeing. At the hit the target is at height $17.32 - 4.9 \approx 12.42 > 0$, and indeed $2u\sin\alpha(u\sin\alpha - v\sin\beta) \approx 340.2 > bg \approx 96.2$. A computer check over many thousands of random launches confirmed that the inequality holds exactly when the target is still airborne at the hit, and that every hit had $\alpha > \beta$.

Common slips

- Forgetting that the two particles must be at the same place at the *same* time, and solving for two different times.
- Multiplying through by $u\sin\alpha - v\sin\beta$ without first establishing that it is positive; it is, because $t > 0$ and $b > 0$.

- Arguing $\alpha > \beta$ from $u \sin\alpha > v \sin\beta$ alone, which says nothing about the angles unless the speeds are compared too.

Beyond STEP

The cancellation of gravity is a case of Galilean relativity: in a frame falling freely with the bullet, the target moves in a straight line at constant velocity, so "aim straight at it" works. This is the classic "monkey and hunter" demonstration, and the idea of an inertial (or freely falling) frame is the starting point of first-year mechanics and, later, of the equivalence principle in general relativity. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops relative motion and reference frames carefully.

WORKED SOLUTION

STEP II 2016, Question 12

Probability and Statistics • Inclusion-Exclusion and Derangements

STEP II 2016, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q12

This question has a full guided treatment in *Beyond STEP*, Section 9.8, *Nobody Gets Their Own Shirt*: https://beyondstep.keithmcnulty.org/ch09/derangements.html

The key idea

For three events, treat $A \cup B$ as a single event and apply the two-event rule twice. The same alternating pattern (add singles, subtract pairs, add triples, ...) holds for any number of events. Applied to the events "card i is in place i", every term with r events has the same probability, so each layer of the sum collapses to a single fraction. For "exactly one", fix which card is the match and ask for the rest to have no match at all.

Solution

Three events

Write $A \cup B \cup C = (A \cup B) \cup C$ and use the two-event rule with $A \cup B$ in place of A and C in place of B:

$$\mathrm{P}(A \cup B \cup C) = \mathrm{P}(A \cup B) + \mathrm{P}(C) - \mathrm{P}\big((A \cup B) \cap C\big).$$

The first term is $\mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$. For the last term, the distributive law gives $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$, and applying the two-event rule once more,

$$\begin{aligned}\mathrm{P}\big((A \cap C) \cup (B \cap C)\big) &= \mathrm{P}(A \cap C) + \mathrm{P}(B \cap C) - \mathrm{P}\big((A \cap C) \cap (B \cap C)\big) \\ &= \mathrm{P}(A \cap C) + \mathrm{P}(B \cap C) - \mathrm{P}(A \cap B \cap C).\end{aligned}$$

Substituting both,

$$\mathrm{P}(A \cup B \cup C) = \mathrm{P}(A) + \mathrm{P}(B) + \mathrm{P}(C) - \mathrm{P}(A \cap B) - \mathrm{P}(B \cap C) - \mathrm{P}(C \cap A) + \mathrm{P}(A \cap B \cap C),$$

as required.

Four events

Add the four single probabilities, subtract the six pairwise intersections, add the four triple intersections and subtract the quadruple one:

$$\begin{aligned}\mathrm{P}(A\cup B\cup C\cup D) = {} & \mathrm{P}(A)+\mathrm{P}(B)+\mathrm{P}(C)+\mathrm{P}(D)\\ & -\mathrm{P}(A\cap B)-\mathrm{P}(A\cap C)-\mathrm{P}(A\cap D)\\ & -\mathrm{P}(B\cap C)-\mathrm{P}(B\cap D)-\mathrm{P}(C\cap D)\\ & +\mathrm{P}(A\cap B\cap C)+\mathrm{P}(A\cap B\cap D)+\mathrm{P}(A\cap C\cap D)+\mathrm{P}(B\cap C\cap D)\\ & -\mathrm{P}(A\cap B\cap C\cap D).\end{aligned}$$

Parts (i)–(iii)

All $n!$ orderings are equally likely. If r particular cards are required to sit in their own positions, the remaining $n-r$ cards can be arranged in $(n-r)!$ ways, so the probability is $(n-r)!/n!$. Hence

$$\text{(i) } \mathrm{P}(E_i)=\frac{1}{n}, \qquad \text{(ii) } \mathrm{P}(E_i\cap E_j)=\frac{1}{n(n-1)}, \qquad \text{(iii) } \mathrm{P}(E_i\cap E_j\cap E_k)=\frac{1}{n(n-1)(n-2)}.$$

(Equivalently, for (ii): card i lands in place i with probability $\frac{1}{n}$, and then card j lands in place j among the $n-1$ places left with probability $\frac{1}{n-1}$.)

At least one match

At least one card matches exactly when $E_1\cup E_2\cup\cdots\cup E_n$ occurs. The pattern above extends to n events: the rth layer is $(-1)^{r+1}$ times the sum of the probabilities of all intersections of r distinct events.

> **Ask yourself** How can a sum over all choices of r events be evaluated without knowing which events they are?

By symmetry every intersection of r distinct events has the same probability, $(n-r)!/n!$, and there are $\binom{n}{r}$ ways to choose the r events. So the rth layer contributes

$$(-1)^{r+1}\binom{n}{r}\frac{(n-r)!}{n!}=(-1)^{r+1}\frac{n!}{r!\,(n-r)!}\cdot\frac{(n-r)!}{n!}=\frac{(-1)^{r+1}}{r!}.$$

Summing over $r=1,2,\ldots,n$,

$$\mathrm{P}(\text{at least one match})=1-\frac{1}{2!}+\frac{1}{3!}-\cdots+(-1)^{n+1}\frac{1}{n!},$$

as required. (For $r=1,2,3$ this is just $n\cdot\frac{1}{n}$, $\binom{n}{2}\cdot\frac{1}{n(n-1)}$ and $\binom{n}{3}\cdot\frac{1}{n(n-1)(n-2)}$ from parts (i)–(iii).)

Exactly one match

Write p_m for the probability that none of m shuffled cards is in its own place. By the previous result,

$$p_m = 1 - \left(1 - \frac{1}{2!} + \cdots + (-1)^{m+1}\frac{1}{m!}\right) = \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^m \frac{1}{m!}.$$

The event "exactly one match" is the disjoint union over i of the events "card i matches and no other card does". Given E_i, the other $n-1$ cards are equally likely to be in any order in the other $n-1$ positions, so this is a fresh shuffle of $n-1$ cards, and the probability that none of them matches is p_{n-1}. Therefore

$$\mathrm{P}(\text{exactly one match}) = \sum_{i=1}^{n} \mathrm{P}(E_i)\, p_{n-1} = n \cdot \frac{1}{n} \cdot p_{n-1} = p_{n-1},$$

that is,

$$\mathrm{P}(\text{exactly one match}) = \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^{n-1}\frac{1}{(n-1)!} \qquad (n \geq 2),$$

and 1 when $n = 1$. Equivalently, it is $\sum_{r=0}^{n-1}(-1)^r/r!$, which covers $n = 1$ too.

Checks

For $n = 3$ the orderings are 123, 132, 213, 231, 312, 321. Four of them have a match, giving $\frac{4}{6} = \frac{2}{3} = 1 - \frac{1}{2} + \frac{1}{6}$; and three (the swaps 132, 213, 321) have exactly one match, giving $\frac{1}{2}$, which agrees with the formula. For $n = 2$, exactly one match is impossible (if one card is in place, so is the other), and the formula gives 0. A computer enumeration of all orderings for $n \leq 7$ agrees with both formulas; for example, for $n = 7$ exactly one match has probability $\frac{53}{144} \approx 0.368$. As $n \to \infty$ both answers tend to familiar limits: at least one match tends to $1 - \mathrm{e}^{-1} \approx 0.632$, and exactly one match tends to e^{-1}.

Common slips

- Writing $\mathrm{P}(E_i \cap E_j) = \frac{1}{n^2}$: the events are not independent, because once card i is in place i only $n-1$ places remain for card j.
- Answering "exactly one" with $n \cdot \frac{1}{n} = 1$ or with $\mathrm{P}(E_1)$: we need card i in place *and* no other card in place.
- Forgetting that the term in the sum for "at least one" with $r = 1$ is $n \cdot \frac{1}{n} = 1$, which is why the series starts with 1.

Beyond STEP

This is the classical *derangement* problem (the *problème des rencontres*), and the general inclusion–exclusion formula is proved neatly by induction or with indicator random

variables. The number of matches turns out to be approximately Poisson with mean 1 for large n: the probability of exactly k matches tends to $\mathrm{e}^{-1}/k!$. Grimmett and Welsh, *Probability: An Introduction*, and Blitzstein and Hwang, *Introduction to Probability*, both treat inclusion–exclusion and matching problems.

WORKED SOLUTION

STEP II 2016, Question 13

Probability and Statistics • Normal Approximation

STEP II 2016, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q13

This question has a full guided treatment in *Beyond STEP*, Section 12.23, *Stirling from Probability*: https://beyondstep.keithmcnulty.org/ch12/stirling-probability.html

The key idea

A single probability $\mathrm{P}(X = k)$ of a discrete distribution is the area of a bar of width 1 centred on k. If the distribution is close to a normal distribution, that area is roughly the normal density at k times the width 1. Applying this at the mean, where the standard normal density is $\frac{1}{\sqrt{2\pi}}$, turns a known exact probability into an approximation. Choosing the distribution cleverly makes the exact probability contain the factorials we want.

Solution

The approximation we use throughout

Suppose W is integer-valued with mean μ and variance σ^2, and is well approximated by $V \sim \mathrm{N}(\mu, \sigma^2)$. The density of V is

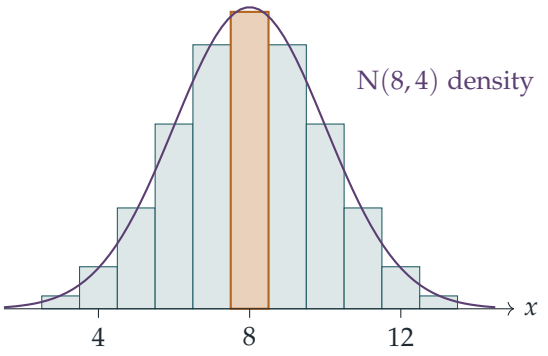
$$f(v) = \frac{1}{\sigma} \cdot \frac{1}{\sqrt{2\pi}} \, \mathrm{e}^{-\frac{1}{2}\left(\frac{v-\mu}{\sigma}\right)^2}.$$

With a continuity correction,

$$\mathrm{P}(W = k) \approx \mathrm{P}\left(k - \tfrac{1}{2} < V < k + \tfrac{1}{2}\right) = \int_{k-1/2}^{k+1/2} f(v)\,\mathrm{d}v \approx 1 \times f(k),$$

the last step because the interval has width 1 and, when σ is not small, f hardly changes across it. At $k = \mu$ the exponential is $\mathrm{e}^0 = 1$, so

$$\mathrm{P}(W = \mu) \approx \frac{1}{\sigma\sqrt{2\pi}}. \qquad (*)$$



Part (i)

For $X \sim \mathrm{B}(16, \frac{1}{2})$,

$$\mathrm{E}(X) = np = 8, \qquad \mathrm{Var}(X) = np(1-p) = 16 \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} = 4,$$

so $\sigma = 2$. Since $p = \frac{1}{2}$ the distribution is symmetric about 8 and n is reasonably large, so $\mathrm{N}(8, 4)$ is a good approximation (the figure shows the bars against the normal curve). By $(*)$, the shaded bar has area

$$\mathrm{P}(X = 8) \approx \frac{1}{2\sqrt{2\pi}},$$

as required.

Part (ii)

Let $Y \sim \mathrm{B}(2n, \frac{1}{2})$. Then $\mathrm{E}(Y) = n$ and $\mathrm{Var}(Y) = 2n \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{n}{2}$, so $\sigma = \sqrt{n/2}$. Exactly,

$$\mathrm{P}(Y = n) = \binom{2n}{n} \left(\tfrac{1}{2}\right)^n \left(\tfrac{1}{2}\right)^n = \frac{(2n)!}{(n!)^2\, 2^{2n}}.$$

For large n the normal approximation applies, and $(*)$ gives

$$\mathrm{P}(Y = n) \approx \frac{1}{\sqrt{n/2}\,\sqrt{2\pi}} = \frac{1}{\sqrt{n\pi}}.$$

Equating the two expressions and multiplying by $2^{2n}(n!)^2$:

$$(2n)! \approx \frac{2^{2n}(n!)^2}{\sqrt{n\pi}},$$

which is the required result. (Part (i) is the case $n = 8$: $\sqrt{8\pi} = 2\sqrt{2\pi}$.)

Part (iii)

Let $Z \sim \mathrm{Po}(n)$, so $\mathrm{E}(Z) = \mathrm{Var}(Z) = n$ and $\sigma = \sqrt{n}$. Exactly,

$$\mathrm{P}(Z = n) = \frac{\mathrm{e}^{-n} n^n}{n!}.$$

Ask yourself Why may we approximate a Poisson distribution by a normal one here?

A $\mathrm{Po}(n)$ variable has the same distribution as the sum of n independent $\mathrm{Po}(1)$ variables, so by the central limit theorem it is approximately $\mathrm{N}(n, n)$ when n is large. This is the standard rule that $\mathrm{Po}(\lambda) \approx \mathrm{N}(\lambda, \lambda)$ for large λ.

So by $(*)$,

$$\frac{\mathrm{e}^{-n} n^n}{n!} \approx \frac{1}{\sqrt{n}\sqrt{2\pi}} = \frac{1}{\sqrt{2\pi n}},$$

and rearranging,

$$n! \approx \sqrt{2\pi n}\,\mathrm{e}^{-n}\, n^n,$$

as required.

Checks

- Part (i): exactly, $\mathrm{P}(X = 8) = \binom{16}{8}/2^{16} = 12870/65536 \approx 0.1964$, while $\frac{1}{2\sqrt{2\pi}} \approx 0.1995$. That is within about 1.6%.
- Part (ii) with $n = 10$: $\binom{20}{10}/2^{20} \approx 0.1762$ against $1/\sqrt{10\pi} \approx 0.1784$. The ratio of the two sides of the factorial approximation tends to 1 as n grows (about 0.9988 at $n = 100$).
- Part (iii) with $n = 10$: $10! = 3\,628\,800$, and $\sqrt{20\pi}\,\mathrm{e}^{-10}\,10^{10} \approx 3\,598\,696$, under 1% too small.
- Consistency: substituting the result of (iii) for both $(2n)!$ and $n!$ into (ii) gives $\sqrt{4\pi n}\,\mathrm{e}^{-2n}(2n)^{2n}$ on both sides, so the two approximations agree.

Common slips

- Using the standard normal density $\frac{1}{\sqrt{2\pi}}$ without dividing by σ. The density of $\mathrm{N}(\mu, \sigma^2)$ at its mean is $\frac{1}{\sigma\sqrt{2\pi}}$.
- Using the standard deviation $\frac{n}{2}$ instead of $\sqrt{n/2}$ in part (ii), or getting the variance of $\mathrm{B}(2n, \frac{1}{2})$ wrong.
- Forgetting to say why the Poisson distribution may be approximated by a normal one in part (iii).

Beyond STEP

The result of part (iii) is Stirling's formula, and the argument is a heuristic form of the central limit theorem applied at a single point, a "local limit theorem". Making it rigorous needs care: knowing that a distribution function is close to the normal one

does not by itself say that individual probabilities are close to the density. Stirling's formula can also be proved directly, for example by comparing $\ln n! = \sum \ln k$ with $\int \ln x \, \mathrm{d}x$, or from the gamma function by Laplace's method. Blitzstein and Hwang, *Introduction to Probability*, covers the central limit theorem and the Poisson–normal link; Grimmett and Welsh, *Probability: An Introduction*, is a good companion.

WORKED SOLUTION

STEP III 2016, Question 1

Pure • Reduction Formulae

STEP III 2016, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q1

The key idea

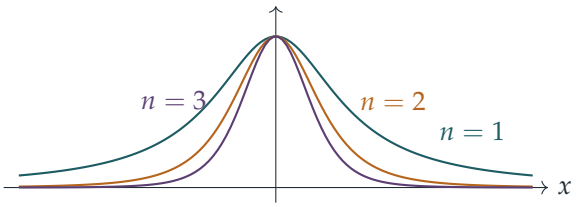
Completing the square turns the quadratic into $(x+a)^2+c^2$, with $c^2 = b-a^2 > 0$, and then the tangent substitution makes I_1 immediate. For the reduction formula, the trick is to integrate by parts *against* 1: differentiating $(x^2+2ax+b)^{-n}$ raises the power to $n+1$, which is exactly what we want to link I_n to I_{n+1}. The induction is then just tidying binomial coefficients.

Solution

Throughout, write

$$Q(x) = x^2+2ax+b = (x+a)^2+c^2, \qquad c = \sqrt{b-a^2} > 0.$$

Since $Q(x) \geq c^2 > 0$ for all x, the integrand is continuous, and since $Q(x)^n$ grows like x^{2n} with $2n \geq 2$, every I_n converges.



The integrands $1/Q^n$ for $a=0$, $b=1$: as n grows they become narrower, so we expect I_n to shrink when $c=1$.

Part (i)

Put $x+a = c\tan u$, so $\mathrm{d}x = c\sec^2 u\,\mathrm{d}u$. As x runs from $-\infty$ to ∞, u runs from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$, and

$$Q(x) = c^2\tan^2 u + c^2 = c^2\sec^2 u.$$

Hence
$$I_1 = \int_{-\pi/2}^{\pi/2} \frac{c\sec^2 u}{c^2\sec^2 u}\,\mathrm{d}u = \frac{1}{c}\int_{-\pi/2}^{\pi/2}\mathrm{d}u = \frac{\pi}{c} = \frac{\pi}{\sqrt{b-a^2}},$$
as required.

Part (ii)

Integrate I_n by parts, taking $u = Q^{-n}$ and $\mathrm{d}v = \mathrm{d}x$. It is cleverer to choose $v = x + a$ rather than $v = x$ (any antiderivative will do), because $Q'(x) = 2(x+a)$. Then
$$\frac{\mathrm{d}}{\mathrm{d}x}\,Q^{-n} = -2n(x+a)\,Q^{-n-1},$$
so
$$I_n = \Big[\,\frac{x+a}{Q^n}\,\Big]_{-\infty}^{\infty} + 2n\int_{-\infty}^{\infty}\frac{(x+a)^2}{Q^{n+1}}\,\mathrm{d}x.$$

? **Ask yourself** Why does the boundary term vanish?

The numerator grows like $|x|$ while the denominator grows like x^{2n}, and $2n \geq 2$ because $n \geq 1$. So $(x+a)/Q^n \to 0$ as $x \to \pm\infty$, and the bracket is zero.

Now use $(x+a)^2 = Q - c^2$ to split the remaining integral:
$$\begin{aligned} I_n &= 2n\int_{-\infty}^{\infty}\frac{Q-c^2}{Q^{n+1}}\,\mathrm{d}x \\ &= 2n\,I_n - 2nc^2\,I_{n+1}. \end{aligned}$$
Rearranging, $2nc^2 I_{n+1} = (2n-1)I_n$, that is,
$$2n(b-a^2)\,I_{n+1} = (2n-1)\,I_n,$$
as required. (Splitting is legitimate because I_n and I_{n+1} both converge.)

Part (iii)

Let $P(n)$ be the statement
$$I_n = \frac{\pi}{2^{2n-2}\,c^{2n-1}}\binom{2n-2}{n-1},$$
where $c^{2n-1} = (b-a^2)^{n-\frac{1}{2}}$.

Base case. For $n = 1$ the right-hand side is $\dfrac{\pi}{2^0 c^1}\binom{0}{0} = \dfrac{\pi}{c}$, which equals I_1 by part (i). So $P(1)$ holds.

Inductive step. Suppose $P(n)$ holds for some $n \geq 1$. By part (ii),
$$I_{n+1} = \frac{2n-1}{2nc^2}\,I_n = \frac{2n-1}{2n}\cdot\frac{\pi}{2^{2n-2}\,c^{2n+1}}\binom{2n-2}{n-1}.$$

We want to show that this equals $\dfrac{\pi}{2^{2n}c^{2n+1}}\binom{2n}{n}$, which is $P(n+1)$. The link between the two binomial coefficients is

$$\binom{2n}{n} = \frac{(2n)!}{n!\,n!} = \frac{(2n)(2n-1)}{n\cdot n}\cdot\frac{(2n-2)!}{(n-1)!\,(n-1)!} = \frac{2(2n-1)}{n}\binom{2n-2}{n-1}.$$

Ask yourself How do we make the powers of 2 match?

Write $\dfrac{2n-1}{2n} = \dfrac{4}{2^2}\cdot\dfrac{2n-1}{2n} = \dfrac{1}{4}\cdot\dfrac{2(2n-1)}{n}$. Then

$$\begin{aligned} I_{n+1} &= \frac{1}{4}\cdot\frac{2(2n-1)}{n}\cdot\frac{\pi}{2^{2n-2}\,c^{2n+1}}\binom{2n-2}{n-1} \\ &= \frac{\pi}{2^{2n}\,c^{2n+1}}\cdot\frac{2(2n-1)}{n}\binom{2n-2}{n-1} = \frac{\pi}{2^{2n}\,c^{2n+1}}\binom{2n}{n}. \end{aligned}$$

Since $2n = 2(n+1)-2$, $n = (n+1)-1$ and $c^{2n+1} = (b-a^2)^{(n+1)-\frac{1}{2}}$, this is exactly $P(n+1)$.

By induction, $P(n)$ holds for every positive integer n:

$$I_n = \frac{\pi}{2^{2n-2}(b-a^2)^{n-\frac{1}{2}}}\binom{2n-2}{n-1}.$$

Checks

- With $a=0$, $b=1$ and $n=2$ the formula gives $I_2 = \frac{\pi}{4}\binom{2}{1} = \frac{\pi}{2} \approx 1.5708$. A numerical evaluation of $\int_{-\infty}^{\infty}(1+x^2)^{-2}\,\mathrm{d}x$ agrees to ten decimal places.
- The tangent substitution also gives $I_n = c^{1-2n}\int_{-\pi/2}^{\pi/2}\cos^{2n-2}u\,\mathrm{d}u$, a standard Wallis integral, whose known value matches the formula.
- For several random a and b with $b > a^2$, numerical integration confirmed the formula for $n = 1,\ldots,5$, and hence also the recurrence of part (ii).

Common slips

- Forgetting to justify that the boundary term in part (ii) vanishes; it needs $n \geq 1$.
- Choosing $v = x$ instead of $v = x+a$ in the integration by parts. This still works, but it leaves an extra $a\int(x+a)Q^{-n-1}\,\mathrm{d}x$ term to deal with.
- Losing track of the power of $b-a^2$ in the inductive step: each step multiplies by $1/c^2$, so the exponent of c goes up by exactly 2, from $2n-1$ to $2n+1$.

Beyond STEP

Integrals over the whole real line of rational functions are a showcase for complex analysis: the residue theorem evaluates I_n in a few lines by looking at the single pole of $1/Q^n$ in the upper half-plane, at $z = -a + ic$. The central binomial coefficients that appear here also turn up in the Wallis product for π and in random walks. Needham's *Visual Complex Analysis* gives a beautiful, geometric account of residues.

WORKED SOLUTION

STEP III 2016, Question 2

Pure • Normals to a Parabola

STEP III 2016, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q2

The key idea

Write down the normal at a general point $(at^2, 2at)$ and ask which values of t give a normal through P. That condition is a cubic in t, and one root, $t = p$, is always there (the normal at P passes through P). Removing it leaves a quadratic whose two roots are exactly q and r. From then on everything is driven by the sum and product of the roots, $q + r$ and qr.

Solution

Part (i)

On $y^2 = 4ax$ with $x = at^2$, $y = 2at$, we have $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2a}{2at} = \dfrac{1}{t}$, so the normal at parameter t has gradient $-t$:

$$y - 2at = -t\,(x - at^2), \qquad \text{that is} \qquad y + tx = 2at + at^3.$$

(This form also covers the vertex, $t = 0$, where the normal is the x-axis.)

The normal at t passes through $P(ap^2, 2ap)$ exactly when

$$2ap + tap^2 = 2at + at^3.$$

Dividing by $a > 0$ and rearranging,

$$\begin{aligned} 0 &= 2(p - t) + t(p^2 - t^2) \\ &= (p - t)\big(2 + t(p + t)\big) \\ &= (p - t)\,(t^2 + pt + 2). \end{aligned}$$

? **Ask yourself** Why can we throw away the factor $p - t$?

The root $t = p$ is the trivial one: the normal at P itself goes through P. We are told $Q \neq P$, so $q \neq p$, and the factor $p - q$ is non-zero. Hence

$$q^2 + qp + 2 = 0,$$

as required. By the same argument $r^2 + rp + 2 = 0$.

Part (ii)

So q and r are both roots of $t^2 + pt + 2 = 0$. Since $Q \neq R$ we have $q \neq r$, so they are its *two* roots, and

$$q + r = -p, \qquad qr = 2.$$

(In particular the discriminant is positive: $p^2 > 8$. Note also that $q = p$ would need $2p^2 + 2 = 0$, which is impossible, so Q and R are automatically different from P.)

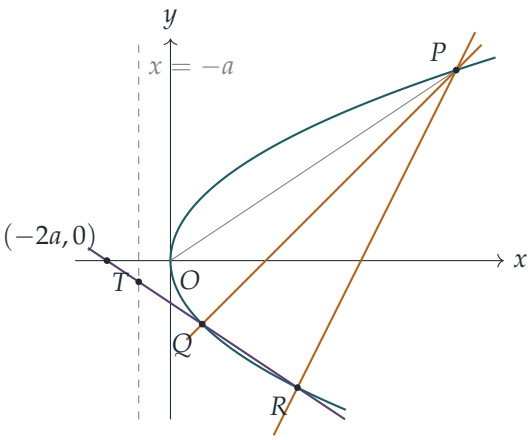
The chord through $(aq^2, 2aq)$ and $(ar^2, 2ar)$ has gradient $\dfrac{2a(q-r)}{a(q^2-r^2)} = \dfrac{2}{q+r}$, and its equation is

$$(q + r)\, y = 2x + 2aqr.$$

(Check: at Q the left side is $2aq^2 + 2aqr$, which equals the right side; similarly at R.) Substituting $q + r = -p$ and $qr = 2$:

$$-p\, y = 2x + 4a.$$

Whatever the value of p, the point $(-2a, 0)$ satisfies this equation. So QR always passes through $(-2a, 0)$, which does not depend on the choice of P.



The figure shows the case $a = 1$, $p = 3$, where $q = -1$ and $r = -2$: the two normals (amber) meet at P, the chord QR (plum) passes through $(-2, 0)$, and T lies on the dashed line $x = -1$.

Part (iii)

Since $p^2 > 8$, $p \neq 0$, so $P \neq O$ and the line OP is $y = \dfrac{2ap}{ap^2}\,x = \dfrac{2}{p}\,x$, or $x = \frac{1}{2}py$. (It is not parallel to QR, whose gradient is $-2/p$.)

Substituting $x = \frac{1}{2}py$ into the equation of QR:

$$-py = py + 4a \quad \Longrightarrow \quad y = -\frac{2a}{p}, \qquad x = \tfrac{1}{2}p\left(-\frac{2a}{p}\right) = -a.$$

So $T = \left(-a, -\frac{2a}{p}\right)$, which always lies on the line $x = -a$ (the directrix of the parabola), independent of the choice of P.

The distance from the x-axis to T is $\dfrac{2a}{|p|}$. From part (ii), the existence of two distinct normals through P forces $p^2 > 8$, i.e. $|p| > 2\sqrt{2}$. Since $a > 0$,

$$\frac{2a}{|p|} < \frac{2a}{2\sqrt{2}} = \frac{a}{\sqrt{2}},$$

which completes the proof. The inequality is strict because $p^2 = 8$ would make $q = r$.

Checks

- In the figure's case $a = 1$, $p = 3$: the roots of $t^2 + 3t + 2$ are -1 and -2; the normals $y = x - 3$ and $y = 2x - 12$ both pass through $P(9, 6)$; the chord through $(1, -2)$ and $(4, -4)$ is $-3y = 2x + 4$, which passes through $(-2, 0)$; and $T = (-1, -\frac{2}{3})$, with $\frac{2}{3} < \frac{1}{\sqrt{2}} \approx 0.707$.
- Letting $|p| \to 2\sqrt{2}$ from above, Q and R merge and the height of T tends to $a/\sqrt{2}$, so the bound is sharp. A computer check over 20,000 random choices of a and p confirmed every result, with the height of T reaching 0.9996 of $a/\sqrt{2}$ but never exceeding it.

Common slips

- Forgetting to justify dividing by $p - q$: it is the condition $Q \neq P$ that allows it.
- Not using $q \neq r$ to say that q and r are *both* roots of the same quadratic, which is what gives $q + r = -p$ and $qr = 2$.
- Proving the final bound without saying where $p^2 > 8$ comes from: it is the requirement that the quadratic has two distinct real roots.

Beyond STEP

The cubic in t behind part (i) is a special case of a classical fact: for a general point (h, k) the condition is $at^3 + (2a - h)\,t - k = 0$, so from a point in the plane there are up

to three normals to a parabola, and the boundary between "one normal" and "three normals" is a curve called the *evolute* (here a semicubical parabola). Evolutes, envelopes and curvature are standard topics in a first course on the differential geometry of curves. Siklos, *Advanced Problems in Mathematics*, has more STEP problems in this coordinate-geometry style.

WORKED SOLUTION

STEP III 2016, Question 3

Pure • Integration and Polynomials

STEP III 2016, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q3

This question has a full guided treatment in *Beyond STEP*, Section 4.6, *Integrals With No Formula*: https://beyondstep.keithmcnulty.org/ch04/no-formula.html

The key idea

An equation of the form $\int g(x)\,\mathrm{e}^x\,\mathrm{d}x = \frac{P}{Q}\,\mathrm{e}^x + \text{constant}$ is just a statement about a derivative, so differentiate the right-hand side and clear denominators. That turns the question into a polynomial identity, and polynomial identities can be attacked in two ways: by substituting a well-chosen value (here $x = -1$) and by comparing degrees.

Solution

Throughout, write P, Q for $P(x)$, $Q(x)$. By the quotient and product rules,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{P}{Q}\,\mathrm{e}^x\right) = \frac{PQ + P'Q - PQ'}{Q^2}\,\mathrm{e}^x.$$

So the integral equation holds exactly when this equals the integrand (with Q not the zero polynomial).

Part (i)

Equating the derivative with $\frac{x^3-2}{(x+1)^2}\,\mathrm{e}^x$, cancelling $\mathrm{e}^x \neq 0$ and multiplying up gives the polynomial identity

$$(x+1)^2\left(PQ + P'Q - PQ'\right) = (x^3 - 2)\,Q^2. \qquad (*)$$

Q has a factor $x+1$. Put $x = -1$ in $(*)$. The left-hand side is 0, and the right-hand side is $-3\,Q(-1)^2$. Hence $Q(-1) = 0$, and by the factor theorem $x+1$ divides Q.

Degrees. P is not the zero polynomial, since otherwise the derivative would be 0, not the integrand. Let $\deg P = p$ and $\deg Q = q$, with leading coefficients $a \neq 0$ and

$b \neq 0$. In $PQ + P'Q - PQ'$, the term PQ has degree $p+q$ with leading coefficient ab, while $P'Q$ and PQ' each have degree at most $p+q-1$. So $PQ + P'Q - PQ'$ has degree exactly $p+q$, and the left-hand side of $(*)$ has degree $p+q+2$. The right-hand side has degree $3+2q$. Therefore

$$p+q+2 = 2q+3 \quad \Longrightarrow \quad p = q+1,$$

so the degree of P is exactly one more than that of Q.

The case $Q = x+1$. Now P has degree 2, say $P = ax^2 + bx + c$. With $Q = x+1$ and $Q' = 1$, $(*)$ becomes

$$(x+1)^2\big((x+1)P + (x+1)P' - P\big) = (x^3-2)(x+1)^2,$$

and cancelling $(x+1)^2$ gives

$$xP + (x+1)P' = x^3 - 2.$$

Since $xP = ax^3 + bx^2 + cx$ and $(x+1)P' = (x+1)(2ax+b) = 2ax^2 + (2a+b)x + b$,

$$ax^3 + (2a+b)x^2 + (2a+b+c)x + b = x^3 - 2.$$

Comparing coefficients: $a = 1$; the constant term gives $b = -2$; then the x^2 term, $2a+b = 0$, is consistent; and the x term gives $c = -(2a+b) = 0$. Hence

$$P(x) = x^2 - 2x, \qquad \int \frac{x^3-2}{(x+1)^2}\,\mathrm{e}^x\,\mathrm{d}x = \frac{x^2-2x}{x+1}\,\mathrm{e}^x + \text{constant}.$$

Part (ii)

Suppose, for a contradiction, that such P and Q exist with no common factor. As before, differentiating and clearing denominators gives

$$(x+1)\big(PQ + P'Q - PQ'\big) = Q^2. \qquad (**)$$

Putting $x = -1$ gives $Q(-1)^2 = 0$, so $x+1$ divides Q. Write

$$Q = (x+1)^n R, \qquad n \geq 1, \quad R(-1) \neq 0,$$

so that n is the exact power of $x+1$ in Q. Then $Q' = n(x+1)^{n-1}R + (x+1)^n R'$, and

$$\begin{aligned} PQ + P'Q - PQ' &= (x+1)^n\big(PR + P'R - PR'\big) - n(x+1)^{n-1}PR \\ &= (x+1)^{n-1}\Big[(x+1)\big(PR + P'R - PR'\big) - nPR\Big]. \end{aligned}$$

Substituting into $(**)$ and cancelling the common factor $(x+1)^n$ from both sides (the right-hand side is $(x+1)^{2n}R^2$):

$$(x+1)\big(PR + P'R - PR'\big) - nPR = (x+1)^n R^2.$$

Ask yourself Why go to the trouble of pulling out the exact power of $x+1$, rather than just substituting $x=-1$ into $(**)$ again?

Substituting $x=-1$ straight into $(**)$ only tells us $Q(-1)=0$; every term then vanishes and we learn nothing about P. Removing all the factors of $x+1$ from Q leaves a term, $-nPR$, that does *not* automatically vanish at $x=-1$.

Now put $x=-1$. Since $n\geq 1$, the right-hand side is 0, and so is the first term on the left. Hence

$$-n\,P(-1)\,R(-1)=0.$$

But $n\neq 0$ and $R(-1)\neq 0$, so $P(-1)=0$. Then $x+1$ is a factor of P as well as of Q, contradicting the assumption that they have no common factor. So no such polynomials exist. ■

(The restriction to coprime P, Q costs nothing: any common factor could be cancelled from P/Q without changing the function.)

Checks

- Differentiate the answer to (i) directly: $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{x^2-2x}{x+1}\right)=\frac{x^2+2x-2}{(x+1)^2}$, and adding $\frac{x^2-2x}{x+1}=\frac{x^3-x^2-2x}{(x+1)^2}$ gives $\frac{x^3-2}{(x+1)^2}$, as required.
- Numerically, $\int_0^1 \frac{x^3-2}{(x+1)^2}\,\mathrm{e}^x\,\mathrm{d}x\approx -1.359141$, which matches $\left[\frac{x^2-2x}{x+1}\mathrm{e}^x\right]_0^1=-\frac{\mathrm{e}}{2}$.
- In (i) the degrees agree: P has degree 2, one more than $Q=x+1$.

Common slips

- Forgetting that $P'Q-PQ'$ has degree *less* than $p+q$, because the leading terms cancel; it is PQ alone that fixes the degree.
- In (ii), substituting $x=-1$ only into $(**)$ and stopping once $Q(-1)=0$ is found. The contradiction needs the exact power of $x+1$ in Q.

Beyond STEP

Part (ii) says that $\int \frac{\mathrm{e}^x}{x+1}\,\mathrm{d}x$, essentially the exponential integral Ei, is not "elementary" in this restricted sense. The full theory, which decides when an integral can be written using elementary functions at all, is Liouville's theorem and the Risch algorithm of differential algebra; the argument here is a small instance of it. For a careful treatment of the polynomial algebra (factor theorem, divisibility, coprimality) used above, see Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP III 2016, Question 4

Pure • Telescoping Series

STEP III 2016, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q4

The key idea

The suggested difference of two fractions turns out to be a constant multiple of the general term, so the sum telescopes: almost everything cancels and only the first and last pieces survive. For part (ii), writing sech in terms of exponentials and pulling out a factor makes each term look exactly like the terms of part (i) with $x = \mathrm{e}^{-2y}$. The doubly infinite sum then folds back onto the one-sided sum, because sech is an even function.

Solution

Part (i)

Putting the suggested difference over a common denominator,

$$\frac{1}{1+x^r} - \frac{1}{1+x^{r+1}} = \frac{x^{r+1}-x^r}{(1+x^r)(1+x^{r+1})} = (x-1)\,\frac{x^r}{(1+x^r)(1+x^{r+1})}.$$

Since $|x| \neq 1$ we have $x \neq 1$, so we may divide by $x-1$. (The condition also guarantees $x \neq -1$, so no denominator $1+x^r$ vanishes: $x^r = -1$ would force $|x| = 1$.) Hence

$$\frac{x^r}{(1+x^r)(1+x^{r+1})} = \frac{1}{x-1}\left(\frac{1}{1+x^r} - \frac{1}{1+x^{r+1}}\right).$$

Summing from $r = 1$ to N, the second fraction of each bracket cancels the first fraction of the next, leaving

$$\begin{aligned}\sum_{r=1}^{N} \frac{x^r}{(1+x^r)(1+x^{r+1})} &= \frac{1}{x-1}\left(\frac{1}{1+x} - \frac{1}{1+x^{N+1}}\right)\\ &= \frac{x\,(x^N-1)}{(x-1)(1+x)(1+x^{N+1})},\end{aligned}$$

where the second form comes from combining the two fractions: the numerator is $x^{N+1} - x = x(x^N - 1)$.

Now suppose $|x| < 1$. Then $x^{N+1} \to 0$ as $N \to \infty$, so the partial sums tend to

$$\frac{1}{x-1}\left(\frac{1}{1+x} - 1\right) = \frac{1}{x-1} \cdot \frac{-x}{1+x} = \frac{x}{1-x^2},$$

which is the required value of the infinite sum.

Part (ii)

For any real t,

$$\operatorname{sech} t = \frac{2}{e^t + e^{-t}} = \frac{2e^{-t}}{1+e^{-2t}}.$$

Let $x = e^{-2y}$; since $y > 0$ we have $0 < x < 1$. Then $\operatorname{sech}(ry) = 2e^{-ry}/(1+x^r)$, and

$$\operatorname{sech}(ry)\operatorname{sech}\big((r+1)y\big) = \frac{4\,e^{-(2r+1)y}}{(1+x^r)(1+x^{r+1})} = 4e^{-y}\,\frac{x^r}{(1+x^r)(1+x^{r+1})},$$

because $e^{-2ry} = x^r$.

> **Ask yourself** Why split $e^{-(2r+1)y}$ as $e^{-y} \cdot e^{-2ry}$?

We want the numerator to be x^r, a power of the *same* x that appears in the denominator, so that part (i) applies term by term; the leftover factor e^{-y} does not depend on r and comes outside the sum.

Since $|x| < 1$, part (i) gives

$$\sum_{r=1}^{\infty} \operatorname{sech}(ry)\operatorname{sech}\big((r+1)y\big) = 4e^{-y}\,\frac{x}{1-x^2} = \frac{4e^{-y}}{x^{-1}-x} = \frac{4e^{-y}}{e^{2y}-e^{-2y}} = \frac{4e^{-y}}{2\sinh 2y},$$

which is $2e^{-y}\operatorname{cosech}(2y)$, as required. Call this sum S.

The doubly infinite sum

Split the sum over all integers r into three pieces: $r \geq 1$, $r \in \{-1, 0\}$, and $r \leq -2$.

> **Ask yourself** What do the terms with negative r look like?

For $r \leq -2$ put $s = -r - 1$, so $s \geq 1$ and $r + 1 = -s$, $r = -(s+1)$. As sech is even,

$$\operatorname{sech}(ry)\operatorname{sech}\big((r+1)y\big) = \operatorname{sech}\big((s+1)y\big)\operatorname{sech}(sy),$$

and as r runs over $-2, -3, \ldots$, s runs over $1, 2, \ldots$. So the terms with $r \leq -2$ also sum to S. (Both one-sided sums converge, so splitting the doubly infinite sum like this is legitimate.)

The two middle terms are

$$r = -1:\ \operatorname{sech}(-y)\operatorname{sech} 0 = \operatorname{sech} y, \qquad r = 0:\ \operatorname{sech} 0 \operatorname{sech} y = \operatorname{sech} y.$$

Therefore, using $\sinh 2y = 2\sinh y \cosh y$,

$$\sum_{r=-\infty}^{\infty} \operatorname{sech}(ry)\operatorname{sech}\big((r+1)y\big) = 2S + 2\operatorname{sech} y = \frac{2\mathrm{e}^{-y}}{\sinh y\cosh y} + \frac{2\sinh y}{\sinh y\cosh y}$$
$$= \frac{2(\mathrm{e}^{-y}+\sinh y)}{\sinh y\cosh y}.$$

But $\mathrm{e}^{-y} + \sinh y = \mathrm{e}^{-y} + \frac{1}{2}(\mathrm{e}^{y} - \mathrm{e}^{-y}) = \frac{1}{2}(\mathrm{e}^{y} + \mathrm{e}^{-y}) = \cosh y$, so the $\cosh y$ cancels and

$$\sum_{r=-\infty}^{\infty} \operatorname{sech}(ry)\operatorname{sech}\big((r+1)y\big) = \frac{2}{\sinh y} = 2\operatorname{cosech} y.$$

Checks

- With $N = 1$ the finite formula gives $\dfrac{x(x-1)}{(x-1)(1+x)(1+x^2)} = \dfrac{x}{(1+x)(1+x^2)}$, which is the single term $r = 1$.
- With $x = \frac{1}{2}$: the infinite sum should be $\frac{1/2}{3/4} = \frac{2}{3}$, and the partial sums do approach 0.6667.
- With $y = 1$: $2\mathrm{e}^{-1}\operatorname{cosech} 2 \approx 0.202864$ and $2\operatorname{cosech} 1 \approx 1.701836$, and both agree with direct summation of the series to many decimal places (confirmed by computer for many values of y).
- As $y \to \infty$ only the terms $r = -1, 0$ survive, each about $2\mathrm{e}^{-y}$, giving about $4\mathrm{e}^{-y}$; and $2\operatorname{cosech} y \approx 4\mathrm{e}^{-y}$ too.

Common slips

- Forgetting the factor $\frac{1}{x-1}$, or getting its sign wrong: the difference equals $(x-1)$ times the term, not $(1-x)$ times it.
- In the doubly infinite sum, doubling S and forgetting the two terms $r = -1$ and $r = 0$, which involve $\operatorname{sech} 0 = 1$ and are not part of either one-sided sum.

Beyond STEP

Rearranging and splitting an infinite sum is safe here because every term is positive (the series converges absolutely); for conditionally convergent series, Riemann's rearrangement theorem says reordering can change the sum to anything you like, a striking result met in a first analysis course. Sums of products like $\operatorname{sech}(ry)\operatorname{sech}((r+1)y)$ over all integers also appear in the theory of elliptic functions, where such bilateral series have beautiful closed forms. For the analysis of series, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2016, Question 5

Pure • Products of Primes

STEP III 2016, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q5

This question has a full guided treatment in *Beyond STEP*, Section 1.2, *How Thinly Spread Are the Primes?*: https://beyondstep.keithmcnulty.org/ch01/prime-distribution.html

The key idea

The middle binomial coefficient $\binom{2m+1}{m}$ is small (it and its twin $\binom{2m+1}{m+1}$ share the total 2^{2m+1} with other terms) but it is divisible by every prime between $m+1$ and $2m+1$. So the product of those primes is small too. Splitting the primes up to $2m+1$ into "up to $m+1$" and "above $m+1$" then sets up a strong induction, and even numbers cost nothing because they are never prime (apart from 2).

Solution

Part (i)

Put $x = 1$ in the binomial expansion of $(1+x)^{2m+1}$:

$$\sum_{k=0}^{2m+1} \binom{2m+1}{k} = 2^{2m+1}.$$

Every term on the left is positive. By symmetry, $\binom{2m+1}{m} = \binom{2m+1}{m+1}$, and these are two different terms of the sum. Since $m \geq 1$, the term $\binom{2m+1}{0} = 1$ is a third, different term. Hence

$$2\binom{2m+1}{m} = \binom{2m+1}{m} + \binom{2m+1}{m+1} < 2^{2m+1},$$

so $\binom{2m+1}{m} < 2^{2m}$, as required.

Part (ii)

Write

$$\binom{2m+1}{m} = \frac{(2m+1)!}{m!\,(m+1)!}, \qquad \text{so} \qquad m!\,(m+1)!\binom{2m+1}{m} = (2m+1)!.$$

Let p be a prime with $m+1 < p \leq 2m+1$. Then p is one of the factors of $(2m+1)!$, so p divides the right-hand side. But every factor of $m!\,(m+1)!$ is at most $m+1 < p$, so the prime p divides none of them, and therefore $p \nmid m!\,(m+1)!$. A prime that divides a product ab must divide a or b, so $p \mid \binom{2m+1}{m}$.

> **Ask yourself** Each such prime divides the coefficient. Why does their *product* divide it?

The primes are distinct, so they are pairwise coprime, and a number divisible by pairwise coprime numbers is divisible by their product. (Equivalently, by unique factorisation each of these primes appears in the factorisation of $\binom{2m+1}{m}$.) So $P_{m+1,2m+1}$ divides $\binom{2m+1}{m}$. If there are no such primes, $P_{m+1,2m+1} = 1$, which divides everything.

A positive divisor of a positive integer is no larger than it, so by part (i)

$$P_{m+1,2m+1} \leq \binom{2m+1}{m} < 2^{2m}.$$

Part (iii)

Since $m \geq 1$, we have $1 < m+1 < 2m+1$. Every prime $p \leq 2m+1$ satisfies either $p \leq m+1$ or $m+1 < p \leq 2m+1$, but not both, so

$$P_{1,2m+1} = P_{1,m+1}\,P_{m+1,2m+1}.$$

Also $2 \leq m+1 \leq 2m$ (because $m \geq 1$), so $k = m+1$ is one of the values covered by the hypothesis, giving $P_{1,m+1} < 4^{m+1}$. Together with part (ii),

$$P_{1,2m+1} < 4^{m+1} \cdot 2^{2m} = 4^{m+1} \cdot 4^m = 4^{2m+1}.$$

Part (iv)

We use strong induction on n. Let $S(n)$ be the statement $P_{1,n} < 4^n$.

Base case. $P_{1,2} = 2 < 16 = 4^2$.

Inductive step. Let $n \geq 3$ and suppose $S(k)$ holds for $k = 2, 3, \ldots, n-1$.

- If n is odd, write $n = 2m+1$ with $m \geq 1$. Then $n-1 = 2m$, so $S(k)$ holds for $k = 2, \ldots, 2m$. Part (iii) gives $P_{1,2m+1} < 4^{2m+1}$, which is $S(n)$.
- If n is even, then $n \geq 4$ is even and greater than 2, so it is not prime. Hence the primes up to n are exactly the primes up to $n-1$, and

$$P_{1,n} = P_{1,n-1} < 4^{n-1} < 4^n.$$

Ask yourself Why can't we just use part (iii) for every n?

Part (iii) only reaches odd numbers $2m+1$. For even n we need a separate argument, and the one above is easy because an even number above 2 can't add a new prime to the product.

So $S(n)$ holds in both cases, and by strong induction $P_{1,n} < 4^n$ for all $n \geq 2$.

Checks

- $m = 3$: $\binom{7}{3} = 35 < 64 = 2^6$. The primes in $(4, 7]$ are 5 and 7, and $P_{4,7} = 35$ divides 35 exactly.
- $m = 5$: $\binom{11}{5} = 462 = 2 \cdot 3 \cdot 7 \cdot 11$, which is divisible by $P_{6,11} = 77$. Also $462 < 1024$.
- $P_{1,10} = 2 \cdot 3 \cdot 5 \cdot 7 = 210$, far below $4^{10} = 1\,048\,576$. A computer check confirms $P_{1,n} < 4^n$ for every n from 2 to 1500.

Common slips

- Writing $\leq$ instead of $<$ in part (i): strictness needs a third positive term, such as $\binom{2m+1}{0}$, besides the two middle ones.
- In part (iii), forgetting to check that $m+1$ lies in the range $2, \ldots, 2m$ where the hypothesis applies.
- In part (iv), only treating odd n, or forgetting the base case $n = 2$.

Beyond STEP

This is Erdős's elementary proof of the bound $\prod_{p \leq n} p < 4^n$. Taking logarithms, it says Chebyshev's function $\theta(n) = \sum_{p \leq n} \ln p$ is less than $n \ln 4$. That is one half of Chebyshev's theorem that the number of primes up to n is of order $n / \ln n$. Erdős used the same kind of binomial-coefficient argument to prove Bertrand's postulate: there is always a prime between n and $2n$. Davenport's *The Higher Arithmetic* is a good next step in elementary number theory.

WORKED SOLUTION

STEP III 2016, Question 6

Pure • Hyperbolic Functions

STEP III 2016, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q6

The key idea

Multiplying $\operatorname{sech} x = a\tanh x + b$ through by $\cosh x$ gives $a\sinh x + b\cosh x = 1$, which is exactly the combination in the opening part. So the first job is the hyperbolic version of the familiar "$R\cos(x+\alpha)$" trick. Unlike the trigonometric case, the form you get depends on how B compares with $\pm A$, and each form tells you at once how many solutions there are.

Solution

The forms of $A\sinh x + B\cosh x$

Expanding,

$$R\cosh(x+\gamma) = R\sinh\gamma\,\sinh x + R\cosh\gamma\,\cosh x,$$

so we need $R\sinh\gamma = A$ and $R\cosh\gamma = B$. Since $\cosh^2\gamma - \sinh^2\gamma = 1$, this forces $R^2 = B^2 - A^2$ and $\tanh\gamma = A/B$. When $B > A > 0$ both of these can be met: take

$$R = \sqrt{B^2 - A^2} > 0, \qquad \gamma = \operatorname{artanh}\frac{A}{B},$$

which exists because $0 < A/B < 1$. Then $\cosh\gamma = 1/\sqrt{1 - A^2/B^2} = B/R$ and $\sinh\gamma = \tanh\gamma\cosh\gamma = A/R$, so $R\cosh\gamma = B$ and $R\sinh\gamma = A$, as required. Hence $A\sinh x + B\cosh x = R\cosh(x+\gamma)$.

The other cases, with $A > 0$:

- $-A < B < A$: matching $R\sinh(x+\gamma) = R\cosh\gamma\sinh x + R\sinh\gamma\cosh x$ instead gives

$$A\sinh x + B\cosh x = \sqrt{A^2 - B^2}\,\sinh(x+\gamma), \qquad \gamma = \operatorname{artanh}\frac{B}{A}.$$

- $B = A$: $A(\sinh x + \cosh x) = Ae^x$.

- $B = -A$: $A(\sinh x - \cosh x) = -Ae^{-x}$.
- $B < -A$: applying the first result to $-A\sinh x - B\cosh x$ (with $-B > A$ and $-A < 0$; the working above only needed $|A| < B$) gives

$$A\sinh x + B\cosh x = -\sqrt{B^2 - A^2}\,\cosh(x + \gamma), \qquad \gamma = \operatorname{artanh}\frac{A}{B}.$$

The intersection equation

Since $\cosh x > 0$, the curves meet where

$$\operatorname{sech} x = a\tanh x + b \iff 1 = a\sinh x + b\cosh x. \qquad (*)$$

Part (i)

With $b > a > 0$, write $R = \sqrt{b^2 - a^2}$; then $(*)$ becomes

$$\cosh\left(x + \operatorname{artanh}\tfrac{a}{b}\right) = \frac{1}{\sqrt{b^2 - a^2}}.$$

If the curves intersect, this has a solution, so the right-hand side is at least 1. The solutions of $\cosh u = c$ with $c \geq 1$ are $u = \pm\operatorname{arcosh} c$, so

$$x = \pm\operatorname{arcosh}\left(\frac{1}{\sqrt{b^2 - a^2}}\right) - \operatorname{artanh}\frac{a}{b},$$

which is the required form.

Part (ii)

With $a > b > 0$ we are in the sinh case: $(*)$ becomes $\sqrt{a^2 - b^2}\,\sinh\left(x + \operatorname{artanh}\frac{b}{a}\right) = 1$. Since sinh is a bijection from $\mathbb{R}$ to $\mathbb{R}$, there is always exactly one intersection, at

$$x = \operatorname{arsinh}\left(\frac{1}{\sqrt{a^2 - b^2}}\right) - \operatorname{artanh}\frac{b}{a}.$$

Part (iii)

Count the solutions of $(*)$ in each case of the opening part (with $A = a > 0$, $B = b$):

- $b < -a$: the left-hand side $-\sqrt{b^2 - a^2}\cosh(x + \gamma)$ is negative, so there is no solution.
- $b = -a$: $-ae^{-x} = 1$ has no solution.
- $-a < b < a$: exactly one solution, as in (ii); the argument there works for any b in this range.
- $b = a$: $ae^{x} = 1$ has the single solution $x = -\ln a$.
- $b > a$: $\cosh(x + \gamma) = 1/\sqrt{b^2 - a^2}$ has two solutions if $1/\sqrt{b^2 - a^2} > 1$, one if it equals 1, and none if it is less than 1.

Ask yourself Why can't the cases with $b \le a$ ever give two points?

In those cases the left-hand side of $(*)$ is a strictly monotonic function of x (a multiple of $\sinh(x+\gamma)$ or of $e^{\pm x}$), or is always negative, so it takes the value 1 at most once. Only the cosh form, which goes down and then up again, can reach 1 twice.

So the curves meet at two distinct points if and only if $b > a$ and $b^2 - a^2 < 1$, that is,

$$a < b < \sqrt{a^2+1}.$$

Part (iv)

Since $\operatorname{sech} x - (a\tanh x + b) = \operatorname{sech} x\,(1 - a\sinh x - b\cosh x)$ and $\operatorname{sech} x \neq 0$, the curves touch exactly where $g(x) = a\sinh x + b\cosh x - 1$ has a repeated zero, i.e. $g(x) = g'(x) = 0$. In the sinh and exponential cases $g'(x)$ is a multiple of $\cosh(x+\gamma)$ or e^x, never zero, so there are no repeated zeros. In the case $b > a$, $g(x) = R\cosh(x+\gamma) - 1$ and $g'(x) = R\sinh(x+\gamma)$, which vanish together only when $x + \gamma = 0$ and $R = 1$. Hence the curves touch if and only if

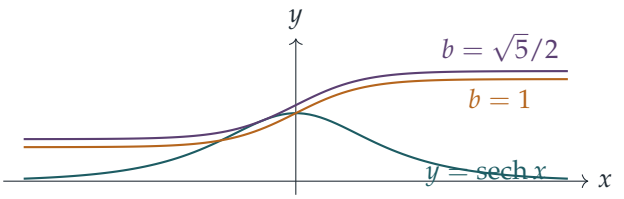
$$b > a \quad \text{and} \quad b^2 - a^2 = 1, \qquad \text{i.e.} \quad b = \sqrt{1+a^2}.$$

The point of contact has $x = -\gamma = -\operatorname{artanh}(a/b)$, so

$$y = \operatorname{sech}\left(\operatorname{artanh}\tfrac{a}{b}\right) = \sqrt{1 - \tfrac{a^2}{b^2}} = \frac{\sqrt{b^2-a^2}}{b} = \frac{1}{b} = \frac{1}{\sqrt{1+a^2}},$$

using $\operatorname{sech}^2 u = 1 - \tanh^2 u$ and $\operatorname{sech} u > 0$. As a check, the other curve gives $y = -a\cdot\frac{a}{b} + b = \frac{b^2-a^2}{b} = \frac{1}{b}$ too.

The picture below shows $a = \frac{1}{2}$: with $b = 1$ (amber) there are two intersections, and with $b = \sqrt{5}/2$ (plum) the curves touch at height $2/\sqrt{5}$.



Checks

Take $a = \frac{3}{4}$, $b = 1$, so that $b^2 - a^2 = \frac{7}{16} < 1$ and (iii) predicts two points. Then $\operatorname{arcosh}(4/\sqrt{7}) = \ln\big((4+3)/\sqrt{7}\big) = \ln\sqrt{7}$ and $\operatorname{artanh}\frac{3}{4} = \frac{1}{2}\ln 7 = \ln\sqrt{7}$, so the formula in (i) gives $x = 0$ and $x = -\ln 7 \approx -1.946$. At $x = 0$ both curves pass through $(0,1)$, as they should. A computer check of $(*)$ over many random a and b also confirmed the root counts in every case, and that the tangency condition gives a double root.

Common slips

- Forgetting the boundary cases $b = \pm a$, where neither cosh nor sinh form exists but the expression is a single exponential.
- In (iii), giving only $b^2 - a^2 < 1$ and leaving out $b > a$: for $b < a$ the curves still meet, but only once.
- Getting the sign of γ wrong: the solutions are $\pm \operatorname{arcosh}(\cdot)$ *minus* $\operatorname{artanh}(a/b)$.

Beyond STEP

The case split is governed by the sign of $B^2 - A^2$: the point (A, B) lies inside one of the regions bounded by the lines $B = \pm A$, or on those lines. This is the geometry of special relativity, where Lorentz boosts are hyperbolic "rotations" preserving $t^2 - x^2$ just as ordinary rotations preserve $x^2 + y^2$, and $B = \pm A$ plays the role of the light cone, the one boundary a boost can never move you across. For more hyperbolic functions and problems of this kind, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 2016, Question 7

Pure • Roots of Unity

STEP III 2016, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q7

This question has a full guided treatment in *Beyond STEP*, Section 5.4, *Distances in a Regular Polygon*: https://beyondstep.keithmcnulty.org/ch05/polygon-distances.html

The key idea

Distances are moduli, and a product of moduli is the modulus of a product. If we put the polygon in the Argand diagram so that its vertices are rotated nth roots of unity, the product of distances becomes $|w^n - 1|$ for a suitable w, by the opening identity. The only real work is choosing the coordinates well: put O at the origin and make the perpendicular bisector of X_0X_1 the real axis, so that P is a real number.

Solution

The factorisation

For $k = 0, 1, \ldots, n-1$ we have $(\omega^k)^n = \mathrm{e}^{2\pi \mathrm{i} k} = 1$, so each ω^k is a root of $z^n - 1$. These n numbers are distinct, because their arguments $2\pi k/n$ are distinct numbers in $[0, 2\pi)$. Both sides of the identity are monic polynomials of degree n, so their difference

$$D(z) = (z-1)(z-\omega)\cdots(z-\omega^{n-1}) - (z^n - 1)$$

has degree at most $n-1$. But D vanishes at the n distinct points $1, \omega, \ldots, \omega^{n-1}$, and a non-zero polynomial of degree at most $n-1$ has at most $n-1$ roots. Hence $D \equiv 0$, and

$$\prod_{k=0}^{n-1}(z - \omega^k) = z^n - 1 \quad \text{for every complex } z.$$

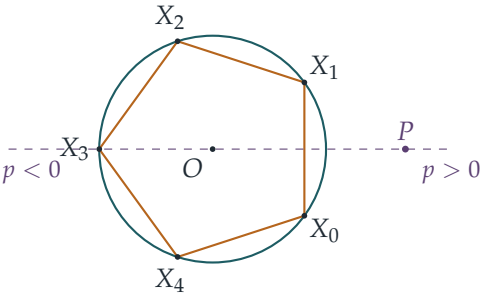
Part (i)

We take the vertices to be labelled in order round the circle, so X_0 and X_1 are adjacent and $\angle X_0 O X_1 = 2\pi/n$. Rotating or reflecting the figure changes no distances, so we

may place O at the origin with

$$X_k = \mathrm{e}^{\mathrm{i}(2k-1)\pi/n} = \mathrm{e}^{-\mathrm{i}\pi/n}\,\omega^k, \qquad k = 0, 1, \ldots, n-1.$$

Then $X_0 = \mathrm{e}^{-\mathrm{i}\pi/n}$ and $X_1 = \mathrm{e}^{\mathrm{i}\pi/n}$ are reflections of each other in the real axis, so the perpendicular bisector of X_0X_1 is the real axis. The points equidistant from X_0 and X_1 are exactly the points of this line, so P is a real number p, and $|OP| = |p|$. The midpoint of X_0X_1 is $\cos(\pi/n) > 0$, so $p > 0$ means P is on the same side of O as the side X_0X_1.



(The figure shows $n = 5$; for odd n the line through O also passes through the opposite vertex.)

Now, using $|ab| = |a||b|$ and $|\mathrm{e}^{-\mathrm{i}\pi/n}| = 1$,

$$\begin{aligned}|PX_0|\,|PX_1|\cdots|PX_{n-1}| &= \prod_{k=0}^{n-1}|p - \mathrm{e}^{-\mathrm{i}\pi/n}\omega^k| = \prod_{k=0}^{n-1}|\mathrm{e}^{-\mathrm{i}\pi/n}|\,|p\,\mathrm{e}^{\mathrm{i}\pi/n} - \omega^k| \\ &= \left|\prod_{k=0}^{n-1}(p\,\mathrm{e}^{\mathrm{i}\pi/n} - \omega^k)\right| = \left|(p\,\mathrm{e}^{\mathrm{i}\pi/n})^n - 1\right|,\end{aligned}$$

by the factorisation with $z = p\,\mathrm{e}^{\mathrm{i}\pi/n}$. Since $\mathrm{e}^{\mathrm{i}\pi} = -1$,

$$|PX_0|\,|PX_1|\cdots|PX_{n-1}| = |-p^n - 1| = |p^n + 1|.$$

Ask yourself Why does the sign of p matter for odd n but not for even n?

If n is even, then $p^n = |p|^n \geq 0$ whatever the sign of p, so $|p^n + 1| = |p|^n + 1$, and

$$|PX_0| \times |PX_1| \times \cdots \times |PX_{n-1}| = |OP|^n + 1,$$

as required. If n is odd, then p^n has the same sign as p, and there are two cases.

- If P lies on the ray from O through the midpoint of X_0X_1 ($p \geq 0$), then $p^n = |OP|^n$ and the product is $|OP|^n + 1$.
- If P lies on the opposite ray ($p < 0$), which for odd n is the ray from O through the vertex $X_{(n+1)/2}$ (at $\mathrm{e}^{\mathrm{i}\pi} = -1$), then $p^n = -|OP|^n$ and the product is

$$\big|\,|OP|^n - 1\big|.$$

The modulus signs matter in the second case: the value is $1 - |OP|^n$ when P is inside the circle, and it is 0 when $P = X_{(n+1)/2}$, as it must be, since one of the distances is then zero.

Part (ii)

This time rotate so that $X_k = \omega^k$, so $X_0 = 1$. For $z \neq 1$ the factorisation gives

$$(z-\omega)(z-\omega^2)\cdots(z-\omega^{n-1}) = \frac{z^n-1}{z-1} = 1 + z + z^2 + \cdots + z^{n-1}.$$

Both ends are polynomials that agree at infinitely many points (every $z \neq 1$), so they are equal for all z, including $z = 1$. Putting $z = 1$:

$$(1-\omega)(1-\omega^2)\cdots(1-\omega^{n-1}) = \underbrace{1+1+\cdots+1}_{n} = n.$$

Taking moduli, and using $|1-\omega^k| = |X_0X_k|$,

$$|X_0X_1| \times |X_0X_2| \times \cdots \times |X_0X_{n-1}| = n,$$

which is the required result.

Checks

- Square ($n = 4$) with $P = O$: every distance is 1, so the product is $1 = 0^4 + 1$. With P far away, the product behaves like $|OP|^n$, as the formula says.
- Equilateral triangle ($n = 3$) with P on the far side at distance 2 from O: the distances are 1 (to the opposite vertex at -1) and $\sqrt{(-2-\frac{1}{2})^2 + \frac{3}{4}} = \sqrt{7}$ twice, giving $7 = 2^3 - 1$.
- Part (ii) for the square: $|X_0X_1|\,|X_0X_2|\,|X_0X_3| = \sqrt{2} \times 2 \times \sqrt{2} = 4$. All three results were also confirmed by computer for random n, rotations and positions of P.

Common slips

- Placing X_0 at 1 in part (i): then the perpendicular bisector of X_0X_1 is the line at angle π/n, not the real axis, and the algebra is messier. Choose coordinates to suit P.
- Forgetting the modulus in the odd case with P on the far side: $|OP|^n - 1$ is negative when P is inside the circle.
- Substituting $z = 1$ straight into $\dfrac{z^n-1}{z-1}$; the polynomial identity is what justifies the value n.

Beyond STEP

Part (ii) is the statement that the derivative of $z^n - 1$ at $z = 1$ is n: in general, if $f(z) = \prod(z - a_k)$ then $f'(a_0) = \prod_{k \neq 0}(a_0 - a_k)$. Products over roots of unity such as these appear in the theory of cyclotomic polynomials and in computing discriminants, and part (i) is essentially a classical result of Roger Cotes on distances from a point on a line of symmetry of a regular polygon. Needham's *Visual Complex Analysis* is a beautiful place to see this geometry of complex multiplication developed.

WORKED SOLUTION

STEP III 2016, Question 8

Pure • Functional Equations

STEP III 2016, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q8

The key idea

Each equation links the unknown function at x to its value at some transformed point. If applying the transformation again (or a third time) brings you back to x, then substituting the transformed point into the equation gives a *second* equation in the same unknowns. You now have simultaneous linear equations, and you can eliminate the values you don't want, just as you would for ordinary unknowns.

Solution

Part (i)

The equation holds for *all* x, so it holds with x replaced by $-x$:

$$\mathrm{f}(-x) + (1+x)\,\mathrm{f}(x) = (-x)^2 = x^2,$$

which is the required result.

Now we have two equations in the two unknowns $\mathrm{f}(x)$ and $\mathrm{f}(-x)$. The second gives $\mathrm{f}(-x) = x^2 - (1+x)\mathrm{f}(x)$, and substituting this into the original equation gives

$$\begin{aligned} \mathrm{f}(x) + (1-x)\big(x^2 - (1+x)\mathrm{f}(x)\big) &= x^2 \\ \mathrm{f}(x)\big(1 - (1-x^2)\big) &= x^2 - (1-x)x^2 \\ x^2\,\mathrm{f}(x) &= x^3. \end{aligned}$$

For $x \neq 0$ we may divide by x^2 to get $\mathrm{f}(x) = x$. At $x = 0$ the original equation reads $\mathrm{f}(0) + \mathrm{f}(0) = 0$, so $\mathrm{f}(0) = 0$ as well. Hence

$$\mathrm{f}(x) = x \quad \text{for all } x.$$

Verification: $x + (1-x)(-x) = x - x + x^2 = x^2$, as required.

Ask yourself Why does the question insist that we verify the answer?

Our argument shows only that *if* a function satisfies the equation, it must be $\mathrm{f}(x) = x$. It doesn't show that such a function exists: the manipulations might have thrown away information, and then no function would work. Substituting back closes the gap.

Part (ii)

First note that $\mathrm{K}(x) \neq 1$ for every $x \neq 1$, since $\frac{x+1}{x-1} = 1$ would need $x+1 = x-1$. So $\mathrm{K}(\mathrm{K}(x))$ is defined, and multiplying top and bottom by $x-1$,

$$\mathrm{K}(\mathrm{K}(x)) = \frac{\dfrac{x+1}{x-1}+1}{\dfrac{x+1}{x-1}-1} = \frac{(x+1)+(x-1)}{(x+1)-(x-1)} = \frac{2x}{2} = x.$$

The equation for g is $\mathrm{g}(x) + x\,\mathrm{g}(\mathrm{K}(x)) = x$. Because $\mathrm{K}(x) \neq 1$, we may replace x by $\mathrm{K}(x)$, and since $\mathrm{K}(\mathrm{K}(x)) = x$ this gives

$$\mathrm{g}(\mathrm{K}(x)) + \mathrm{K}(x)\,\mathrm{g}(x) = \mathrm{K}(x).$$

Write K for $\mathrm{K}(x)$. Then $\mathrm{g}(K) = K - K\,\mathrm{g}(x)$, and substituting into the original equation,

$$\mathrm{g}(x) + x\big(K - K\,\mathrm{g}(x)\big) = x \quad\Longrightarrow\quad (1 - xK)\,\mathrm{g}(x) = x - xK.$$

Now

$$1 - xK = \frac{(x-1) - x(x+1)}{x-1} = -\frac{x^2+1}{x-1},$$
$$x - xK = x \cdot \frac{(x-1)-(x+1)}{x-1} = -\frac{2x}{x-1}.$$

Since $x^2 + 1 > 0$, the factor $1 - xK$ is never zero, so we can divide:

$$\mathrm{g}(x) = \frac{-2x/(x-1)}{-(x^2+1)/(x-1)} = \frac{2x}{x^2+1} \qquad (x \neq 1),$$

as required. (One can check that this function does satisfy the equation; for instance at $x = 2$, $\mathrm{K}(2) = 3$ and $\frac{4}{5} + 2 \cdot \frac{6}{10} = 2$.)

Part (iii)

Let $\mathrm{M}(x) = \dfrac{1}{1-x}$. The equation is

$$\mathrm{h}(x) + \mathrm{h}(\mathrm{M}(x)) = 1 - x - \mathrm{M}(x),$$

and the right-hand side is written in terms of x and $\mathrm{M}(x)$ only, which is encouraging.

This time M is not its own inverse, so look at repeated applications. For $x \neq 0, 1$:

$$\mathrm{M}(\mathrm{M}(x)) = \frac{1}{1 - \frac{1}{1-x}} = \frac{1-x}{-x} = \frac{x-1}{x}, \qquad \mathrm{M}(\mathrm{M}(\mathrm{M}(x))) = \frac{1}{1 - \frac{x-1}{x}} = \frac{x}{1} = x.$$

So M has period three. We also need M to keep us inside the domain: $\mathrm{M}(x)$ is never 0, and $\mathrm{M}(x) = 1$ only when $x = 0$, so if $x \neq 0, 1$ then $\mathrm{M}(x) \neq 0, 1$ too.

Ask yourself Two applications of M don't return to x. How do we get enough equations?

Use all three points of the cycle. Write $a = x$, $b = \mathrm{M}(x)$, $c = \mathrm{M}(b) = \frac{x-1}{x}$, so that $\mathrm{M}(c) = a$. Applying the equation at a, b and c in turn:

$$\begin{aligned} \mathrm{h}(a) + \mathrm{h}(b) &= 1 - a - b, \\ \mathrm{h}(b) + \mathrm{h}(c) &= 1 - b - c, \\ \mathrm{h}(c) + \mathrm{h}(a) &= 1 - c - a. \end{aligned}$$

Adding all three and halving,

$$\mathrm{h}(a) + \mathrm{h}(b) + \mathrm{h}(c) = \tfrac{3}{2} - (a + b + c).$$

Subtracting the second equation leaves

$$\mathrm{h}(a) = \tfrac{3}{2} - a - b - c - (1 - b - c) = \tfrac{1}{2} - a.$$

Hence

$$\mathrm{h}(x) = \tfrac{1}{2} - x \qquad (x \neq 0,\ x \neq 1).$$

Verification: $\left(\frac{1}{2} - x\right) + \left(\frac{1}{2} - \frac{1}{1-x}\right) = 1 - x - \frac{1}{1-x}$, as required.

Checks

- In (i), try $x = 3$: $3 + (-2)(-3) = 9 = 3^2$.
- In (ii), try $x = -1$: $\mathrm{K}(-1) = 0$, and $\mathrm{g}(-1) + (-1)\,\mathrm{g}(0) = -1 - 0 = -1$. Also $\mathrm{K}(0) = -1$, confirming the period two.
- In (iii), try $x = 2$: the cycle is $2 \to -1 \to \frac{1}{2} \to 2$, and $\mathrm{h}(2) + \mathrm{h}(-1) = -\frac{3}{2} + \frac{3}{2} = 0$, while $1 - 2 - (-1) = 0$.

All three functions were also confirmed by a computer check of each equation at two thousand random rational values, using exact arithmetic.

Common slips

- Dividing by x^2 in (i) and forgetting $x = 0$, which needs its own (one-line) argument.

- Substituting $\mathrm{K}(x)$ or $\mathrm{M}(x)$ for x without checking that it stays in the domain where the equation holds.
- In (iii), substituting only once and getting stuck with three unknowns: you need the whole cycle.

Beyond STEP

The maps $x \mapsto -x$, K and M are all Möbius transformations $x \mapsto \frac{ax+b}{cx+d}$, and these form a group under composition. A transformation of finite order generates a finite cyclic group, and the method here is to sum (or eliminate) over that group, an idea that reappears in group representation theory. Möbius maps correspond to 2×2 matrices, and you can find $\mathrm{M}^3 =$ identity by cubing a matrix. Needham's *Visual Complex Analysis* has a beautiful treatment of Möbius transformations.

WORKED SOLUTION

STEP III 2016, Question 9

Mechanics • Elastic Springs and SHM

STEP III 2016, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q9

The key idea

Put the origin at the centre of the triangle and use symmetry. The equilibrium position is the centre, where the three equal tensions cancel. When the particle moves along the line towards P, the springs QX and RX stay mirror images of each other, so their sideways pulls cancel and the motion stays on that line. Finding RX needs the cosine rule. For the period we write down the exact restoring force and then keep only the terms that are linear in the small displacement.

Solution

We model X as a particle on a smooth table and the springs as light springs obeying Hooke's law, so a spring of length r has tension $\frac{\lambda}{l}(r-l)$.

Equilibrium

Let O be the centre of the triangle. Each vertex is at distance

$$c = \frac{2}{3} \times (\text{height}) = \frac{2}{3} \cdot \sqrt{3}\, a = \frac{2a}{\sqrt{3}}$$

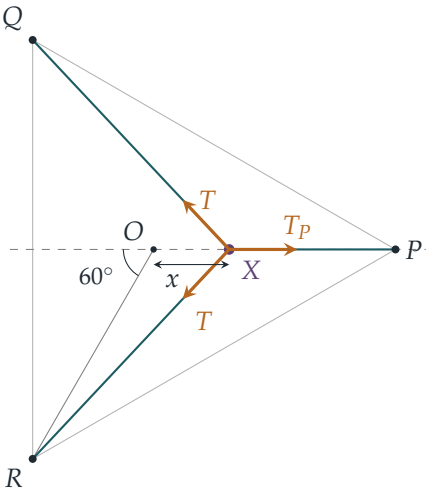
from O, and the lines OP, OQ, OR are at 120° to each other. With X at O, each spring has length c, so all three tensions are equal. Three equal forces at 120° to each other add to zero, so O is an equilibrium position. This is the equilibrium we use. Since $l < \frac{2}{\sqrt{3}}a$, each spring is stretched, with extension

$$\frac{2}{\sqrt{3}}\, a - l,$$

as required.

The tension in RX

Take the x-axis along OP, with P in the positive direction. Let X be at displacement x from O towards P. Then $OX = x$, $OR = c$ and $\angle XOR = 180° - 60° = 120°$.



By the cosine rule in triangle XOR,

$$RX^2 = c^2 + x^2 - 2cx\cos 120° = c^2 + cx + x^2 = \frac{4a^2}{3} + \frac{2ax}{\sqrt{3}} + x^2.$$

So by Hooke's law the tension in RX is

$$T = \frac{\lambda}{l}\left(\sqrt{\frac{4a^2}{3} + \frac{2ax}{\sqrt{3}} + x^2} - l\right),$$

as required. By symmetry in the axis OP, $QX = RX$, so QX has the same tension T.

The equation of motion

Write $r = RX$. The spring PX has length $c - x$, so its tension is $T_P = \frac{\lambda}{l}(c - x - l)$, acting towards P (in the $+x$ direction).

The point R is at x-coordinate $-c\cos 60° = -\frac{c}{2}$, so the displacement from X to R has x-component $-\left(x + \frac{c}{2}\right)$. The force T along XR therefore has x-component $-T\,\dfrac{x + c/2}{r}$. The force from QX has the same x-component, and the two perpendicular components cancel by symmetry. So there is no force across the line OP, and X moves along it.

Ask yourself How do we get a linear equation out of this square root?

First write the exact force, then expand for small x. Resolving along OP,

$$m\ddot{x} = T_P - 2T\,\frac{x + c/2}{r} = \frac{\lambda}{l}\left[(c - x - l) - (2x + c)\left(1 - \frac{l}{r}\right)\right]$$
$$= \frac{\lambda}{l}\left[-3x - l + \frac{l(2x + c)}{r}\right].$$

This equation is exact. Now $r^2 = c^2\left(1 + \frac{x}{c} + \frac{x^2}{c^2}\right)$. Using the binomial expansion and dropping terms in x^2 and higher,

$$\frac{1}{r} \approx \frac{1}{c}\left(1 - \frac{x}{2c}\right), \qquad \frac{2x + c}{r} \approx \left(1 + \frac{2x}{c}\right)\left(1 - \frac{x}{2c}\right) \approx 1 + \frac{3x}{2c}.$$

Substituting,

$$m\ddot{x} \approx \frac{\lambda}{l}\left[-3x - l + l + \frac{3lx}{2c}\right] = -\frac{3\lambda}{l}\left(1 - \frac{l}{2c}\right)x.$$

Since $\frac{l}{2c} = \frac{\sqrt{3}\,l}{4a}$,

$$\ddot{x} \approx -\omega^2 x, \qquad \omega^2 = \frac{3\lambda(4a - \sqrt{3}\,l)}{4mla}.$$

This ω^2 is positive because $\sqrt{3}\,l < 2a < 4a$. So the force is restoring, and for small x the particle performs approximate simple harmonic motion about O with period

$$\frac{2\pi}{\omega} = 2\pi\sqrt{\frac{4mla}{3(4a - \sqrt{3}\,l)\lambda}},$$

as required.

Checks

- At $x = 0$ the exact force is $\frac{\lambda}{l}\left[-l + \frac{lc}{c}\right] = 0$, which agrees with the equilibrium found first.
- Units: λ is a force and l, a are lengths, so $\frac{mla}{a\lambda}$ has units $\mathrm{kg\,m/N} = \mathrm{s}^2$, and the period is a time.
- A second route to ω^2. For a small displacement, a spring contributes stiffness $\frac{\lambda}{l}\cos^2\phi$ from its stretching and $\frac{T_0}{c}\sin^2\phi$ from turning, where ϕ is its angle to OP and $T_0 = \frac{\lambda}{l}(c - l)$ is the equilibrium tension. With $\phi = 0°, 60°, 60°$ the total stiffness is $\frac{3}{2}\cdot\frac{\lambda}{l} + \frac{3}{2}\cdot\frac{T_0}{c} = \frac{3\lambda}{l}\left(1 - \frac{l}{2c}\right)$, as before.
- With $a = \lambda = m = 1$ and $l = 0.8$ the formula gives a period of about 2.838. A computer simulation of the full two-dimensional motion, starting from a small displacement towards P, gives the same value to four significant figures.

Common slips

- Using $\cos 120^\circ = +\frac{1}{2}$, or the wrong angle in the cosine rule. This gives $-\frac{2ax}{\sqrt{3}}$ inside the square root.
- Treating the tension in QX and RX as constant. The change in their *direction* and the change in their *size* both contribute terms linear in x.
- Writing the side of the triangle as a rather than $2a$ when finding the distance from the centre to a vertex.

Beyond STEP

Near any stable equilibrium, the force is approximately linear in the displacement. Here the restoring force is the gradient of the elastic potential energy, and $m\omega^2$ is its second derivative. In two dimensions that second derivative becomes a 2×2 matrix (the Hessian), and its eigenvectors are the *normal modes* of small oscillation. By the threefold symmetry, this system oscillates with the same period in every direction about O, not just along OP. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers small oscillations and coupled systems.

WORKED SOLUTION

STEP III 2016, Question 10

Mechanics • Circular Motion on a Plane

STEP III 2016, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q10

The key idea

While the string is taut and the particle stays on the plane, P is always a distance a from A and lies in the plane, so it moves on a circle *in the plane*, centred at the foot of the perpendicular from A. Within the plane this is just a vertical circle under a reduced gravity $g\sin\alpha$; out of the plane, the string pulls P away from the surface, and the normal reaction must stay positive. So we find the tension, find the reaction, and ask where each is smallest.

Solution

Setting up

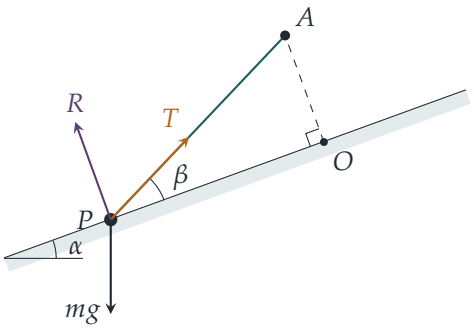
Model P as a particle, the string as light and inextensible, and the plane as smooth. Let O be the foot of the perpendicular from A to the plane. Since the string has length a and makes angle β with the plane,

$$AO = a\sin\beta, \qquad OP = a\cos\beta.$$

In equilibrium the only force with a component along the plane besides gravity is the tension, so the string must lie along a line of greatest slope, with P directly down the slope from O. Resolving along and perpendicular to the plane (tension T_0, reaction R_0):

$$T_0\cos\beta = mg\sin\alpha, \qquad R_0 + T_0\sin\beta = mg\cos\alpha,$$

so $R_0 = mg(\cos\alpha - \sin\alpha\tan\beta) = mg\cos(\alpha+\beta)/\cos\beta$. For the equilibrium to exist at all we therefore need $\cos(\alpha+\beta) > 0$.



The figure is the cross-section through a line of greatest slope at the start. AO is perpendicular to the plane.

Once the impulse is applied, as long as the string stays taut and P stays on the plane, $AP = a$ and $AO = a\sin\beta$, so by Pythagoras $OP = a\cos\beta$ always: P moves on a circle of radius

$$r = a\cos\beta$$

in the plane, centred at O. The string always makes angle β with the plane, and its projection onto the plane points from P to O. The impulse is horizontal and parallel to the plane, so it is tangent to this circle at its lowest point.

Let θ be the angle OP makes with the downward line of greatest slope through O, so $\theta = 0$ at the start, and let v be the speed. The weight has component $mg\sin\alpha$ down the slope (in the plane) and $mg\cos\alpha$ into the plane.

Perpendicular to the plane P has no acceleration, so

$$R + T\sin\beta = mg\cos\alpha. \tag{1}$$

Towards O, within the plane, the acceleration is v^2/r. The tension contributes $T\cos\beta$ and the down-slope weight component contributes $-mg\sin\alpha\cos\theta$:

$$T\cos\beta - mg\sin\alpha\cos\theta = \frac{mv^2}{a\cos\beta}. \tag{2}$$

Energy: the reaction and tension do no work (each is perpendicular to the velocity), and P rises a distance $r(1-\cos\theta)$ along the slope, i.e. a height $r(1-\cos\theta)\sin\alpha$. Hence

$$v^2 = u^2 - 2ga\cos\beta\sin\alpha\,(1-\cos\theta). \tag{3}$$

Not leaving the plane immediately

Just after the impulse, $\theta = 0$ and $v = u$. From (2), $T\cos\beta = mg\sin\alpha + \dfrac{mu^2}{a\cos\beta}$, and then (1) gives

$$\begin{aligned} R &= mg\cos\alpha - \tan\beta\left(mg\sin\alpha + \frac{mu^2}{a\cos\beta}\right) \\ &= \frac{m}{\cos\beta}\left(g\cos\alpha\cos\beta - g\sin\alpha\sin\beta\right) - \frac{mu^2\tan\beta}{a\cos\beta} \\ &= \frac{m}{a\cos\beta}\Big(ag\cos(\alpha+\beta) - u^2\tan\beta\Big). \end{aligned}$$

Since $\cos\beta > 0$, the reaction is positive, and the particle does not immediately leave the plane, if

$$ag\cos(\alpha+\beta) > u^2\tan\beta,$$

as required.

A complete circle

Now use (3) in (2) to get the tension at a general position:

$$T\cos\beta = \frac{mu^2}{a\cos\beta} - 2mg\sin\alpha + 3mg\sin\alpha\cos\theta,$$

and so, from (1),

$$R = mg\cos\alpha - \tan\beta\left(\frac{mu^2}{a\cos\beta} - 2mg\sin\alpha + 3mg\sin\alpha\cos\theta\right).$$

Ask yourself Where on the circle is the particle most likely to leave the plane: at the top or at the bottom?

Intuition from a vertical circle says "the top", but here that's the danger point for the *string*, not the plane. The string pulls P away from the plane with force $T\sin\beta$, and the tension is greatest at the bottom, where P is moving fastest and gravity in the plane also pulls away from O. Indeed R is smallest when $\cos\theta$ is largest, i.e. at $\theta = 0$, and T is smallest when $\cos\theta = -1$, at the top. So for a complete circle in contact we need two things.

- **Contact at the bottom.** $R > 0$ at $\theta = 0$, which is exactly the condition of the first part: $u^2\tan\beta < ag\cos(\alpha+\beta)$.
- **String taut at the top.** $T \geq 0$ at $\theta = \pi$: $\dfrac{u^2}{a\cos\beta} - 5g\sin\alpha \geq 0$, i.e. $u^2 \geq 5ag\sin\alpha\cos\beta$. (This also makes $v^2 > 0$ at the top, by (3).)

Both can hold only if, multiplying the second by $\tan\beta > 0$,

$$5ag\sin\alpha\cos\beta\tan\beta \le u^2\tan\beta < ag\cos(\alpha+\beta),$$

that is,

$$5\sin\alpha\sin\beta < \cos\alpha\cos\beta - \sin\alpha\sin\beta \iff 6\sin\alpha\sin\beta < \cos\alpha\cos\beta.$$

Dividing by $\cos\alpha\cos\beta > 0$ gives the necessary condition

$$6\tan\alpha\tan\beta < 1.$$

(It is also sufficient: if it holds, any u with $5ag\sin\alpha\cos\beta \le u^2 < ag\cos(\alpha+\beta)/\tan\beta$ works, because R and T are then positive everywhere on the circle.)

Checks

- With $u = 0$ the reaction from the first part becomes $mg\cos(\alpha+\beta)/\cos\beta$, matching the equilibrium value R_0.
- If $\beta \to 0$ (string almost along the plane) the string cannot lift the particle, and the condition $6\tan\alpha\tan\beta < 1$ holds automatically, as it should.
- With $\alpha = 30°$, $\beta = 5°$ (so $6\tan\alpha\tan\beta \approx 0.30$), $a = 1\,\text{m}$ and $g = 9.8\,\text{m}\,\text{s}^{-2}$, the allowed range is $24.4 \le u^2 < 91.8$ (in $\text{m}^2\,\text{s}^{-2}$). Taking $u^2 = 50$ gives a smallest reaction of about $0.37mg$ at the bottom and a smallest tension of about $2.63mg$ at the top. We confirmed by computer, by solving Newton's second law in three dimensions all round the circle, that R and T stay positive, with the values above.

Common slips

- Using the full string length a as the radius of the circle: the radius is $a\cos\beta$.
- Assuming the particle is most likely to leave the plane at the top of the circle. The reaction is smallest at the bottom.
- Using g instead of $g\sin\alpha$ in the energy equation: the particle rises only $r(1-\cos\theta)\sin\alpha$ vertically.

Beyond STEP

This is motion under constraints: the plane and the string each supply a reaction force whose size is whatever is needed to keep the constraint, and the motion stays valid only while those forces have the right sign. In a first-year course you'd learn to handle this systematically, using Lagrangian mechanics to remove the constraints and recover the reactions afterwards. Kleppner and Kolenkow, *An Introduction to Mechanics*, has many problems of exactly this "when does it lose contact?" kind.

WORKED SOLUTION

STEP III 2016, Question 11

Mechanics • Power and Resistance

STEP III 2016, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

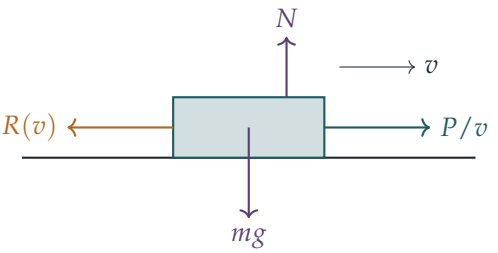
step.maths.org/questions/16-s3-q11

The key idea

An engine working at constant rate P exerts a driving force P/v at speed v. Since we want *distance* as a function of speed, not time, we write the acceleration as $v\frac{\mathrm{d}v}{\mathrm{d}x}$. That turns Newton's second law into a separable equation for x in terms of v. The fact that the acceleration vanishes at $4U$ fixes the resistance constant in terms of P and U, and each distance then comes down to a single definite integral.

Solution

We model the car as a particle of mass m moving along a horizontal line, with tractive force P/v forwards and resistance $R(v)$ backwards. Vertically the weight and normal reaction balance and play no further part.



Using $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$, the horizontal equation of motion is

$$mv\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{P}{v} - R(v), \qquad \text{so} \qquad \frac{\mathrm{d}x}{\mathrm{d}v} = \frac{mv^2}{P - vR(v)}.$$

In both models $vR(v)$ increases with v and equals P at $v = 4U$. So for $U \le v \le 2U$ the denominator is positive, and the car really is accelerating throughout, which means x increases with v and the distance is

$$X = \int_U^{2U} \frac{mv^2}{P - vR(v)}\,\mathrm{d}v.$$

Part (i)

Let $R = kv$. Zero acceleration at $4U$ gives $\dfrac{P}{4U} = 4kU$, so $k = \dfrac{P}{16U^2}$. Then

$$P - kv^2 = k\,(16U^2 - v^2), \qquad \frac{m}{k} = \frac{16mU^2}{P} = \frac{1}{\lambda U}.$$

Hence

$$\lambda X_1 = \frac{1}{U}\int_U^{2U} \frac{v^2}{16U^2 - v^2}\,\mathrm{d}v = \int_1^2 \frac{w^2}{16 - w^2}\,\mathrm{d}w,$$

substituting $v = Uw$. Writing $\dfrac{w^2}{16 - w^2} = -1 + \dfrac{16}{16 - w^2} = -1 + \dfrac{2}{4 - w} + \dfrac{2}{4 + w}$,

$$\begin{aligned}\lambda X_1 &= \Big[-w + 2\ln\frac{4 + w}{4 - w}\Big]_1^2 \\ &= -1 + 2\ln 3 - 2\ln\tfrac{5}{3} = 2\ln\tfrac{9}{5} - 1,\end{aligned}$$

as required.

Part (ii)

Now let $R = cv^2$. Zero acceleration at $4U$ gives $\dfrac{P}{4U} = 16cU^2$, so $c = \dfrac{P}{64U^3}$. Then

$$P - cv^3 = c\,(64U^3 - v^3), \qquad \frac{m}{c} = \frac{64mU^3}{P} = \frac{4}{\lambda}.$$

Ask yourself Is there a neat antiderivative here, or do we need partial fractions again?

The numerator v^2 is, up to a constant, the derivative of the denominator $64U^3 - v^3$. So this one is a logarithm straight away:

$$\begin{aligned}\lambda X_2 &= 4\int_U^{2U} \frac{v^2}{64U^3 - v^3}\,\mathrm{d}v = 4\Big[-\tfrac{1}{3}\ln\big(64U^3 - v^3\big)\Big]_U^{2U} \\ &= \tfrac{4}{3}\ln\frac{63U^3}{56U^3} = \tfrac{4}{3}\ln\tfrac{9}{8},\end{aligned}$$

as required.

Part (iii)

Since $\lambda > 0$, $X_1 > X_2$ exactly when $\lambda X_1 - \lambda X_2 > 0$. We only have bounds for $\ln 24$ and $\ln 5$, so we rewrite the difference in terms of those:

$$\begin{aligned}\lambda(X_1 - X_2) &= 2\ln 9 - 2\ln 5 - 1 - \tfrac{4}{3}\ln 9 + \tfrac{4}{3}\ln 8 \\ &= \tfrac{4}{3}\ln 3 + 4\ln 2 - 2\ln 5 - 1 \\ &= \tfrac{4}{3}(\ln 3 + 3\ln 2) - 2\ln 5 - 1 = \tfrac{4}{3}\ln 24 - 2\ln 5 - 1.\end{aligned}$$

Ask yourself Which way round should we use each bound?

To show the difference is positive we need a *lower* bound, so we use the lower bound for $\ln 24$ (which appears with a plus sign) and the upper bound for $\ln 5$ (which appears with a minus sign):

$$\lambda(X_1 - X_2) > \tfrac{4}{3}(3.17) - 2(1.61) - 1 = 4.22\dot{6} - 3.22 - 1 = 0.00\dot{6} > 0.$$

Therefore $X_1 > X_2$: the car covers more ground getting from U to $2U$ when the resistance is proportional to speed than when it is proportional to speed squared.

This makes physical sense. Both resistances agree at $v = 4U$, but for $v < 4U$ we have $cv^2 = kv \cdot \frac{v}{4U} < kv$. So the squared law resists *less* at the speeds in question, the car accelerates harder, and it reaches $2U$ sooner.

Checks

- Numerically, $\lambda X_1 = 2\ln 1.8 - 1 \approx 0.1756$ and $\lambda X_2 = \frac{4}{3}\ln 1.125 \approx 0.1570$, so the difference is about 0.0185, which is positive, as found. A computer simulation stepping the equation of motion in time from $v = U$ reproduces both values to five decimal places.
- Units: $P/(mU^3)$ has units $\mathrm{W}/(\mathrm{kg\,m^3\,s^{-3}}) = \mathrm{m^{-1}}$, so λX is dimensionless, as it should be.
- The physical argument at the end of part (iii) predicts the same ordering independently.

Common slips

- Using $a = \frac{\mathrm{d}v}{\mathrm{d}t}$, which gives time rather than distance. Here we need $v\,\frac{\mathrm{d}v}{\mathrm{d}x}$.
- Using the bounds the wrong way round in part (iii). Using 3.18 and 1.60 only gives an upper bound, which proves nothing about the sign.
- Forgetting that the margin is tiny ($0.00\dot{6}$), so rounding $\frac{4}{3}(3.17)$ to 4.2 loses the result.

Beyond STEP

Writing $a = v\,\mathrm{d}v/\mathrm{d}x$ is the first step towards the energy method of university mechanics. Multiplying the equation of motion by v gives the rate of change of kinetic energy, which equals power in minus power dissipated. Speeds like $4U$, where the two balance, are *equilibria* of a first-order differential equation, and the study of how solutions approach them is the beginning of dynamical systems. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats work, power and velocity-dependent forces in this style.

WORKED SOLUTION

STEP III 2016, Question 12

Probability and Statistics • Chebyshev's Inequality

STEP III 2016, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q12

The key idea

Chebyshev's inequality bounds the probability of being *far* from the mean, so a statement that something is *close* to the mean comes from the complement. In each part the job is to spot the right random variable, find its mean and standard deviation, and choose k so that $k\sigma$ is exactly the distance we care about. In part (ii) the sum $\sum n^r/r!$, sitting next to e^n, is the fingerprint of a Poisson distribution.

Solution

Part (i)

Let X be the number of heads in the $100n$ throws. Assuming the throws are independent, each landing heads with probability 0.2, we have $X \sim \mathrm{B}(100n, 0.2)$, so

$$\mu = 100n \times 0.2 = 20n, \qquad \sigma^2 = 100n \times 0.2 \times 0.8 = 16n, \qquad \sigma = 4\sqrt{n}.$$

The event $16n \le X \le 24n$ is exactly $|X - 20n| \le 4n$, so

$$\alpha = \mathrm{P}\big(|X-\mu| \le 4n\big) = 1 - \mathrm{P}\big(|X-\mu| > 4n\big).$$

We want $k\sigma = 4n$, that is $k \cdot 4\sqrt{n} = 4n$, so $k = \sqrt{n}$, which is positive. Chebyshev's inequality gives

$$\mathrm{P}\big(|X-\mu| > 4n\big) = \mathrm{P}\big(|X-\mu| > \sqrt{n}\,\sigma\big) \le \frac{1}{(\sqrt{n})^2} = \frac{1}{n},$$

and therefore

$$\alpha \ge 1 - \frac{1}{n},$$

as required. (The condition $n > 1$ just makes the bound say something: for $n = 1$ it reads $\alpha \ge 0$.)

Part (ii)

Ask yourself Which random variable has probabilities that look like $n^r/r!$?

If $Y \sim \mathrm{Po}(n)$ then $\mathrm{P}(Y = r) = \mathrm{e}^{-n} n^r / r!$ for $r = 0, 1, 2, \ldots$, so

$$\mathrm{e}^{-n}\left(1 + n + \frac{n^2}{2!} + \cdots + \frac{n^{2n}}{(2n)!}\right) = \mathrm{P}(Y \le 2n).$$

The required inequality, after dividing by $\mathrm{e}^n > 0$, is therefore equivalent to

$$\mathrm{P}(Y \le 2n) \ge 1 - \frac{1}{n}.$$

For a Poisson distribution the mean and variance are both n, so $\mu = n$ and $\sigma = \sqrt{n}$.

Ask yourself Chebyshev's inequality is two-sided, but $Y \le 2n$ is a one-sided event. How do they match up?

Because $Y \ge 0$, we always have $Y - n \ge -n$, so $Y - n < -n$ is impossible. Hence

$$|Y - n| > n \iff Y - n > n \iff Y > 2n.$$

Taking $k = \sqrt{n}$, so that $k\sigma = n$, Chebyshev's inequality gives

$$\mathrm{P}(Y > 2n) = \mathrm{P}\big(|Y - n| > \sqrt{n}\,\sigma\big) \le \frac{1}{n},$$

so $\mathrm{P}(Y \le 2n) = 1 - \mathrm{P}(Y > 2n) \ge 1 - \frac{1}{n}$. Multiplying by e^n,

$$1 + n + \frac{n^2}{2!} + \cdots + \frac{n^{2n}}{(2n)!} \ge \left(1 - \frac{1}{n}\right)\mathrm{e}^n,$$

which is the result.

Checks

The bounds are true but generous. For $n = 2$ the exact binomial probability in (i) is $\alpha \approx 0.868$, against the bound 0.5; for $n = 10$ it is $\alpha \approx 0.9986$, against 0.9. In (ii) with $n = 2$ the left-hand side is $1 + 2 + 2 + \frac{4}{3} + \frac{2}{3} = 7$, while $\frac{1}{2}\mathrm{e}^2 \approx 3.69$. We checked both inequalities exactly for every n from 2 to 30 in (i), and from 1 to 59 in (ii).

Common slips

- Using the variance $16n$ in place of the standard deviation $4\sqrt{n}$ when choosing k.
- Forgetting that Chebyshev bounds the probability of being *outside* the interval, so we need the complement to get α.
- In (ii), not explaining why the lower tail $Y < 0$ contributes nothing.

Beyond STEP

Chebyshev's inequality is the simplest example of a *concentration inequality*, and applying it to an average of m independent trials gives the weak law of large numbers: the spread shrinks like $1/\sqrt{m}$. Part (ii) also says that the tail of the exponential series beyond $n^{2n}/(2n)!$ is at most e^n/n, which is an analytic fact proved by probability. Sharper tools such as Chernoff bounds show the true probabilities are exponentially close to 1. For more, see Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP III 2016, Question 13

Probability and Statistics • Kurtosis of Sums

STEP III 2016, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q13

The key idea

Kurtosis is built only from deviations from the mean, so anything that shifts the variable leaves it unchanged; that lets us centre every variable at 0 and work with plain moments. For the Normal distribution, the fourth moment comes from integrating by parts, peeling off one factor of x to pair with the exponential. For a sum of independent mean-zero variables, expanding T^4 and taking expectations kills every term that contains some variable to the first power. This is the whole of part (ii), and it feeds straight into part (iii).

Solution

Shifting. Let $W = X - a$. By linearity of expectation, $\mathrm{E}(W) = \mu - a$, so

$$W - \mathrm{E}(W) = (X - a) - (\mu - a) = X - \mu.$$

The deviation of W from its mean is exactly the deviation of X from its mean. So W has the same variance $\mathrm{E}\big((X-\mu)^2\big) = \sigma^2$, the same fourth central moment $\mathrm{E}\big((X-\mu)^4\big)$, and hence the same kurtosis. In particular, taking $a = \mu$, every random variable has the same kurtosis as its centred version $X - \mu$, which has mean 0.

Part (i)

Let $X \sim \mathrm{N}(0, \sigma^2)$, with density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\,\mathrm{e}^{-x^2/(2\sigma^2)}, \qquad \text{so that} \qquad f'(x) = -\frac{x}{\sigma^2}\,f(x).$$

Since $\mu = 0$, we need $\mathrm{E}(X^4) = \int_{-\infty}^{\infty} x^4 f(x)\,\mathrm{d}x$.

Ask yourself How do we integrate $x^4\mathrm{e}^{-x^2/(2\sigma^2)}$ when $\mathrm{e}^{-x^2/(2\sigma^2)}$ has no elementary antiderivative?

The function $xf(x)$ *does* have an antiderivative, namely $-\sigma^2 f(x)$. So write $x^4 f(x) = x^3 \cdot xf(x)$ and integrate by parts with $u = x^3$ and $\mathrm{d}v = xf(x)\,\mathrm{d}x$, $v = -\sigma^2 f(x)$:

$$\int_{-\infty}^{\infty} x^4 f(x)\,\mathrm{d}x = \Big[-\sigma^2 x^3 f(x)\Big]_{-\infty}^{\infty} + 3\sigma^2 \int_{-\infty}^{\infty} x^2 f(x)\,\mathrm{d}x = 3\sigma^2 \int_{-\infty}^{\infty} x^2 f(x)\,\mathrm{d}x.$$

The boundary term vanishes because $x^3 \mathrm{e}^{-x^2/(2\sigma^2)} \to 0$ as $x \to \pm\infty$ (the exponential decays faster than any power grows). The same step with $u = x$ gives

$$\int_{-\infty}^{\infty} x^2 f(x)\,\mathrm{d}x = \Big[-\sigma^2 x f(x)\Big]_{-\infty}^{\infty} + \sigma^2 \int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = \sigma^2,$$

since f is a probability density and integrates to 1. (This confirms that the variance is σ^2.) Hence $\mathrm{E}(X^4) = 3\sigma^2 \cdot \sigma^2 = 3\sigma^4$, and

$$\kappa = \frac{3\sigma^4}{\sigma^4} - 3 = 0.$$

By the shifting result, every Normal distribution, whatever its mean, has kurtosis 0.

Part (ii)

Expand $T^4 = (Y_1 + \cdots + Y_n)^4$ by the multinomial theorem. Each term is a product of four factors Y_r, and the terms fall into five types according to how the four factors are shared among distinct indices:

$$Y_r^4, \qquad 4\,Y_r^3 Y_s, \qquad 6\,Y_r^2 Y_s^2, \qquad 12\,Y_r^2 Y_s Y_t, \qquad 24\,Y_r Y_s Y_t Y_u,$$

where the different letters stand for different indices. The coefficients count the arrangements: for example, $Y_r^2 Y_s^2$ arises from choosing which 2 of the 4 brackets supply Y_r, giving $\binom{4}{2} = 6$ ways. Each unordered pair $\{r, s\}$ appears once in the $Y_r^2 Y_s^2$ type.

Now take expectations. The Y's are independent, so the expectation of a product of powers of different Y's is the product of their expectations. In the types $Y_r^3 Y_s$, $Y_r^2 Y_s Y_t$ and $Y_r Y_s Y_t Y_u$, some variable appears to the first power only, and its factor $\mathrm{E}(Y_s) = 0$ makes the whole expectation zero. For example,

$$\mathrm{E}(Y_r^3 Y_s) = \mathrm{E}(Y_r^3)\,\mathrm{E}(Y_s) = 0.$$

Only the first and third types survive, and $\mathrm{E}(Y_r^2 Y_s^2) = \mathrm{E}(Y_r^2)\,\mathrm{E}(Y_s^2)$ for $r \neq s$, again by independence. Listing the pairs as $r < s$,

$$\mathrm{E}(T^4) = \sum_{r=1}^{n} \mathrm{E}(Y_r^4) + 6 \sum_{r=1}^{n-1} \sum_{s=r+1}^{n} \mathrm{E}(Y_r^2)\,\mathrm{E}(Y_s^2),$$

as required. (Note that this part used independence and zero means, but not that the Y's are identically distributed.)

Part (iii)

Let each X_r have mean μ and standard deviation σ, and let $S = X_1 + \cdots + X_n$. Put $Y_r = X_r - \mu$. The Y_r are independent and identically distributed with mean 0, and

$$\mathrm{E}(Y_r^2) = \sigma^2, \qquad \mathrm{E}(Y_r^4) = (\kappa + 3)\sigma^4,$$

the second by rearranging the definition of κ. Then $T = \sum Y_r = S - n\mu$. By the shifting result, S has the same kurtosis as T, so we work with T, which has mean 0.

By independence, $\operatorname{Var}(T) = \sum \operatorname{Var}(Y_r) = n\sigma^2$. The double sum in part (ii) has $\binom{n}{2} = \frac{1}{2}n(n-1)$ terms, each equal to σ^4, so

$$\begin{aligned} \mathrm{E}(T^4) &= n(\kappa+3)\sigma^4 + 6\cdot\tfrac{1}{2}n(n-1)\sigma^4 \\ &= n\sigma^4\big(\kappa + 3 + 3(n-1)\big) = n\sigma^4(\kappa+3n). \end{aligned}$$

Therefore the kurtosis of T, and so of S, is

$$\frac{\mathrm{E}(T^4)}{(\operatorname{Var} T)^2} - 3 = \frac{n\sigma^4(\kappa+3n)}{n^2\sigma^4} - 3 = \frac{\kappa}{n} + 3 - 3 = \frac{\kappa}{n},$$

as required.

Checks

- Take $Y = \pm 1$ with probability $\frac{1}{2}$ each. Then $\mathrm{E}(Y^4) = \sigma^4 = 1$, so $\kappa = 1 - 3 = -2$. For $n = 2$, the sum takes values $-2, 0, 2$ with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$, so $\mathrm{E}(T^4) = 8$ and $\operatorname{Var} T = 2$. This gives kurtosis $8/4 - 3 = -1 = \kappa/2$, as predicted.
- With $n = 1$, part (iii) says nothing changes, as it must.
- Since a sum of independent Normals is Normal, part (iii) with $\kappa = 0$ must give 0 again, and it does.
- We confirmed by computer, using exact fractions, that the kurtosis of a sum of up to six independent copies of a three-valued variable is exactly κ/n. We also checked numerically that $\int x^4 f(x)\,\mathrm{d}x = 3\sigma^4$ for several values of σ.

Common slips

- Getting the coefficient of $Y_r^2 Y_s^2$ wrong: it is $\binom{4}{2} = 6$ per unordered pair (or 3 per ordered pair), not 4 or 12.
- Forgetting to centre the X_r before using part (ii), which needs mean 0.
- Dividing by σ^4 instead of $(n\sigma^2)^2$: the variance of the sum is $n\sigma^2$.

Beyond STEP

Kurtosis as defined here is the fourth *cumulant* divided by σ^4. Cumulants are the coefficients in the expansion of the logarithm of the moment generating function, and they simply add for independent sums. That is why the standardised fourth cumulant of a sum shrinks like $1/n$, and the third (skewness) like $1/\sqrt{n}$. This is a quantitative glimpse of the central limit theorem: the sum looks more and more Normal,

whose cumulants beyond the second are all zero. Grimmett and Welsh, *Probability: An Introduction*, develops generating functions and the central limit theorem at this level.

Beyond STEP

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